

Characterizing network-specific pathophysiology to guide targeted therapeutic interventions in parkinsonian motor syndromes

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Preface

*Dedicated to Anna Ajoba, Aiji, Aba, and Suneeta Aiji, for laying the foundation
on which we could build our dreams*

कर्मण्येवाधिकारस्ते मा फलेषु कदाचन।
मा कर्मफलहेतुर्भूर्मा ते सङ्गोऽस्त्वकर्मणि॥

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Characterizing network-specific pathophysiology to guide targeted therapeutic interventions for motor recovery in parkinsonian motor syndromes

By

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Abstract

Parkinson's disease (PD) is a debilitating movement disorder that affects more than 10 million people worldwide. Symptoms that include tremor, bradykinesia and gait difficulties, severely limit the quality of life of patients. Treatments like dopamine-replacement medications and deep brain stimulation effectively ameliorate motor symptoms. However, there is considerable variability in the symptom-profiles and therapeutic responses, limiting therapeutic optimization. We used this as an opportunity to tackle both mechanisms and therapies that engage circuits that underlie motor symptoms seen in PD. In this thesis we describe two main undertakings: 1) investigating the neural activity in the thalamus of two models of Parkinson's pathology that show either tremor or bradykinesia and 2) using insights from optogenetically induced cell-type modulation to identify patterns and targets of deep brain stimulation (DBS) that enhance the duration of motor recovery in a dopamine depletion model showing bradykinesia.

In the introduction we outline the circuitry underlying normal motor control, and its dysfunction in the context of Parkinson's disease. We summarize the neuromodulation approaches studied in pre-clinical and clinical models of movement disorders and highlight the need for circuit-inspired treatments that can interrupt pathological loops of neural activity. The following chapters discuss the neurophysiology underlying synuclein-induced tremor and dopamine depletion-induced bradykinesia, providing a mechanistic framework to design future symptom-specific therapeutic targeting. Finally, we provide a translatable electrical stimulation paradigm for the long-lasting improvement of dopamine depletion-induced bradykinesia that can impact the standard of care for Parkinson's disease patients by maximizing their therapeutic duration, minimizing side effects, and potentially altering their pathological circuitry.

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Chapter 1

1 Introduction

1.1 The Circuitry implicated in Parkinson's disease (PD)

Translating thought into action requires a coordinated effort from multiple regions in the brain that fine-tune their activity towards this common goal. Parkinson's disease is a neurodegenerative movement disorder characterized by symptoms like slowing of movement, difficulty walking, and a predominant tremor, caused by dysfunction in this system. Understanding the neural circuitry underlying these symptoms, requires us to establish the role of various nodes in the brain that are involved in healthy motor control. This includes the basal ganglia, motor cortex, thalamus, and the cerebellum. Neurons in these regions of the brain are structurally and functionally connected to create specific sub-circuits governing movement initiation, vigor, execution, and error correction.

In a disease state, these circuits become susceptible to cell loss, aberrant firing patterns, or structural changes that can affect their functional output. Previously the brain regions involved in disease pathology were thought to be homogenously altered in the context of disease, however growing evidence has led to the understanding of complex cell-type specific changes that might differentially encode pathological states. Characterization of these cell-type or projection-specific alterations in the disease state can guide therapeutic interventions that go beyond masking symptoms, and instead attempt to knock these circuits out of their pathological functional states. Here, we will describe the parallel loops originating from either the basal ganglia or the cerebellum that pass through the thalamus updating top-down motor commands, to initiate and execute movements in healthy and disease states.

1.1.1 Role of the cortico-basal ganglia-thalamo-cortical circuit in PD

The cortico-basal ganglia-thalamo-cortical loop starts in the motor cortex, which sends a command to the basal ganglia which relays the updated signal back to the cortex via the thalamus.

The basal ganglia are a set of interconnected subcortical nuclei (groups of neurons responsible for specific functions), that form the neural substrate for appropriate motor control. The classical simplified model of basal ganglia motor control involves two primary pathways, the direct and indirect pathway, originating in the striatum which is the input nucleus and ending in the substantia nigra pars reticulata (SNr) or the globus pallidus interna (GPi), which are the output nuclei. The direct pathway comprises of inhibitory projections from the D1 neurons in the striatum directly to the output nuclei. Whereas the indirect pathway has intermediate synapses in the globus pallidus externa (GPe) and the subthalamic nucleus (STN). In the normal state, the midbrain dopaminergic neurons from the substantia nigra pars compacta (SNc) modulate the two cell-types (D1 and D2) in the striatum such that the direct pathway facilitates desired movement, whereas the indirect pathway suppresses undesired movement. The output nucleus of the basal ganglia pathways is primarily the SNr in rodents, and a combination of the SNr and GPi in primates. These output projections send an inhibitory input to the basal ganglia-recipient thalamus which loops this information back to the motor cortex. The basal ganglia are part of motor, associative and limbic parallel circuits to and from the cortex via the thalamus (Alexander, 1986; Haber, 2003; Parent & Hazrati, 1995). The motor loop is involved in automatic execution of learned motor plans (Foster et al., 2021)). The thalamus serves as an integrator of top-down information from the motor cortex, as well as bottom-up information from the SNr regarding specific aspects of movement encoding.

The primary pathology in Parkinson's disease is thought to be the loss of dopaminergic neurons in the SNc of the midbrain that produces a dopamine-deficient state (Albin et al., 1989), leading to a disinhibition of the inhibitory output nuclei, hence producing an overall state of reduced movement or akinesia (Lang, 1998) as predicted by the rate model of basal ganglia function. Hence therapeutic approaches have focused on stimulating the GPe or functionally lesioning the STN, or GPi/SNr (Okun & Vitek, 2004; Wichmann & DeLong, 2016) to circumvent the excessive inhibition in the pathological state. The pathological effects of dopamine deficiency have been reflected via a change, not only in the rate, but also the pattern of firing of nodes in the basal ganglia circuit, including a dampened activity in the GPe, pathologically synchronous firing in the SNr, and pathological oscillations in the STN (Bosch-Bouju et al., 2013). Hence, there is growing evidence for studying basal

ganglia structures not just as singular entities but by looking at specific cell-types within them or projection patterns from the basal ganglia to the thalamocortical or brainstem pathways (Dodson et al., 2015; Jeon et al., 2022, 2022; Lilascharoen et al., 2021; Mastro et al., 2014; McElvain et al., 2021). Previous work from our lab has demonstrated the importance of differentially modulating the PV and Lhx6 cell-types within the GPe in the context of improving parkinsonian therapies (Mastro et al., 2017). Our understanding of the collateral input from D1 neurons to the GPe as well as the nuances of projection pattern from GPe cell-types to the different parts of the SNr have led to a more complex version of the classic circuit (Fig 1.1). While there is some evidence for mild changes in thalamic activity, like inhibition in the firing rate and an increase in the bursts seen in thalamic firing patterns in primates (Magnin et al., 2000; Pessiglione et al., 2005; Schneider & Rothblat, 1996; Zirh et al., 1998), the role of the thalamus in the pathophysiology of the akinesia or bradykinesia induced by the dopamine loss remains unclear.

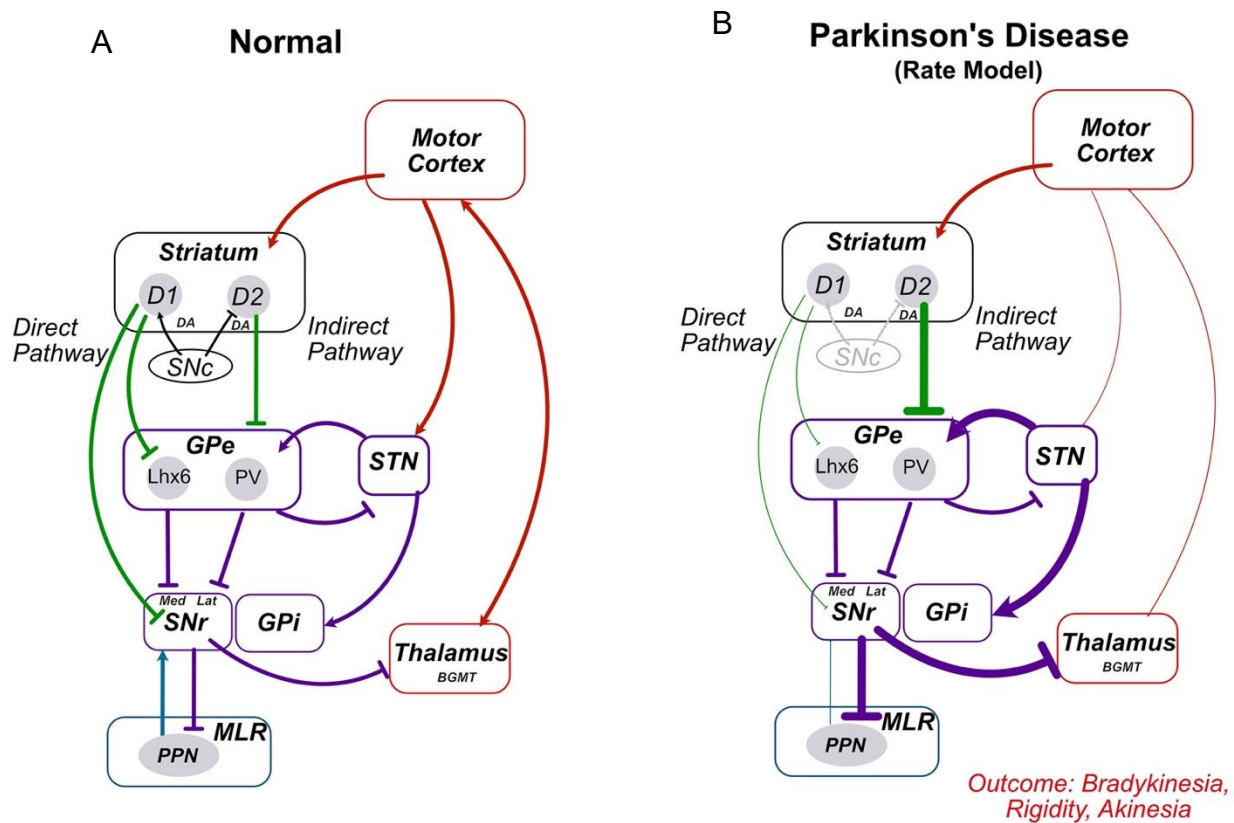


Figure 1.1 The cortico-basal ganglia-thalamo-cortical circuit in normal and disease states (updated classical model).

A, In the healthy state, motor commands from the motor cortex are sent to the striatum which contains D1 and D2 neurons that are modulated by dopamine released from the midbrain SNc dopamine neurons. The D1 forms the direct pathway by sending inhibitory input directly to the basal ganglia output nuclei SNr, GPi. Collaterals from D1 neurons are sent to the GPe that might be critical for cell-type specific modulation. In the indirect pathway, D2 neurons send inhibitory

input to the GPe which makes reciprocal connections with the STN (excitatory). The GPe then projects to the SNr (some evidence for cell-type specific projections to medial vs lateral SNr). The SNr is reciprocally connected to the MLR for downstream movement pathways and sends inhibitory signals to the basal ganglia recipient thalamus which then projects back reciprocally to the premotor/motor cortex. **B**, Rate model predictions in a parkinsonian state, where the loss of dopaminergic input from the SNc, produces a weakened direct pathway and strong indirect pathway leading to overall disinhibition of the SNr, and an increased inhibition of the MLR as well as the thalamocortical projections from the SNr leading to symptoms like rigidity, bradykinesia, and akinesia.

Motor thalamic regions (sub-nuclei including the VM, and the VA) are interconnected with associative and premotor cortices (Bosch-Bouju et al., 2013) based on primate studies. Tracing studies from rats have established that the basal ganglia recipient motor thalamus (BGMT) comprising of both the VM and rostroventral VA-VL complex, receive inhibitory inputs from the SNr and the entopeduncular nucleus (EPN) and send projections to layer I of multiple cortical areas. The BGMT projects to layer I of motor cortical areas including M1, M2 and sensorimotor cortices, as well as the neostriatum. Specifically, the VM within the BGMT projects to layer I in a more widespread rostro-caudal cortical extent including to non-motor regions like the orbital and cingulate cortices. (Kuramoto et al., 2015a, Herkenham, 1979). This pathway is thought to be involved in the pre-motor preparatory phase wherein the basal ganglia convey information to the apical dendrites of layer II/III or layer V pyramidal neurons to modulate the gain of the pyramidal cell response. (Kuramoto et al., 2009; Larkum et al., 2004)(Fig 1.2).

1.1.2 The cerebello-thalamo-cortical circuit and its association with tremor

In the context of motor control, the cerebellum integrates somatosensory and proprioceptive inputs from ascending spinocerebellar tracts and the efference copy it receives from the cortex via the brainstem (pons) as a form of the descending motor commands to predict errors in motor performance (Ishikawa et al., 2014, 2016; Tomatsu et al., 2016). Movement-related activity in the deep cerebellar nuclei (DCN) that send projections to the cerebellar-recipient thalamus are thought to coordinate activity across joints and encode proprioceptive information and motor learning without directly controlling movement (Hewitt et al., 2011; Horne & Butler, 1995; Thach, 2025). The caudo-dorsal part of the VA-VL complex, established as the cerebellar-recipient motor thalamus (VAL) receives excitatory inputs from the cerebellum and sends projections to intermediate layers (III-V) of motor cortical areas including M1, M2, somatosensory areas and not layer I or the neostriatum like the BGMT. Layer IV pyramidal neurons in motor areas receive these cerebellar motor commands and

transfer it to layer V corticospinal neurons. This may serve as a ‘go’ signal that activates downstream motor execution signals to the spinal cord (Kuramoto et al., 2009)(Fig. 1.2). Hence the cerebellar-recipient thalamus forms reciprocal connections with the motor cortex to fine-tune movement execution (Kuramoto et al., 2015b). In the context of movement-related changes in activity, the basal ganglia and cerebellar territories of the motor thalamus exhibit similar movement-related changes in activity involving an increase in firing preceding and at the onset of movement initiation (Anderson & Turner, 1991; Nambu et al., 1991).

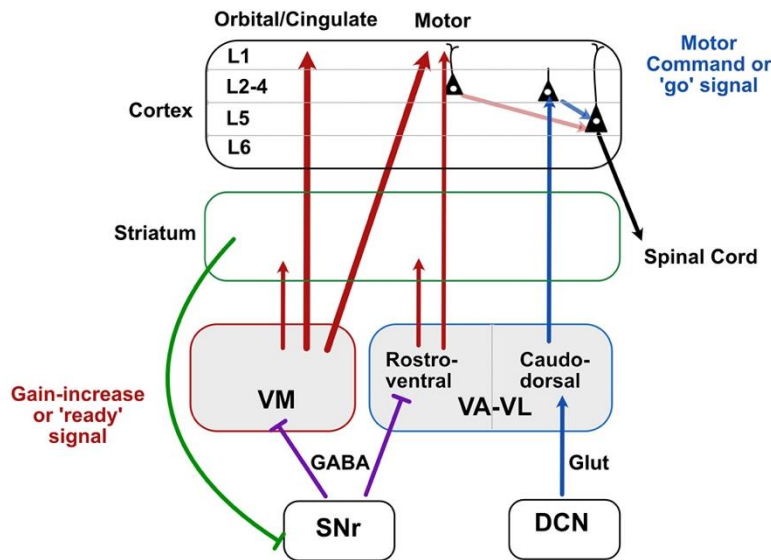


Figure 1.2 Projections from the basal-ganglia and cerebellar thalamus to the cortex.

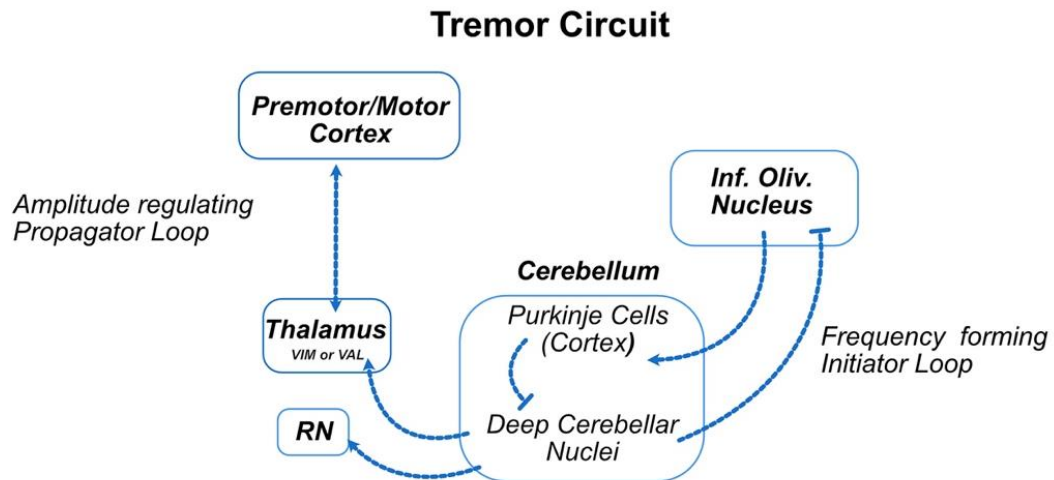
(Adapted from Kuramoto et al, 2009, 2015). VM and rostroventral VAL receive inhibitory inputs from the SNr and project to Layer 1 of the cortex and the neostriatum forming a feedback loop that is essential for a gain increase or preparatory signal. Caudo-dorsal VAL receives excitatory inputs from the DCN and projects to Layers 3-5 of the motor cortex and not to the neostriatum. These projections send motor commands to Layer 5 corticospinal tract neurons to execute movement forming the go-signal.

*For the purposes of this thesis, VM will be used to refer to the BGMT defined based on projections and includes both VM and rostroventral VA-VL. VAL refers to the cerebellar-recipient thalamus and includes the caudodorsal VA-VL.

While the role of the cerebellum in encoding motor dysfunction in various movement disorders including PD remains unclear, there is evidence for distinct pathological signatures in the cerebellum for disorders like dystonia, tremor, and ataxia (Van Der Heijden et al., 2024). Pathophysiological changes observed in the cerebellum and the cerebellar-recipient thalamus in various tremor models, implicates this pathway in the production of tremor (Haubenberger & Hallett, 2018; Hewitt et al., 2011). The regular tremor observed in essential tremor (ET) is thought to arise

from pathological oscillations in the dentato-olivo-cerebellar loop (Bazzigaluppi et al., 2012; Handforth & Lang, 2021; Kakei et al., 2021) (Fig. 1.3A) which is the frequency forming aspect of the circuit. The thalamus is thought to act as a gating resistor that regulates the amplitude and severity of the tremor (Pan, 2025). In the context of Parkinson's disease, the dimmer-switch model suggests that the basal ganglia triggers the tremor, whereas the cerebellothalamocortical network produces the tremor signal (Helmich & Hallett, 2012)(Fig. 1.3B).

A



B

Parkinson's Disease: Tremor and Bradykinesia

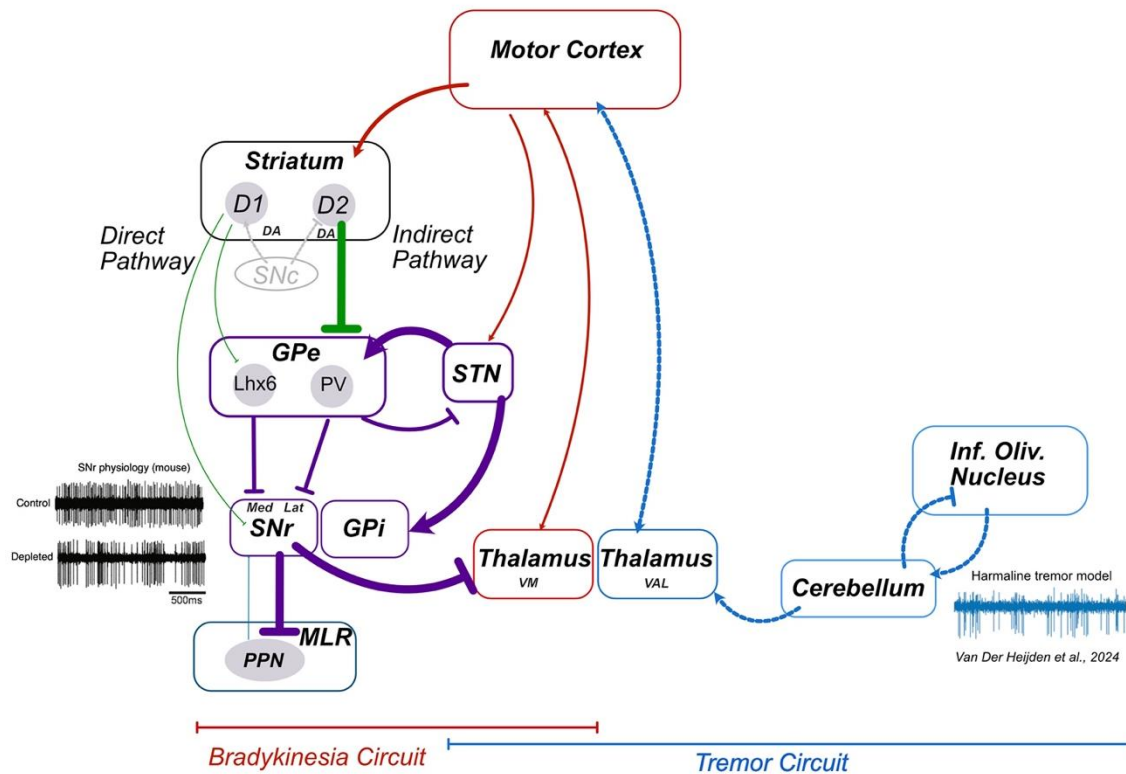


Figure 1.3 The cerebello-thalamo-cortical circuit and tremor.

A, The connectivity between the inferior olivary nucleus and the cerebellum propagates oscillatory activity and encodes the frequency of tremor in animal models of essential tremor and patients with tremor. These oscillations are transmitted from the DCN to the red nucleus (not depicted) and the cerebellar-recipient thalamus (VIM in humans, VAL in rodents) to the motor cortex wherein the thalamus serves as a gating resistor. This has been referred to as the brain circuit primarily responsible for the generation and propagation of tremor. **B**, In PD bradykinesia, in addition to the rate model predictions of suppressed activity in the basal-ganglia-thalamo-cortical circuit, aberrant firing patterns seen in the basal ganglia - example shown for a bursty SNr neuron. In PD rest tremor, the cerebello-thalamic tremor circuit interacts with the pathological basal ganglia circuit to produce a rest tremor. Hyperconnectivity between the basal ganglia output and motor cortex is thought to trigger the tremor which is then produced by the aberrant cerebello-thalamic circuit (dimmer-switch model, (Helmich & Hallett, 2012). Example of aberrant activity in the cerebellum from a harmaline-induced tremor model mouse shown.

1.2 Parkinsonian mouse models: symptoms, and pathophysiology

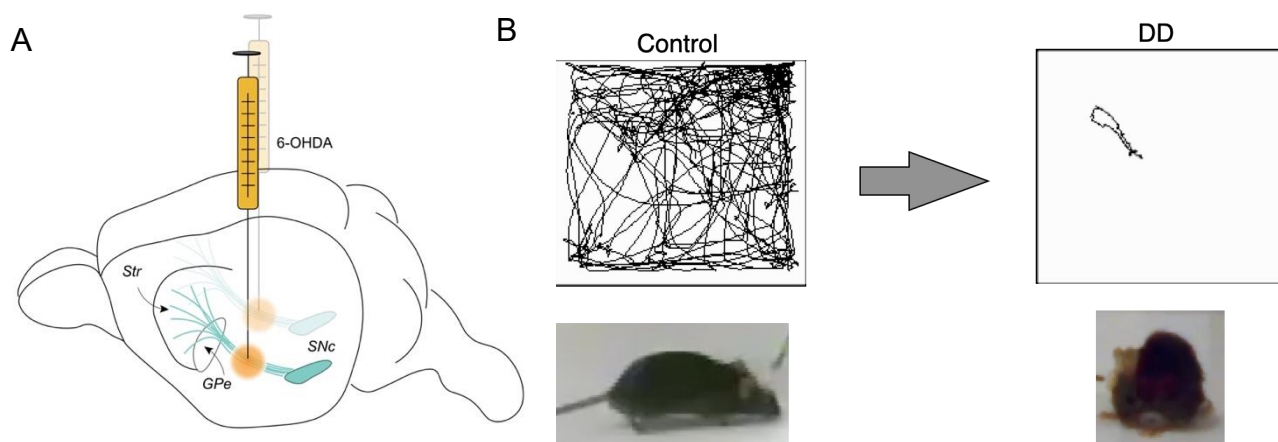
As the prevalence of Parkinson's disease continues to increase in the population (Willis et al., 2022), our understanding of the symptomatology and pathology seen in patients with the disease has broadened. There is also an increasing demand for pre-clinical models of the disease that capture

multiple aspects of the disease phenotype and pathology, to develop accurate treatments that can be translated. Animal models of PD have ranged from toxin-induced models (Blandini et al., 2008; Hernandez-Baltazar et al., 2017), pesticide or drug-induced models, genetic modifications, viral vector injections, as well as alpha-synuclein (aSyn, pathological protein aggregate) induction. There are differences in the features demonstrated by primate and rodent models of PD in their pathology and symptoms. Mouse models of PD provide the unique opportunity to investigate and perturb circuit function using powerful viral and genetic tools, while also studying the motor phenotypes seen in PD. Here we will describe the toxin-induced dopamine depletion model that produces severe acute-onset akinesia, and models of genetic synucleinopathy that show a progressive loss of dopamine, and deficits in motor and non-motor functions.

1.2.1 Acutely induced parkinsonian pathology in mice: Toxin models

Parkinsonian dopamine loss is mimicked in animal models of dopamine depletion by injecting the 6-hydroxydopamine in either the striatum or in the medial forebrain bundle which contains projections from the SNc to the striatum (Fig. 1.4A). The toxin injection produces a dopamine depleted state in rodents by inducing a loss of dopaminergic inputs to the striatum via various cytotoxic mechanisms including free radical generation, respiratory chain dysfunction, and microglial activation (Blum et al., 2001; Schober, 2004). The primary symptom of this model in bilaterally affected mice, is a profound loss in the ability to move i.e. akinesia, and a slowing of any residual movements i.e. bradykinesia. The dopamine depletion also leads to pathological neural activity in the basal ganglia network characterized by synchronous and bursting activity (Parr-Brownlie et al., 2007). While the human Parkinsonian neurophysiology studies have highlighted the presence of beta oscillations (12-30 Hz) in all the basal ganglia nuclei (Dostrovsky, 2004; Hutchison et al., 2004; Levy et al., 2000), there is limited evidence for these oscillations in mouse models of PD. There is evidence for slow-wave oscillations (delta: 0.5-4Hz) and increased pathological bursty firing in the SNr (Mastro et al., 2017; Whalen et al., 2020; Willard et al., 2019) in the SNr. While majority of studies induce a dopamine depleted state unilaterally, which predominantly demonstrates ipsiversive turning (Robinson & Becker, 1983), here we will focus on the bilaterally depleted mouse model which shows profound, acute-onset akinesia (Fig 1.4B), that limits the survivability of these mouse to ~5 days post-injection due to weight loss. The time course of this model is different from idiopathic PD and limits the ability to explore compensatory mechanisms in the brain in response to the dopamine loss or study long-term therapeutic effects.

However, this model produces rapid, specific neurodegeneration of the nigrostriatal system mimicking some of the cellular PD pathology, allowing for the study of behavioral recovery and pathophysiology in the context of dopamine (DA) cell loss.



Schematic: c/o Teresa Spix

Figure 1.4 The toxin model of PD producing acute severe akinesia.

A, Injection of 6-OHDA toxin in the medial forebrain bundle carrying dopaminergic axons from the SNc to the striatum bilaterally. **B**, Control animal pre-injection showing extensive movement in the open field depicted by movement traces (top) and normal posture (bottom). After toxin injection, the mouse demonstrates limited movement in the open-field or akinesia (top, right), and a contracted posture (bottom, right).

1.2.2 Gradual genetically induced parkinsonian pathology in mice: Synuclein models

Parkinson's disease along with other synucleinopathies, like dementia with Lewy bodies, demonstrate an accumulation of α Syn (encoded by SNCA), a protein involved in vesicular exocytosis, into oligomers or fibrils that form insoluble cytotoxic aggregates (Spillantini et al., 1998). These form Lewy bodies which have further downstream effects on the cellular machinery including the mitochondria, endoplasmic reticulum and the Golgi apparatus (Huang et al., 2019). These aggregates occur in midbrain dopaminergic and other cortical areas producing the motor and cognitive dysfunction seen in PD (Surmeier, 2018). Transgenic mouse models with various mutations in the SNCA gene have been utilized to mimic the α Syn pathology. Some of these include 153T, A30P and E46K mutations in the SNCA gene (Dovonou et al., 2023). Mouse line 61 was generated as one of the first transgenic models of α Syn by overexpressing human WT α Syn using a neuron-specific Thymocyte differentiation antigen (Thy1) promoter (Rockenstein et al., 2002). This line was characterized by mitochondrial

dysfunction and inflammation, motor deficits and aSyn inclusions starting at 2 months of age, mild decreases in striatal DA and no SNc DA cell loss (Chesselet et al., 2012). A variant of this model (MI2), expressing a truncated human aSyn with a TH promoter shows significant DA loss due to aggregation in the DA neurons, and motor impairments starting at 9 months (Wegrzynowicz et al., 2019). The E46K mutation within aSyn, observed in a subset of patients with familial autosomal dominant parkinsonism, when induced in mice with a prion protein-promotor led to aSyn inclusions in the spinal cord, brainstem, DCN, motor cortex, and thalamus, but spared the SNc DA neurons. These mice showed motor impairments at the age of 15-16 months (Emmer et al., 2011; Zarranz et al., 2004). Recent work from the Selkoe lab, demonstrated a dose-dependent effect for this E-to-K mutation, by adding 2 homologous E-to-K mutations into adjacent repeats which amplified the shift from stable tetramers to aggregation-prone monomers (Dettmer et al., 2015). The triple mutant mouse, also known as a 3K mouse, produced by inducing these additional mutations (E35K, E61K) along with the E46K (1K mutant), showed more striking features including dopaminergic cell loss, and a progressive motor phenotype of a tremor, limb abnormalities and gait dysfunction transiently responsive to L-DOPA (Nuber et al., 2018) (Fig. 1.5). This thesis will illustrate a more detailed behavioral characterization of this mouse model, while exploring the aSyn aggregation profiles in the Chapter 3. For more information on other PD models see the following review paper: (Dovonou et al., 2023).

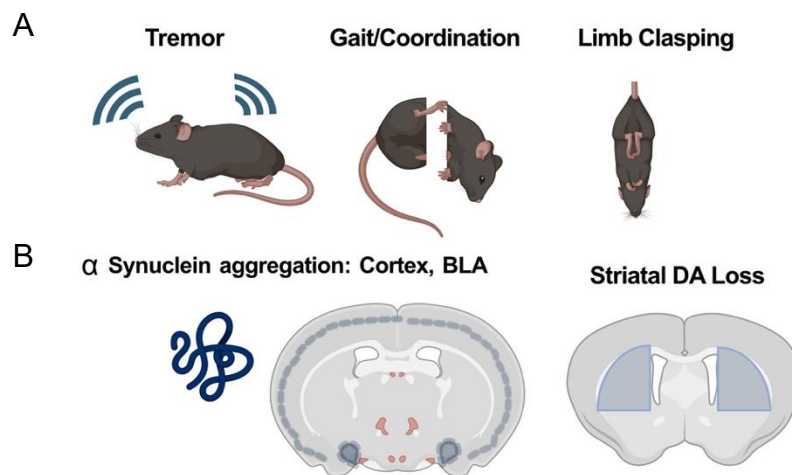


Figure 1.5 The 3K a-synuclein model of PD producing progressive tremor and gait dysfunction.

A, Motor phenotype demonstrating a prominent tremor, issues with gait and coordination depicted by the pole task, and hindlimb clasp when suspended **B**, Pathological a-synuclein aggregation throughout the cortex and in the basolateral amygdala, striatal dopamine loss reported (Nuber et al., 2018).

1.3 Therapeutic interventions in PD

Symptomatic medical treatment in Parkinson's disease is mostly focused on dopamine replacement therapies in the form of levodopa, dopamine agonists that act on dopamine receptors, and monoamine oxidase inhibitors that improve dopamine availability. As a first line of therapy, these treatments improve motor symptoms in patients. However, with disease progression, patients show treatment-resistance in the form of fluctuations from low mobility as the medication wears off, to dyskinesias in response to high-doses of levodopa. This can be debilitating and prove ineffective in managing the disease (Dietrichs & Odin, 2017). This led to the development of more invasive methods of treatment including lesions made in the pathological basal ganglia structures (STN, thalamus, or the GPi). Brain stimulation in the context of pain management and psychiatry provided the opportunity to test the effects of electrical stimulation on subcortical structures of the pathological motor circuit in PD. DBS provides an adaptable, and reversible alternative to lesion studies and has hence gained traction over the last few decades as an alternate treatment option in medication-resistant PD (Wichmann & DeLong, 2016).

1.3.1 Deep-brain stimulation for motor impairments in PD

DBS is an invasive treatment wherein patients undergo neurosurgical implantation of chronic electrodes in the basal ganglia, or thalamus that deliver high-frequency electric current constantly. Building on the effects of pallidotomies and thalamotomies in patients with PD and ET motivated by the role of these structures in producing aberrant pathological firing patterns, the common targets for DBS for motor symptoms included the STN, the GPi, and the motor thalamus (ventral intermediate nucleus/VIM in patients). VIM stimulation showed a suppression of tremor leading to VIM DBS being approved as a site for treatment in both PD tremor and ET (Benabid et al., 1987, Rehncrona et al., 2003). Stimulation delivered in the GPi was found to improve akinesia, on-off fluctuations and dyskinesia (Siegfried, 1994). Based on primate studies showing effect of STN lesions on akinesia and tremor (Bergman, et al., 1990), STN DBS when performed in PD patients showed improvements in akinesia, rigidity and tremor (Limousin et al., 1995). The delivery of electrical stimulation is thought to cause a functional lesion within the pathological loops underlying PD motor symptoms. As described before, the overactive STN or GPi in the context of parkinsonian rigidity and akinesia, or the pathologically oscillatory cerebellar-thalamus in the context of tremor, are thought to be lesioned

using high-frequency electrical stimulation, shifting the activity to normalcy. The exact mechanism of action remains poorly understood, which limits its use to a palliative treatment for symptoms, rather than an intervention with the potential to provide long-lasting therapeutic gains.

1.3.2 Closed-loop DBS approaches based on PD pathophysiology

To enhance the effectiveness of DBS, studies have focused on identifying biomarkers of pathological activity that are altered when stimulation is delivered. The pathological synchronous firing in the basal ganglia, in the form of beta frequency oscillations has been extensively used clinically as a marker of pathological activity. DBS delivered in the STN has demonstrated a suppression of beta frequency power. Similarly, in mouse models of PD, the SNr demonstrates burst firing and synchronous firing which is thought to underlie the parkinsonian symptoms (Rubin et al., 2012). A reduction in this pathology has been shown to correlate with the motor improvements seen in response to optogenetic interventions that produce long-lasting motor recovery (Mastro et al., 2017) as well as clinical DBS (Chen et al., 2020; Feldmann et al., 2022; D. D. Wang et al., 2018). These biomarkers provide an opportunity for real-time feedback and adaptation of stimulation protocols to maximize the suppression of pathology, with subsequent improvement in symptom control. Closed-loop DBS approaches have leveraged such biomarkers by conditionally delivering stimulation based on the detection of these pathological signals. This approach is more targeted and circumvents the difficulties associated with trial-and-error-based clinical programming of optimal stimulation settings. Despite the identification of various biomarkers, closed-loop strategies have struggled with practical issues like computational speed required for real-time feedback, as well as the unreliability and stability of biomarkers in diverse clinical phenotypes. Another stimulation paradigm designed with the goal of disrupting pathological synchrony, is the coordinated-reset stimulation (CR-DBS) (Ebert et al., 2014; Tass, 2003; Tass et al., 2012; Tass & Hauptmann, 2007) which involves the delivery of short bursts of stimulation distributed across the length of the electrode to spatially and temporally isolate stimulation within the target region. These approaches provide the foundation for future exploration of stimulation paradigms designed specifically to break the pathological loops created in a parkinsonian state. This involves the identification of a stable biomarker, and the demonstration of therapeutic interventions that either acutely or persistently abolish this biomarker, to produce symptom relief.

1.3.3 Circuit-inspired DBS targeting of distributed networks or individual cell-types

The responsivity of different regions to DBS is dependent on a variety of factors including the synaptic properties, the axonal connectivity, and the heterogeneity in the cell-types that these regions are comprised of. The variability in the outcomes of DBS, necessitates the exploration of these factors in optimizing circuit-based stimulation paradigms. These paradigms can be studied both in the context of the recruitment of distributed circuits to guide stimulation targeting, as well as in the context of cellular properties that could guide stimulation patterns that improve the efficacy of stimulation. There is growing evidence suggesting that the therapeutic effects of DBS involve stimulation of the white matter tracts around the subcortical nuclei that are targeted. Optogenetic activation of excitatory inputs to the STN from the primary motor cortex has been shown to produce motor recovery in rodent models of PD (Gradinaru et al., 2019; Sanders & Jaeger, 2016). This suggests that one possible therapeutic mechanism for STN DBS might be the antidromic activation of the hyperdirect pathway (from the cortex to the STN). Similar effects have not been established with the GPi stimulation (Johnson et al., 2020). Similarly, in the context of VIM stimulation for tremor, targeting the cerebellothalamic tract outside the VIM, in the posterior subthalamic area (PSA) is shown to be more effective at tremor suppression than direct stimulation in the VIM (Neudorfer et al., 2024). Recent tractography work in the STN, has identified specific tracts associated with tremor, bradykinesia, rigidity, and axial symptoms in patients with PD (Rajamani et al., 2024). This study found that the connectivity between the M1 and the posterior region of the motor STN was associated with tremor improvements, whereas the connectivity between the pre-supplementary motor area and the anterior part of the subthalamic premotor region was associated with rigidity improvements. Tracts correlated with axial symptoms, specifically with gait-related symptoms also included a projection to the pedunculopontine nucleus (PPN) in the locomotor brainstem. A similar symptomatic association between the basal ganglia-thalamo-cortical pathway with bradykinesia, and cerebello-thalamo-cortical pathways with tremor would be predicted at the level of the thalamus. Transsynaptic approaches in mice can allow further exploration of these segregated pathways in the context of their correlated motor symptoms.

To add another layer of complexity to the projection-specific organization of circuits underlying these symptoms, the physiological properties of specific cell-types in the basal ganglia structures could also guide optimization of treatments. Neuronal diversity in the striatum, GPe, STN, as well as the SNr has been explored in the context of their differential effects on behavior. (Courtney

et al., 2023; Cui et al., 2021; Kravitz et al., 2012; Parolari et al., 2021). Recruitment of specific neurons that maximize motor therapeutic outcomes, while minimizing off-target effects is crucial (See Review - (Gittis & Sillitoe, 2024)). While many recent studies characterize cell-types, their projection patterns, and elucidate their causal role in behavioral states, using optogenetic approaches there is still a gap in making these findings translatable using novel electrical stimulation paradigms that can leverage these cellular properties to differentially drive circuits. A similar optogenetic finding from our lab demonstrated that the PV and Lhx6 cell-types in the GPe, when differentially modulated could produce long-lasting motor recovery in dopamine depleted mice (Mastro et al., 2017). In vitro experimentation sought to find the electrical stimulation patterns that could produce similar behavioral effects. A combination of information about synaptic properties of cell-types, and about the colocalization of projections from the D1 striatum and the STN was used to identify the pattern and site of stimulation that would lead to a similar modulation of cell-types (Fig. 1.6). The generalization of this methodology to other brain regions, requires both the exploration of the physiological properties of the circuits to be targeted for specific symptoms (Aim 1), as well as an exploration of the effects of such patterned stimulation on connected nodes in this circuit (Aim 2).

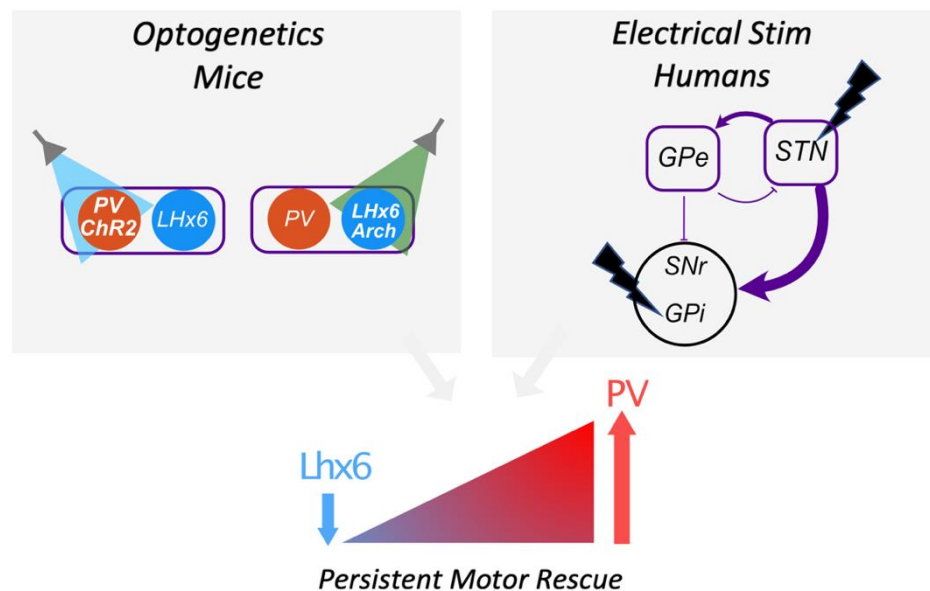


Figure 1.6 Opto-inspired DBS to translate circuit-based insights from optogenetics to clinical therapies using electrical stimulation.

Schematic depicting the finding that differential modulation of cell-types in the GPe using optogenetics produced long-lasting motor recovery (top, left). Schematic depicting the need for identifying electrical stimulation patterns in the clinically relevant DBS targets – the STN or GPi to produce the same modulation of cell-types to improve the therapeutic duration of existing DBS therapies.

1.4 Summary and Aims

The cortico-basal ganglia-thalamo-cortical loop and the cerebello-thalamo-cortical loop form parallel circuits in the brain involved in integrating sensory afferent information and updating top-down motor commands to perform movement initiation, action-selection, and coordinated movement execution. Pathological dysfunction in these circuits aligns with the production of bradykinesia, rigidity or tremor in Parkinson's disease. Rodent models of PD so far have focused on either mimicking the dopamine loss that produces bradykinesia or inducing genetic mutations that lead to the accumulation of pathological α Syn that show limited motor dysfunction. There is a lack of a unifying theory to understand the pathological underpinnings of the various motor symptoms seen in PD. The specific role of the thalamus, which is the central integrator of these inputs, in producing these symptoms remains unclear. Furthermore, DBS techniques have attempted to modulate these pathological loops to produce symptomatic relief, but the optimization of stimulation targeting and patterning based on cell-type and projection specificity is still limited. The contributions of this thesis in addressing the gaps in knowledge are as follows:

- In **Chapter 1**, I provided a summary of the circuitry underlying symptoms of Parkinson's disease with a comprehensive description of the thalamic sub-circuits involved in motor control. I outlined pre-clinical mouse models that mimic parkinsonian motor pathologies to allow for mechanistic and therapeutic exploration. Finally, I summarized the neurosurgical therapeutic approaches in PD. Collectively, optimization of therapeutic outcomes requires a better understanding of the mechanisms and circuits involved in PD.
- In **Chapter 2**, I investigate the role of the thalamus in the production of dopamine depletion-induced bradykinesia and synuclein-induced tremor. Utilizing in vivo electrophysiology in head-fixed mice and transsynaptic viral targeting, I compared the activity of the cerebellar-recipient thalamus and the basal ganglia-recipient thalamus in the two pathological conditions in comparison with healthy animals. In this experiment I show that, in the tremor mice, the neural activity in the cerebellar-recipient thalamus is selectively dysfunctional with dampened movement-responsivity, pathological patterns of firing, and no coherence to the tremor frequency. In the dopamine depleted mice, I show global dysfunction in the basal-ganglia

recipient thalamus and cerebellar-recipient thalamus with suppressed firing rate, and movement-responsivity in both regions. These findings highlight the lack of basal ganglia involvement in the 3K-synuclein tremor model, and the role of the cerebellum in both pathological models of PD.

- In **Chapter 3**, as a follow-up to Chapter 2, I investigate the mechanisms of motor and non-motor dysfunction in the 3K synuclein model further. I elucidate the pattern of spread of the pathological aSyn aggregates throughout the brain, brainstem and spinal cord using an antibody targeting the phosphorylated aSyn. I develop a battery of behavioral tests to demonstrate that these mice show a tremor, changes in limb coordination, as well as changes in novelty-seeking behaviors at baseline. Using a diphtheria toxin-mediated strategy to lesion the SNr, I show that the motor deficits including the tremor are not alleviated, which is consistent with the normal VM activity demonstrated in Chapter 2 suggesting minimal basal-ganglia involvement in the 3K pathophysiology.
- In **Chapter 4**, in a bilaterally dopamine depleted mouse model, I investigate the effects of delivering a specific pattern of electrical stimulation that is inspired by cellular synaptic mechanisms and projections to GPe cell-types. I show that bursts of electrical stimulation delivered in the EPN produces long-lasting improvements in bradykinesia and akinesia. As a follow-up I test clinically relevant patterns and sites of stimulation that could elicit a similar effect on locomotion. I implant the stimulating electrode at various targets along the motor axis spanning the GPe, EPN, STN and the SNr to map the therapeutic efficacy onto this anatomical continuum. I show that burst DBS in the STN produces only an acute effect and that distinct pathways anterior or posteromedial to the STN might be elicited to produce an acute vs. persistent effect on open-field locomotion. Additionally, to identify the neural correlates of the long-lasting effects on mobility demonstrated with EPN DBS, I performed in-vivo electrophysiology recordings in the SNr before and after delivering stimulation in the EPN to measure the attenuation of pathological bursting activity in the SNr as a biomarker for therapeutic efficacy.
- Finally, in **Chapter 5**, I outline future directions for the exploration of mechanistic therapeutic approaches to produce long-lasting motor improvements. I speculate on a unifying framework to understand the pathophysiology of PD motor symptoms and propose a translational approach to taking cellular insights to improve clinical outcomes.

Chapter 2

2 Aberrant Cerebellar-recipient Thalamic Activity in an α -Synuclein Mouse Model with Progressive Motor Deficits

2.1 Abstract

The thalamus serves as an integrator of motor inputs from the basal ganglia and the cerebellum to the motor cortex. The circuits from the basal ganglia to the ventromedial thalamus (VM) and from the cerebellum to the ventral anterior-lateral complex (VAL) of the thalamus remain mostly separate structurally, and form functionally distinct circuits responsible for different aspects of healthy and dysfunctional motor control. The basal ganglia-thalamo-cortical circuit has been implicated in bradykinesia, whereas the cerebello-thalamo-cortical circuit is historically associated with the generation of tremor. The toxin-model of dopamine depletion shows bradykinesia, while a recently developed genetic model (3K) of α -synuclein aggregation shows a prominent tremor in mice. Given the relevance of these symptoms in Parkinson's disease (PD), we investigated their neural correlates using in vivo electrophysiology. Using trans-synaptic viral approaches, we targeted the VM or VAL thalamus and recorded the neural activity in head-fixed mice that showed a tremor dominant or bradykinesia dominant phenotype. We saw selective neural deficits in the VAL of the 3K-tremor mice, with no changes in the VM, suggesting primarily cerebellar dysfunction in this model. In the dopamine depleted mice, we saw deficits in the VM as well as the VAL suggesting dysfunction in both the basal ganglia and cerebellar circuits. These results highlight circuit-specific mechanisms that can be leveraged to target divergent motor symptoms.

2.2 Introduction

Voluntary movement depends on a coordinated effort from various motor regions in the brain including the basal ganglia, motor cortex, thalamus, and the cerebellum. The thalamus serves as an

integrator of proprioceptive and movement-related information from the cerebellum and basal ganglia, forming reciprocal connections with the cortex to effectively execute motor commands. It has been a target of effective DBS in several diseases including dystonia, PD, and ET, emphasizing its integrated connectivity with various motor systems (Koller et al., 2001; Lozano, 2000). Anatomical evidence from non-human primates demonstrates segregation of inputs from the cerebellum and the basal ganglia within the motor thalamus (Haber, 2003; McFarland & Haber, 2002). In rodents, a similar pattern is seen with more enriched projections from the cerebellum to the ventral-anterior lateral complex (VAL) and from the basal ganglia to the ventromedial thalamus (VM) (Alonso-Martínez et al., 2023). Recent evidence showing an overlapping population in mice that receives inputs from both the basal ganglia and the cerebellum is thought to be critical for motor learning (Roth et al., 2024).

Functionally, these two thalamic sub-regions might be involved in distinct motor syndromes. Evidence from clinical and animal studies based on the classical view of PD pathophysiology suggest that the hyperactive basal ganglia output to the VM might be responsible for excessive movement suppression manifesting as akinesia or bradykinesia (Albin et al., 1989; DeLong, 1990). In movement disorders involving tremor, the pathological cerebellar activity transmitted to the VAL is thought to be the source of dysfunction.

Parkinson's disease is a movement disorder associated with both bradykinesia and tremor. Tremor in PD manifests as a rest tremor, and patients with PD often show comorbidity with cerebellar disorders like dystonia and action tremors suggesting a potential involvement of the cerebellum. Animal models that mimic dopamine dysfunction with chemical means or aSyn can drive varying degrees of bradykinesia, but rarely produce tremor. However, a recently developed mouse model of aSyn pathology exhibited progressive motor dysfunction with tremor as the most dominant symptom. This mouse model (3K) developed by (Nuber et al., 2018) expresses three missense mutations in the gene producing aSyn (SNCA), shifted the stable physiological form of aSyn to form aggregation-prone monomers leading to vesicular dysfunction and Lewy-body like aggregates. The physiological correlates of motor symptoms in this mouse remain unknown. Here, we conduct a study of thalamic physiology within sub-circuits relevant for the tremor-dominant mouse model associated with aSyn dysfunction and the bradykinesia-dominant mouse model induced by a depletion of dopamine neurons.

We find that the 3K tremor-dominant model exhibits selective dysfunction of neural activity in the VAL, but not VM, consistent with a hypothesized role of cerebellar-related dysfunction in tremor. Physiological deficits in VAL include a loss of motor encoding in the nucleus and increase in

burst firing and irregularity. An inspection of the distribution of aSyn aggregates in these mice showed strong accumulation in the DCN, but only a weak accumulation in the SNr, accompanied by a mild dopamine loss. These results suggest that the physiological features in the 3K mice are indicative of dysfunction in cerebellar circuits in this model.

In the dopamine depleted mice, we find physiological dysfunction in the VM, comprising of a loss of motor encoding, synchronous firing, and low frequency oscillatory activity, consistent with the role of the basal ganglia in bradykinesia. Interestingly, in addition to the effects in the VM, the physiological features of the VAL seen in the tremor dominant model are also seen in the dopamine depleted mice that exhibited bradykinesia as their primary symptom.

Our results highlight the differences in the regional distribution of physiological dysfunction across territories of cerebellar and basal ganglia motor thalamus in tremor dominant and bradykinesia dominant mouse models. While the causal contribution of these changes cannot be established here, they suggest that changes in the VAL thalamus might be a common feature of PD models which merits a deeper study of cerebellar contributions to motor symptoms in animal models.

2.3 Materials and Methods

2.3.1 Animals:

Experimental procedures were approved by the Carnegie Mellon University Institutional Animal Care and Use Committee and in accordance with the guidelines set by the NIH and SfN Use of Animals in Neuroscience Research. Males and female mice on a C57BL/6J background were used for all experiments. Males and female heterozygotes over the age of 7 months were used from the 3K mouse line. Animals were group housed (2-5 per group) in a 12h/12h light dark cycle until used for experimentation.

2.3.2 Surgical procedures and viral transfection:

All surgical procedures were performed under general anesthesia. Injections of AAV1-EF1a-Flpo (transsynaptic Flp expression), AAV8-CAG-FLPx-rc [ChrimsonR-tdTomato] (Flp-dependent Chrimson), AAV1- pEF1a-DIO-FLPo-WPRE-hGHpA (transsynaptic Cre-dependent Flp), and

pENN-AAV-CMV_s-PI-Cre-rBG (AAV1) (Cre expression) from Addgene were made in wild-type mice, or 3K mice. Anesthesia was induced using 3% isoflurane in an induction chamber and maintained throughout the surgery with 1.5% isoflurane. Mice were placed in a stereotaxic frame, and bilateral holes were drilled in the skull. (DCN: -6.03 posterior, 1.95 lateral, SNr: -2.8 posterior, 1.5 lateral, VM: -0.95 posterior, 0.75 lateral, VAL: -0.95 posterior, 1.2 lateral from bregma). 150 nl of virus was injected with a Nanoject (Drummond Scientific) through a pulled glass pipet (30 μ m tip diameter) whose tip was placed on the top of the dura (DCN: 2.24 mm, SNr: 4.65 mm). 50 nl of virus was injected in the thalamic nuclei (VM: 4.1 mm, VAL: 3.65 mm). The pipet was left in the brain for 5 minutes after the viral injection to prevent backflow. For VM targeting, based on pilot experiments suggesting incomplete transfection from SNr to the VM with just the AAV1-Flp virus, an amplification strategy was utilized wherein the AAV1-DIO-Flp was injected in the SNr, and AAV1-Cre was injected in the VM in the same surgery. 3 weeks later, the Flp-dependent Chrimson was injected in the VM. Recording experiments were performed at least 3 weeks following the opsin injection to allow full viral expression. Post-surgery, animals received 3mg/kg Ketofen and 0.2 ml saline and were placed on a heating pad for recovery, before returning to their home cage.

2.3.3 Head bar implantation and dopamine depletion:

One week before performing in-vivo recordings, the animals underwent a second or third (in the case of VM) surgery to implant the head bars to allow head-fixation during recording. Mice were anesthetized, placed on the stereotaxic frame, and holes were redrilled from the previous viral injection to create bilateral craniotomies over the VM or the VAL. A stainless-steel head bar was affixed to the skull with dental cement (Metabond, Orthojet). A well of dental cement was built around the skull to create a saline pool for grounding, this was filled with a silicon elastomer to secure until the day of recording.

Animals in the dopamine depleted group underwent the depletion procedure in this same surgery. A hole was drilled on both hemispheres over the medial forebrain bundle (MFB: -1.0 posterior, 1.0 lateral). 1 μ L of 5 μ g/ μ L 6-OHDA (TOCRIS) was injected at a depth of 4.9 mm over the course of 5 minutes using a Nanoject with a pulled glass pipet on both sides. The pipet was left in the brain for 5 minutes after the injection. After the dopamine depletion, mice were individually housed and placed on a heating pad, and provided with softened food pellets, and chocolate pellets.

Additionally, they received daily saline injections (0.9% NaCl, IP) for the four-days following the depletion surgery.

2.3.4 In-vivo recording:

On the day of recording, mice were placed on top of the running wheel and allowed ~10 minutes to acclimate to the head position. The silicon elastomer was removed, the craniotomy was cleared, and a 16-channel silicone optrode with sites 50um apart (Neuronexus) and a 100-um fiber terminating 50 um above the tip was attached to the micromanipulator (Sutter Instruments) and centered on the midline marker for each recording site. The probe was advanced slowly until the appropriate depth for the VM or VAL was reached. Some spindle-like activity, with a relatively low-firing rate, and occasional responsiveness to light activation confirmed thalamic recording placement. Post-hoc tissue analysis for viral expression and implant location was used to confirm targeting. The probe was left at the desired depth for about 20 minutes before beginning to record. Saline was added to the well around the craniotomy and the reference ground electrode connected to the ground pin on the head stage was placed in the saline pool. Extracellular recordings for spiking were acquired using the Plexon OmniPlex system bandpass filtered for 150-8000 Hz sampled at 40 kHz. Movement was tracked for the full period of recordings using a rotary encoder attached to the axle that the wheel was rotating on and read through the analog input on the OmniPlex amplifier.

2.3.5 Histological analyses:

At the end of recording experiments, mice were euthanized and perfused with 4% paraformaldehyde (PFA). The brain was extracted and stored in PFA overnight. It was then switched to a 30% sucrose solution for another day. Tissue was sectioned using a freezing microtome (Microm HM 430; Thermo Scientific). Primary antibodies incubations were left in a 4 degree C refrigerator for 24 hours. An RFP antibody (rabbit anti-RFP, 1:500) was used to enhance the red mCherry tag on the Flp-Chrimson opsin in VM/VAL slices. A tyrosine hydroxylase (TH) antibody (rabbit anti-TH, 1:250) was used to label striatal TH as a measure of the level of dopamine depletion in the cohort undergoing bilateral 6OHDA depletion. An Iba1 antibody (rabbit anti-Iba1, 1:500) was used on VM/VAL slices to label microglial activation depicting optrode placement during recordings. Secondary antibodies to the

rabbit primary, included Alexa Fluor 488-conjugated donkey anti-rabbit, Alexa Fluor 568-conjugated donkey anti-rabbit or Alexa Fluor 647-conjugated donkey anti-rabbit (1:500, Life Technologies). Epifluorescent images were taken at 4x magnification (Keyence BZ-X) to identify viral expression, probe placement and 10x magnification to obtain striatal TH windows for quantification. Using the pixel intensity tool in ImageJ, 3 striatal slices from each hemisphere were normalized to their corresponding cortex intensities. Further, the intensity values were compared to healthy control tissue processed in parallel. A threshold of less than 30% remaining TH in the striatum was required for the mouse to be considered depleted compared to controls.

2.3.6 Spiking Analysis:

Spiking data was filtered at 150-8000 Hz. Spike detection was performed in the Plexon Offline Sorter software using principal component analysis to detect units using waveform matching. To be classified as a single unit, the PCA clusters had to be significantly different, and the percent of ISI violations (<1ms) was less than 0.7%. Manual verification of unit quality was performed based on this criterion, in addition to waveform shape relative to noise, and stability of the units over the recording period. Accordingly, units were classified as Grade A, B, or C. Only units in the A grade category were included for the analysis in this study. Following spike-sorting, data was processed with the NeuroExplorer software to further eliminate repeated units across channels using cross-correlation analyses, in addition to custom MATLAB scripts for firing rate analyses.

Rotary Encoder Analysis:

The raw readout from the rotary encoder underwent gaussian smoothing. Combined with the calibration of the number of events recorded per rotation, and sampling frequency of 5000 Hz for the wheel, a smoothed velocity trace was obtained for the entire recording period. A velocity threshold for movement was identified at 0.02 cm/s based on the velocity distributions showing a peak for movement to the right of this threshold value.

Rest Period Analysis:

The first 300 seconds of the recording period involved the mouse sitting passively on the wheel with no induced movement or intervention. To remove the confounds of spontaneous twitches/movement bouts during this period, the movement velocity threshold was applied. Rest

epochs were identified as periods when the velocity remained <0.02 cm/s for longer than 20 seconds. The firing rate, coefficient of variation, and proportion of spikes in bursts were measured for these rest epochs and a weighted average based on the bout duration was computed across the first 300 seconds of the recording session. A longer duration threshold (60s) was used for synchrony measurements to allow for less noisy representations of pairwise cross-correlations. The same procedure was applied to data collected from naïve control, dopamine depleted, and 3K groups.

Movement Analysis:

The longest movement bout was identified by thresholding the velocity at >0.002 cm/s. This corresponded to the forced locomotion movement bout where the wheel was manually displaced by the experimenter (at a variable unknown speed). This was reflected in the increased velocity during this forced locomotion period. All analyses showing movement-related activity were performed for this period of externally induced sustained movement. The binned firing rate (bin size = 5s) was computed for the variable duration of this long (>20 s) movement bout, as well as for 30-seconds before the bout and 60-seconds after the movement bout. The number of positively modulated units was computed by identifying units that showed a statistically significant increase in their firing rate from the pre-movement period to the movement period. Those that were suppressed, i.e. FR statistically higher during the pre-period were categorized as negatively modulated units. All other units were classified as non-modulated units. The binned firing rate was plotted to visualize the dynamics of movement modulation. All angular velocities were converted to linear velocities (multiplying the velocity by the radius of the wheel = 7.5 cm).

Firing pattern Analysis

Irregularity was measured using the coefficient of variance of the interspike interval during the rest epochs as well as the single forced locomotion event. The Poisson surprise method was used for detection of bursts. The surprise threshold was set to $S=3$, with the minimum number of spikes for a burst set to 2. The firing rate threshold was set to 1.5 times the baseline firing rate. For simultaneously recorded units on a given trajectory, synchrony was measured using a modified cross-correlation as an average of cross-correlation values computed over a moving window as described before (Willard et al., 2019). Spike trains were first down-sampled from 40 kHz to 1 kHz and converted to binary vectors before performing the windowed cross-correlation. A bin size of 1 ms was used for 12s long windows with a 4s overlap and 4s unzeroed signal in each window to ensure constant level of zero-padding for

all lags. Each window's cross-correlogram was normalized by the mean correlation from 0.5 to 4 s on either side. An average cross-correlogram was plotted for all pairs within each group. A pair was deemed synchronous if the normalized cross-correlation for a 10ms window around zero-lag was greater than 3-SD's away from the mean of the interval from 0.5s to 4s on each side. The percent of synchronous pairs was then calculated for each group. To calculate synchrony, at least 3 units were required to be recorded simultaneously. Oscillation analysis involved computing the renewal-corrected power spectrum from spiking data, and the phase shift using the method described in (Whalen et al., 2020)). To detect oscillations, frequencies with a high power and low phase shift were identified and the units were labelled as significantly oscillating units in the specified range.

2.3.7 Behavior Analysis:

Tremor Recording and Analysis

Tremor was quantified using a custom-built setup as described previously (Brown et al., 2020). Mice were individually placed in the center of a transparent plastic box (14.9x14.9 cm) suspended by eight bungee cords attached to a metal frame. An accelerometer was mounted at the bottom corner of the box to record tremor-related vibrations. Mice were allowed to move freely during each session, and their locomotor activity was simultaneously monitored from overhead and the side of the box to separate action vs rest tremor. Each session began with a 60–120 s acclimation period, followed by a 12-minute recording. Accelerometer signals were transmitted via BNC cable to an RZ5P real-time processor (Tucker-Davis Technologies) and acquired using Synapse software. Raw signals were zero-centered by subtracting the mean to remove the DC offset and were subsequently downsampled to 2048 Hz. Power spectral density (PSD) was computed over the entire recording session using a fast Fourier transform (FFT) with a Hanning window, focusing on the 0–30 Hz range, which is consistent with physiological and pathological tremor frequencies in rodents and humans. Mean power within this frequency range was also calculated to quantify tremor severity. Body center position was tracked by the top-down-camera (29.97 FPS) using EthoVision XT software (Noldus), and mobility states were classified as either active ($\geq 8.00\%$ mobility) or rest ($\leq 2.00\%$ mobility). PSD and mean power within the 0–30 Hz range were then computed for all active and rest bouts lasting at least 6 seconds and averaged per animal.

Open-field and Hole-Board Analysis

Mouse behavior was analyzed using a top-down camera at a rate of 29.97 frames per second. Positions of the nose, tail, and body center were tracked using Noldus Ethovision XT. All open-field experiments were performed by allowing the animal to move freely in the open field with a hole-board base. Movement was quantified by directly measuring the distance moved divided by the total trial time. Additionally, the type of movement was categorized as ambulation (when the start velocity was >3.2 cm/s and stop velocity > 3 cm/s), immobility when the pixel change was $<0.01\%$ averaged over 60 samples, and fine movements when neither ambulation nor immobility were detected. In addition to movement metrics, the hole-board base allowed for quantification of exploratory behaviors by counting the number of nose-pokes performed by the mouse within the first 2 minutes of every trial. A mobility threshold of $>10\%$ was applied for the inclusion of animals in the nose-poke measurement experiment.

Tail-suspension task

The tail-suspension task involved suspending the mouse by a tape (~ 6 inches long) and measuring the mobility of the mouse over a 5-minute period. This was used as a metric of learned helplessness and therefore a measure of non-motor symptomatology in the pathological groups compared to control animals.

2.3.8 Statistical Analysis:

Statistical analyses were completed using Graphpad Prism 10 (GraphPad Software). Firing rate and pattern data comparisons between the three groups was performed using non-parametric ANOVA (Kruskal-Wallis test with Dunn's correction for multiple comparisons) between units recorded from control vs 3K and control vs depleted animals. The KS-test was used to determine the shift in the cumulative distributions of the change in firing rates during movement. For the categorical analysis of number of movement modulated units between groups, the Fisher's exact test was used to compare naïve controls with either the 3K or the 6OHDA animals in the VM or VAL recording sites. Results of statistical tests can be found in the figure legends. A p-value of <0.05 was considered statistically

2.3.9 Acknowledgements:

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2.4 Results

2.4.1 Behavioral and pathological features of 3K mice suggest cerebello-thalamic dysfunction

The 3K model described in (Nuber et al., 2018) demonstrates a progressive tremor, in addition to gait and postural difficulties. While the behavioral phenotype has been established in the original paper, here we studied older 3K mice, and compared the severity of their motor and non-motor symptoms with a well-established toxin model used in the lab. Movement in the open-field, tremor, exploratory nose-pokes, and measures of depression or learned helplessness were assessed in a cohort of 10 control mice (age = 9 ± 3 mo., mean $\pm$ SD), 11 3K mice (age = 10 ± 3 mo.) and 10 dopamine depleted mice (age = 7 ± 2 mo.).

Tremor was measured using a tremor monitor modeled after the setup described in (Brown et al., 2020) involving an accelerometer attached to the base of a suspended plexiglass box allowing for measurement of changes in acceleration (Fig 2.1A). A fast-Fourier transform of the accelerometer recordings revealed that the dominant frequency of the pathological tremor was centered at ~ 10 Hz (Fig. 2.1B). Power spectral analyses were performed to compute the mean power of the tremor at rest and during movement bouts. Body-center position was tracked using Noldus Ethovision and mobility states were classified as movement ($>8\%$ mobility) or rest ($<2\%$ mobility) with a minimum duration of 6 seconds. The mean spectral power increased with movement across all groups (Fig. 2.1C, p-value < 0.0001). Mean spectral power was then compared between groups for rest bouts (Fig 2.1D) and movement bouts (Fig 2.1E). The 3K mice showed a significantly higher power at rest compared to controls (Fig 2.1D, p-value = 0.004), whereas the dopamine depleted group showed a mean power comparable to controls suggesting no tremor-related activity (Fig 2.1D, p-value = 0.63). The tremor in 3K mice was exaggerated with movement and showed a significantly higher power than control

animals during movement (Fig 2.1E, p -value = 0.0008). This suggests that the 3K mice showed an action tremor, with a postural rest tremor component. Depleted mice had only 1 bout in a single animal that was longer than 6s, hence they were not included in this comparison.

To study bradykinesia, the animals walked freely in the open-field, and their velocity during active ambulatory bouts was measured with Noldus Ethovision tracking (Fig 2.1F). Consistent with the previous study, the 3K mice exhibited reduced movement velocity compared to littermate controls (Fig 2.1G, p -value = 0.048). Further, their open-field movements were categorized as ambulation or fine movements, like grooming or rearing, using a velocity threshold. Akinesia was absent in the 3K model, with the degree of ambulation and fine movements being comparable to controls (Fig 2.1H). As shown in multiple previous studies, dopamine depleted mice exhibited bradykinesia and akinesia relative to controls. (Fig 2.1G, p -value <0.0001, 1H). The 3K mice showed no reduction in mobility and some bradykinesia, which is notably not as bad as the severe toxin model of PD. Their previously reported tremor is characterized to be an action tremor with a predominant frequency of 10Hz.

While the motor symptoms in the 3K and dopamine depleted mice are mostly non-overlapping, cognitive and non-motor symptoms of PD can serve as early biomarkers in the approach to treatment in these models. Novelty seeking behaviors have been seen in patients with PD in conjunction with dopaminergic therapies (Paulk & Neilson, 2024). The open-field arena had a base with holes allowing for exploratory nose-poking behavior (Fig 2.1F). The 3K mice showed a reduction in the number of nose-pokes in 2 minutes (Fig 2.1I, p -value = 0.03), despite comparable mobility metrics between the two groups (Fig 2.1H). The dopamine depleted had insufficient bouts of locomotion, severely limiting the assessment of their novelty-seeking performance. Affective features like depression and learned helplessness were assayed using the immobility of mice in a tail-suspension task (Lopez, 2023). The 3K, and dopamine depleted mice showed no differences relative to controls with respect to the amount of time spent moving in the tail-suspension task (Fig 2.1J, K, p -value 3K > 0.99, Depleted = 0.28).

To identify the neural correlates of these symptoms, we wanted to investigate the pattern of aSyn aggregation across the various components of the motor system including the motor cortex, the cerebellum, basal ganglia, thalamus, and the brainstem in the 3K mice. Immunostaining for the pathological aSyn, which is phosphorylated at Serine 129, performed on coronal slices from these regions revealed aggregates in multiple cortical regions including the motor cortex, in the deep-cerebellar nuclei, and the SNr. No inclusions were evident in the motor thalamus (VM, VAL), the inferior olivary nucleus (ION) or the gigantocellular nucleus of the medulla (Gi) (Fig 1L). aSyn

inclusions were not seen in any brain region in the control and dopamine depleted mice (Appendix B).

In comparison to the toxin model of PD, where a loss of SNc dopamine neurons leads to a dopamine depleted state, evidenced by reduced striatal TH (Fig 2.1L-M avg intensity Control = 100%, Depleted = $2 \pm 1\%$, p -value < 0.0001), we sought to determine whether any of the motor symptoms in the 3K mice were attributable to a loss of dopamine. Striatal TH quantification was performed in parallel on a cohort of control, 3K, and dopamine depleted brains. The 3K mice showed limited non-significant TH loss (avg intensity 3K = $74 \pm 7\%$, p -value = 0.09) relative to control mice.

Taken together, these results suggest that the behavioral phenotype of tremor, mild bradykinesia and reduced exploratory behavior in the 3K mice is likely a result of the pathological aggregates seen in the cerebellum and the motor cortex, rather than dopamine loss.

Figure 1. Nanivadekar, 2025

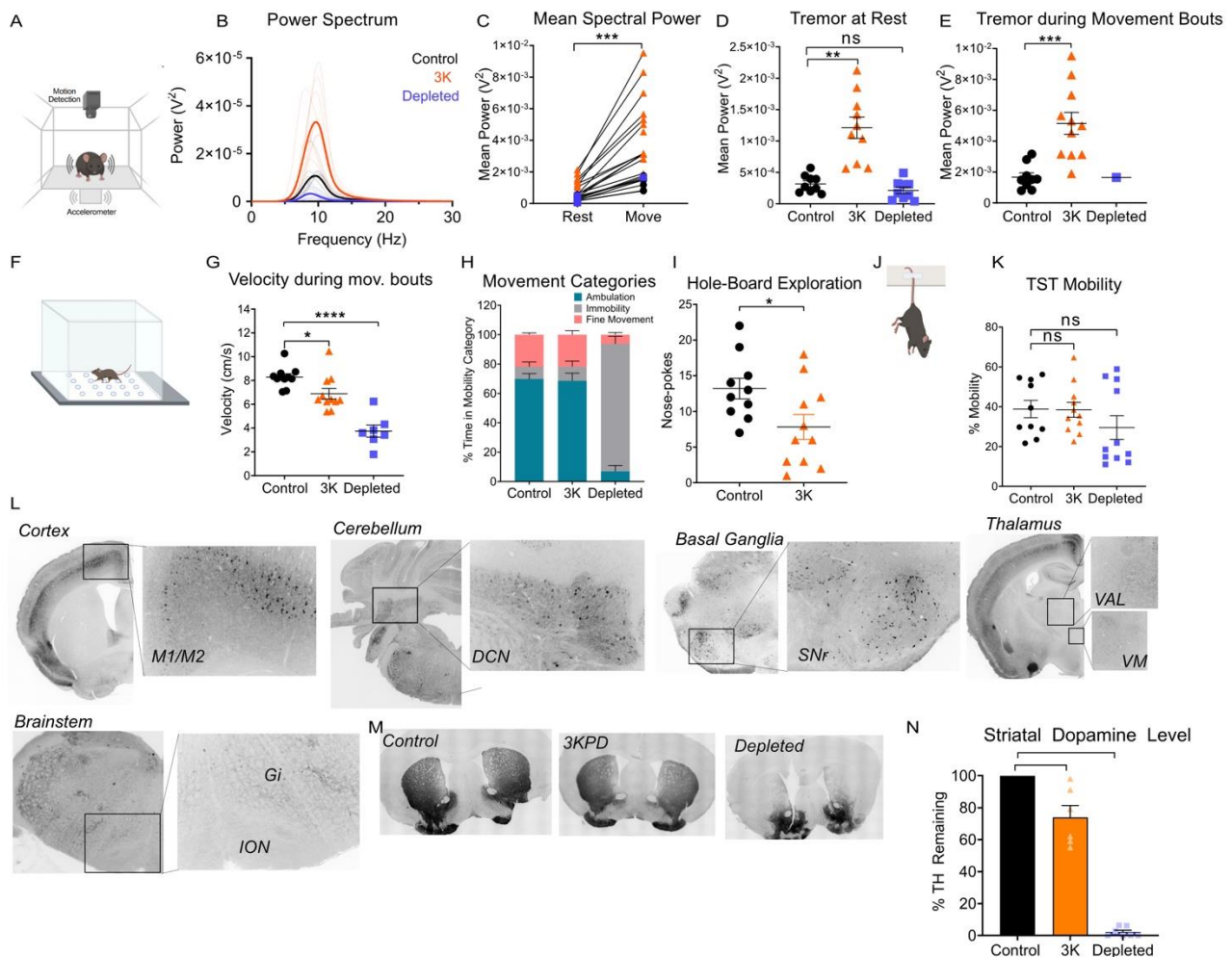


Figure 2.1 Behavioral and pathological features of 3K mice vs control and dopamine depleted mice suggests cerebello-thalamic dysfunction.

A, Tremor monitor setup with accelerometer attached to a suspended plastic box, top-down camera recording video for movement analysis **B**, Power spectral analysis using a FFT with a Hanning window depicted for the 0-30 Hz range revealing 10 Hz tremor in the 3K mice (orange) and physiological tremor in the controls (black), and depleted mice (blue). **C**, Mean spectral power for the 3 groups: control (N=10), 3K (N=11), and depleted (N=10) mice computed for periods of rest (>6 secs) defined by a threshold of < 2 % mobility using Noldus Ethovision tracking, and periods of movement (> 6 secs) defined as > 8% mobility in the mice. (Across groups: Rest = $7.6 \times 10^{-4} \pm 1.3 \times 10^{-4}$, Move = $3.4 \times 10^{-3} \pm 5.7 \times 10^{-4}$, Wilcoxon matched pairs signed rank test, $p < 0.0001$). Depleted mice only had one animal with movement bouts greater than 6 seconds to be included for this analysis. **D**, Average spectral power across all bouts of rest computed per animal, compared between the pathological groups vs. control group (Control = $3.2 \times 10^{-4} \pm 4.7 \times 10^{-5}$, 3K = $1.2 \times 10^{-3} \pm 1.7 \times 10^{-4}$, Depleted = $2.1 \times 10^{-4} \pm 5.1 \times 10^{-5}$, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.004$, Control vs Depleted $p = 0.6318$). **E**, Average power during bouts of movement compared between the pathological groups vs. control group (Control = $1.7 \times 10^{-3} \pm 2.7 \times 10^{-4}$, 3K = $5.1 \times 10^{-3} \pm 7.1 \times 10^{-4}$, Depleted = 1.7×10^{-3} single event, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.0008$). 1 control and 1 3K mouse had missing video files for the movement-analysis. **F**, Square open-field setup with a base containing holes for novelty-seeking exploration task. **G**, Movement velocity (cm/s) depicted for individual animals, recorded in the open-field and measured using Noldus Ethovision tracking of the total distance moved by the animal during a 10-minute recording session. Velocity of 3K and depleted mice compared to control mice. (Control = 8.28 ± 0.28 cm/s, 3K = 6.89 ± 0.44 cm/s, Depleted = 3.74 ± 0.51 cm/s, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.048$, Control vs Depleted $p < 0.0001$) **H**, Movement in the open-field characterized as immobility when pixel change in Noldus animal tracking was < 0.01% averaged over 60 samples, as locomotion when the starting velocity of a bout was >3.2 cm/s and as fine movements when neither of the previous conditions were fulfilled. The breakdown across all animals was as follows: Control = $69 \pm 4\%$ ambulation, $8 \pm 3\%$ immobility, $3 \pm 1\%$ fine movements, 3K = $68 \pm 5\%$ ambulation, $10 \pm 4\%$ immobility, $22 \pm 3\%$ fine movements, Depleted = $7 \pm 4\%$ ambulation, $87 \pm 5\%$ immobility, $6 \pm 2\%$ fine movements. **I**, Number of nose-pokes manually scored for the first 2 minutes of the hole-board session. Depleted mice did not have > 10% mobility during the task, hence no viable nose-poke data. Comparison between novelty-seeking behavior of 3K and Control mice performed. (Control = 13 ± 1 pokes, 3K = 8 ± 2 pokes, Mann-Whitney U test, $p=0.03$). **J**, Tail-suspension task (TST) for learned helplessness and affect. **K**, Using Noldus Ethovision tracking on video collected for 5 minutes during the TST, % mobility was computed for all 3 groups. (Control = $39 \pm 4\%$ mobility, 3K = $38 \pm 4\%$ mobility, Depleted = $30 \pm 6\%$ mobility, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p>0.99$, Control vs Depleted $p = 0.28$). **L**, Immunostaining for Phosphorylated Serine in pathological aSyn protein in the cortex, cerebellum, SNr, and brainstem sections of a 3K mouse, inset: 10x view. **M**, Representative sections of the striatum from a Control, 3K, and Depleted mouse. **N**, Quantification of striatal TH remaining in the 3K (N=6), and depleted (N=7) mice normalized relative to littermate controls (N=7). (Control = 100%, 3K = $74 \pm 7\%$, Depleted = $2 \pm 1\%$, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.09$, Control vs Depleted $p < 0.0001$). ns, not significant; * $p = 0.05-0.01$; ** $p = 0.01-0.001$; *** $p=0.001-0.0001$; **** $p<0.0001$

2.4.2 Cerebellar-recipient and basal ganglia recipient motor thalamic subregions can be probed using a transsynaptic approach

Thalamotomies or informational lesions brought about by DBS delivered in the motor thalamus have led to symptomatic relief of tremor in Parkinson's disease and ET (Krayenbühl et al., 1961, Benabid et al., 1987). This intervention is limited to tremor control and lacks specificity in targeting thalamic sub-domains. Recent tractography-based methodologies for STN DBS targeting have established symptom-specific targeting, tailored to tremor-dominant or rigidity-dominant patients (Rajamani et al., 2024). Functionally segregated pathways in the VM and VAL thalamus could allow for similar

approaches in thalamic interventions. In rodents, cerebellar inputs are concentrated primarily in the VAL regions of the thalamus (Alonso-Martínez et al., 2023; Phillips et al., 2019), with some projections to the VM that might have a role in motor learning (Roth et al., 2024). The proprioceptive information transmitted from the DCN relayed to layers 3-5 of the primary motor cortex (Kuramoto et al., 2015b) via the VAL is thought to be involved in movement execution (Hewitt et al., 2011) and disrupted in the context of tremor. Conversely, the VM integrates information related to movement cue, vigor or motor learning from the basal ganglia and sends distributed projections to the motor cortex, neostriatum, cingulate, and associative cortices (Kuramoto et al., 2015b). It is hence predicted to be disrupted in the parkinsonian state of aberrant basal ganglia output (DeLong, 1990).

To establish the role of these populations within the motor-thalamus in the synuclein-induced tremor model, we used a transsynaptic viral approach to target the projection-specific areas within the thalamus. An AAV1-Flp viral construct was stereotactically injected in either the bilateral DCN or the SNr, in addition to injecting a Flp-dependent opsin (Chrimson) in the corresponding thalamic nucleus, VAL (Fig 2.2) or VM (Fig 2.5) respectively. The AAV1-Flp was transsynaptically expressed in the thalamic projection areas allowing for a localized expression of the Chrimson opsin in either the VAL or VM (Fig 2.2A, Fig 2.5A). In the case of the VM (Fig 2.5), additional amplification was performed by injecting a Cre-dependent AAV1-Flp in the SNr, and an AAV1-Cre construct in addition to the Flp-dependent Chrimson in the VM (Fig. 2.5A). Recordings targeted to a specific thalamic sub-region were confirmed by visualizing the overlap between the viral expression and the recording track postmortem with immunostaining against the microglial marker (Iba1). The viral expression patterns, and recording trajectories are depicted for both thalamic sites in all three groups (Fig 2.2C, 2.5C). Further confirmation of distinct thalamic sub-regions was obtained based on the projection patterns seen with either VAL or VM targeting (Fig 2.2B, Fig 2.5B). This viral strategy allows the unique opportunity to study the role of firing activity in thalamic sub-circuits originating in either the cerebellum or the basal ganglia in the context of tremor or bradykinesia.

Figure 2. Nanivadekar, 2025

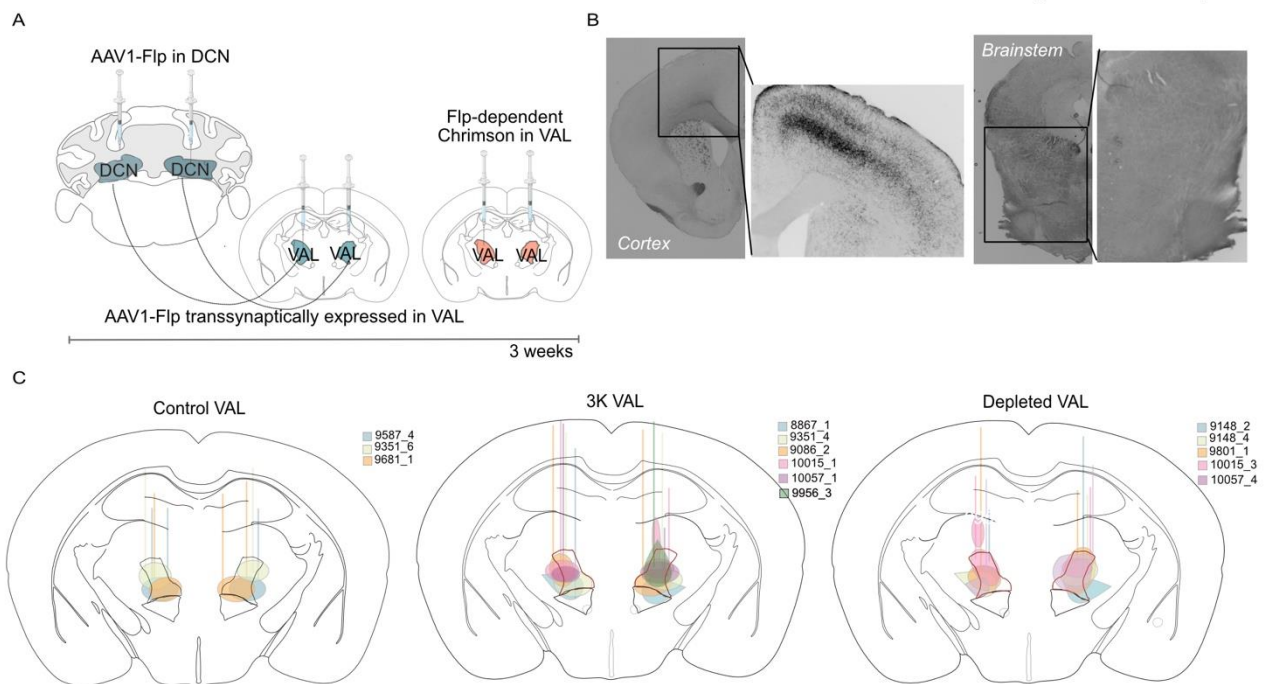


Figure 2.2 Transsynaptic approach to target the cerebellar-recipient thalamus (VAL).

A, Schematic describing the viral approach using an AAV1-Flp injected in the deep cerebellar nucleus that is transsynaptically expressed in the VAL. Followed by an injection of a Flp-dependent Chrimson opsin in the same surgery in the VAL to visualize the DCN-recipient thalamic subregion allowing for post-hoc validation of recording site. Viral expression period was ~ 3-4 weeks. **B**, Representative example of the projection patterns of the VAL to the layers 3-5 of the motor cortex, and no evident projections in pontine reticular nucleus of the brainstem in an animal (9801_1, see panel C) that underwent the viral approach described in A. **C**, Schematic overview of the spread of viral labeling around the VAL and recording trajectories reconstructed from histological slices for animals used in Figures 2.3 and 2.4.

2.4.3 Cerebellar-recipient thalamus in the tremor mice showed loss of movement-related activity

Since the cerebellar-recipient thalamus (VIM in humans) is involved in the propagation of tremor and serves as a therapeutic target for DBS, we wanted to record the neural activity of its rodent counterpart, the VAL, in this synuclein-induced tremor model. We specifically targeted the VAL using the transsynaptic approach described previously (Fig 2.2A). We compared the neural activity of 3K mice (n=77 units, N=8 animals, age = 14 ± 5 months) with age-matched wild-type controls (n=54 units, N=6 animals, age = 13 ± 5 months). Additionally, to study VAL physiology in dopamine depletion, we also recorded from age-matched dopamine depleted mice (n=104 units, N=7 animals, age = 13 ± 5 months).

Recordings were performed in awake, head-fixed mice using the Plexon Omniplex system. Head-fixed mice were placed on top of a running wheel in a position that did not allow much spontaneous locomotion to limit other behavioral confounds, and study neural activity during periods of rest. At rest, the firing rates of VAL units in the 3K mice 6.41 ± 0.7 Hz (mean $\pm$ SEM) showed no differences compared to control mice 5.99 ± 0.8 Hz (Fig 2.3D, $p > 0.999$). The dopamine depleted mice showed a reduction in VAL neuron firing rates 4.03 ± 0.36 Hz relative to controls (Fig 2.3D, $p = 0.052$, Mann-Whitney U test). This was evident from the firing rate distributions as well as representative examples of neurons from these groups (Fig 2.3C-E).

To study, movement-related activity, the wheel was manually displaced by the experimenter for 3-5 minutes. A rotary encoder attached to the axle of the wheel provided a direct readout of this wheel turning, reflected by the increased velocity during the period of induced locomotion (Fig 2.3F). While control, 3K, and dopamine depleted mice, all showed an elevation in firing rates during this period of movement, the degree of movement-related responses was dampened in both pathological groups at the population level (Fig 2.3G). The change in firing rates (Δ_{FR}) from pre-movement to induced movement, was significantly reduced in the 3K mice relative to controls (Fig 2.3H, Control = 4.9 ± 1 Hz, 3K = 2.1 ± 0.37 Hz, p -value = 0.02). This effect was exaggerated in the dopamine depletion condition (Fig 1H, Depleted = 0.94 ± 0.35 Hz p -value < 0.0001). The distribution of Δ_{FR} during movement showed a leftward shift suggesting more units not responding to induced movement, than in the control group (Fig 2.3I, 2.3K $p = 0.0031$, Depleted $p < 0.0001$, K-S test). The narrow distribution of Δ_{FR} suggested a dampening of the degree of modulation of those units that showed movement-related increases in firing. To further explore the mechanism of this population-level dampening of movement-related activity, we categorized the units as positively modulated if they showed a statistically different ($p < 0.05$) firing rate during the movement period compared to the pre-movement period when a t-test was performed on the binned (bin size = 5s) firing rates for these two periods. If the rate was statistically different, and greater during movement, it was classified as positively modulated, if it was statistically different, and lower during movement, it was classified as negatively modulated, statistically indistinguishable units were considered non-modulated. In control mice, 77% of VAL neurons showed a significant change in firing rate during movement with 67% (28 of 42 units) showing an increase in firing rate and 10% (4 of 42 units) showing a decrease. In 3K and 6-OHDA groups, this percentage was reduced to 50% (33 of 66 units) and 62% (56 of 90 units), respectively (Fig. 2.3J, $p < 0.0001$, Chi-square comparing all three groups for the 3 categories). The numbers of excited units (the most common movement-

related response) in the 3K group and 6-OHDA group were significantly lower (45% and 35% respectively) compared to the control group ($p=0.047$ Ctl. vs 3K, $p=0.0013$ Ctl vs. 6-OHDA, Fisher's exact test comparing positively modulated vs non-positively modulated units between two groups). The depleted mice also showed an elevation in the number of negatively modulated units relative to controls. In addition to the changes in the number of responsive units, among those that were considered, positively modulated, the degree of change in firing rate was still lower (Fig 2.3K). Within the population of excited units, ΔFR was significantly attenuated in both 3K and 6-OHDA groups relative to control (Fig. 2.3L, Ctl vs. 3K $p=0.04$, Ctl vs. 6-OHDA $p=0.0006$).

In summary, the 3K mice, while showing normal firing rates at rest, showed a dampened response to movement-related activity both in terms of the number of responsive units, as well as the degree of firing rate modulation among responsive units. These changes were worsened in the more severe toxin model of PD. These similarities could either represent a common mechanism of pathological firing in the VAL thalamus secondary to cerebellar dysfunction in both disease models on a continuum of severity or they might be caused by divergent mechanisms. The suppressed firing rate, and severely reduced movement modulation, could all be the result of top-down cortical influence to the VAL secondary to the spiraling of basal ganglia inputs to the motor cortex in a dopamine depleted state (Bosch-Bouju et al., 2013). In the 3K mice, the cerebellar and motor cortical pathology could impact the VAL, leading to dampened movement modulation. The positively modulated units are specifically studied to understand encoding of movement-related activity. Further investigation is required to understand the functional relevance of units that were suppressed in response to movement. While the firing rate of the VAL is normal at rest, further exploration of the firing patterns could provide neural bases for the motor symptoms seen in the 3K mice.

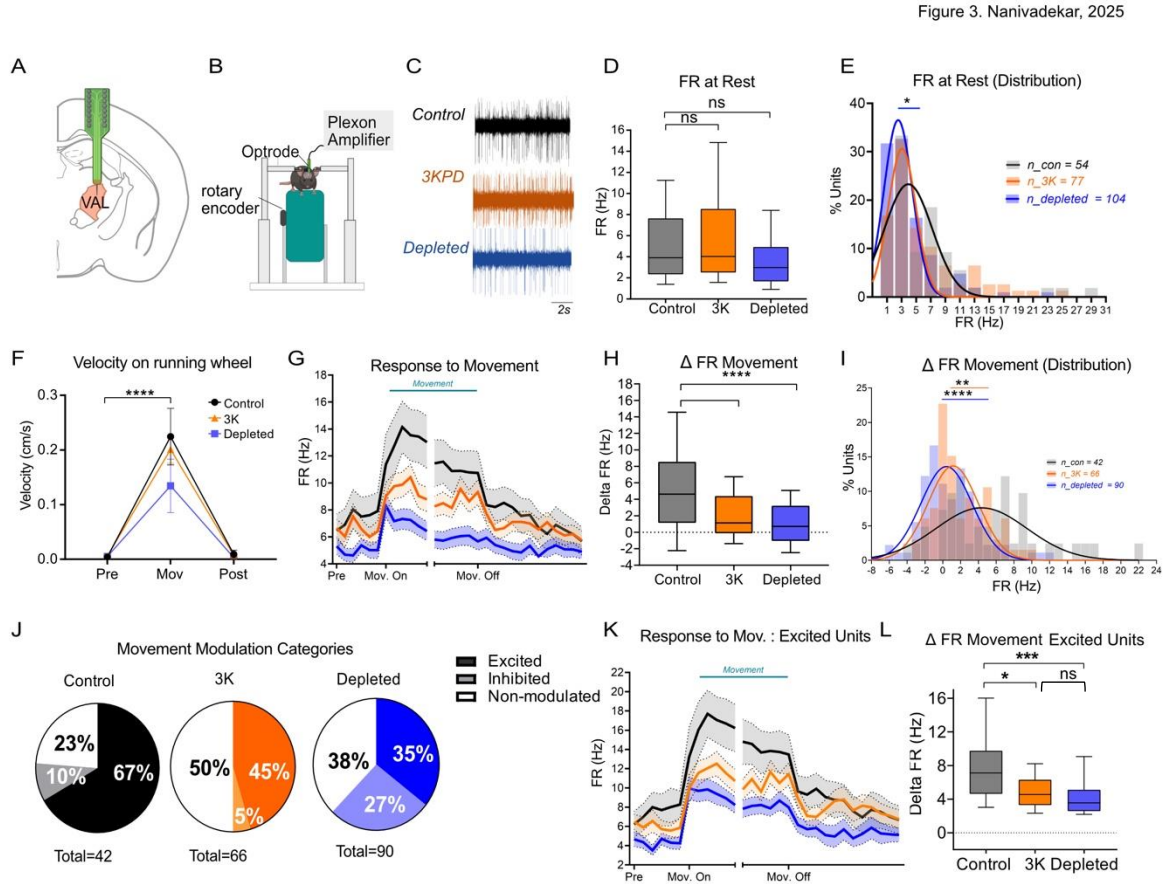


Figure 2.3 VAL in 3K and dopamine depleted mice showed dampened response to movement.

A-B, Schematic showing recording electrode targeting the VAL for awake, head-fixed recordings on a wheel that allows restricted movements and has a rotary encoder attached for movement tracking. **C**, Representative examples of units from a Control (black), 3K (orange), and Depleted (blue) mouse VAL over a 10-second period. **D**, Firing rates of VAL neurons compared between the two pathological groups vs Control animals. (Control = 5.99 ± 0.8 Hz, $n=54$ units, $N=6$; 3K = 6.41 ± 0.7 Hz, $n=77$ units, $N=8$; Depleted = 4.03 ± 0.36 Hz, $n=104$ units, $N=6$; Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p>0.999$, Control vs Depleted $p=0.052$). **E**, Distribution of firing rates of VAL neurons from panel D plotted as a histogram (bin-width = 2Hz, range = 0-32 Hz, relative frequencies depicted) with a Gaussian non-linear curve fit per group. Cumulative distributions compared using the K-S test. (Control vs 3K $p=0.81$, Control vs Depleted $p=0.02$). **F**, Velocity induced during forced locomotion by a manually applied rotational force to the wheel quantified as angular velocity (radius of wheel = 7.5cm). Raw rotary encoder data smoothed with a Gaussian kernel smoothing and converted to velocity traces that are computed for the pre-movement, movement, and post-movement periods. Mean velocity during these periods computed and pre-to-movement comparisons performed for each group. (Control pre = $6.8 \times 10^{-5} \pm 3.7 \times 10^{-5}$ rad/s, mov = $4 \times 10^{-3} \pm 9 \times 10^{-4}$ rad/s, post = $1.6 \times 10^{-4} \pm 1.3 \times 10^{-4}$ rad/s; 3K pre = $1 \times 10^{-4} \pm 5 \times 10^{-5}$ rad/s, mov = $3.6 \times 10^{-3} \pm 4.5 \times 10^{-4}$ rad/s, post = $1.3 \times 10^{-4} \pm 6.2 \times 10^{-5}$ rad/s; Depleted pre = $8.9 \times 10^{-5} \pm 3.1 \times 10^{-5}$ rad/s, mov = $2.4 \times 10^{-3} \pm 9 \times 10^{-4}$ rad/s, post = $5.7 \times 10^{-5} \pm 2.6 \times 10^{-5}$ rad/s, Control pre-vs-mov $p<0.0001$, 3K pre-vs-mov $p<0.0001$, Depleted pre-vs-mov $p=0.0004$, Dunnett's multiple comparison test, 2-way ANOVA. **G**, Peristimulus time-

histogram (PSTH) plotted binned firing rates (bin size = 5s) for 30 seconds pre-movement-onset, for the variable duration of the movement bout (> 20s) and 60 seconds after movement offset for the population of VAL units (black line = Control, orange line = 3K, blue line = depleted, shaded area = SEM). **H**, Changes in firing rate from pre-movement period (30s) to during movement bout was computed for each recorded unit and mean delta FR was depicted for the 3 groups. (Control = 4.9 ± 1 Hz, n=42 units, N=4; 3K = 2.1 ± 0.37 Hz, n=66 units, N=6, Depleted = 0.94 ± 0.35 Hz, n=90 units, N=4, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.02$, Control vs Depleted $p<0.0001$). Movement analysis could be performed only on a subset of total neurons due to missing long duration (>20s) movement bouts. **I**, Cumulative distributions for the delta firing rates during movement depicted for the 3 groups, with a Gaussian non-linear curve, distributions were compared using K-S test. (Control vs 3K $p=0.0031$, Control vs Depleted $p<0.0001$). **J**, Ratios of positively, negatively, and non-modulated units depicted as pie-charts (Control = 28/42 Excited, 4/42 Inhibited, 10/42 Non-modulated; 3K = 30/66 Excited, 3/66 Inhibited, 33/66 Non-modulated; Depleted = 32/90 Excited, 24/90 Inhibited, 34/90 Non-modulated). **K-L**, Similar analyses to panel G-H, performed only for the positively modulated units. PSTH plotted for the subset of neurons that showed excitation in response to movement. Changes in FR from pre to movement periods compared for the Control, 3K, and Depleted units (Control = 7.8 ± 0.94 Hz, n=28 units; 3K = 4.6 ± 0.42 Hz, n=30 units, Depleted = 4.0 ± 0.5 Hz, n=32 units, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.04$, Control vs Depleted $p=0.0006$, 3K vs Depleted $p=0.646$). ns, not significant; * $p=0.05-0.01$; ** $p=0.01-0.001$; *** $p=0.001-0.0001$; **** $p<0.0001$

2.4.4 Cerebellar-recipient thalamus showed pathological bursting in 3K and dopamine depleted mice

Motor thalamic activity shows tonic asynchronous firing pattern in awake animals, and a predominance of slow oscillations and LTS bursts in the context of anesthesia or slow-wave sleep (Hirsch et al., 1983). Thalamic firing displays complex shifts based on the brain state (Rowland et al., 2010) and displays coherence with cortical activity (Nakamura et al., 2014). Similarly, oscillations generated in the cerebellum, have been shown to propagate through the thalamus and encode tremor amplitude (Wang et al., 2024). Hence, we studied specific alterations to firing patterns in the VAL of the 3K mice and dopamine depleted mice. We assessed neural activity patterns at the level of individual neurons, studying irregularity of firing and burstiness, as well as at the population level using measures of synchrony and oscillatory activity emergent from simultaneously recorded units.

The spiking activity of 3K mice and dopamine depleted mice revealed a bursty, irregular pattern, relative to control VAL units (representative examples Fig 2.4A, red underline: bursts). We used the Poisson surprise method for burst detection with a surprise threshold $S=3$, with at least 2 spikes per burst and a firing rate threshold 1.5 times the baseline firing rate. The number of spikes within bursts relative to the total number of spikes was compared for all three groups. The 3K mice and dopamine depleted mice showed a significantly higher proportion of spikes in bursts (Fig 2.4B, p-value relative to controls: 3K = 0.04, Depleted = 0.003). Additionally, the irregularity of firing was studied by computing the coefficient of variation of the interspike interval. The 3K and dopamine

depleted mice showed more irregular firing relative to controls (Fig 2.4C, p-value relative to controls: 3K = 0.01, Depleted = 0.0013). A strong correlation between the irregularity and bursting activity suggests that these alterations in firing likely arose from a common pathological mechanism (Fig 2.4D).

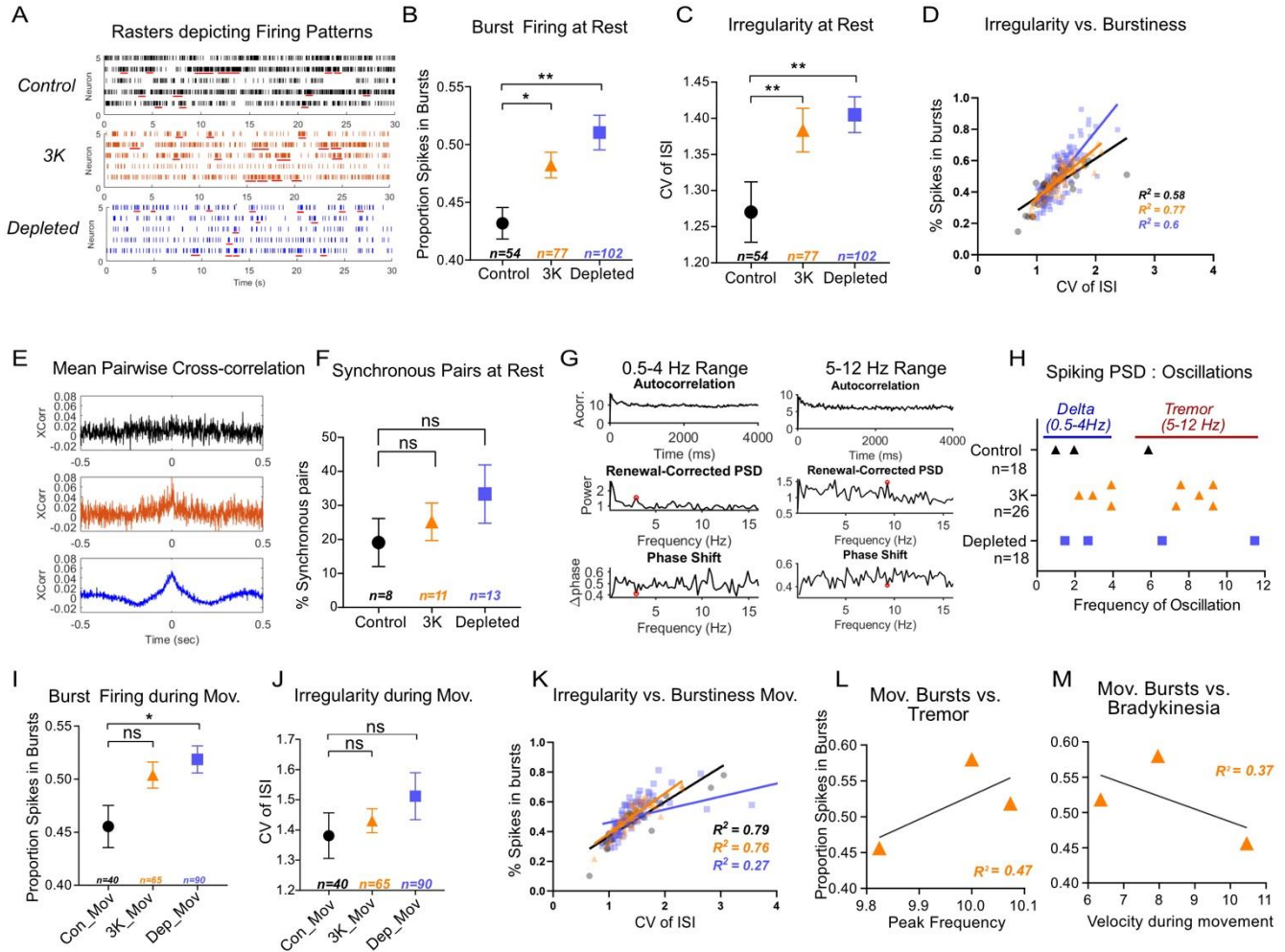
In addition to individual metrics of firing patterns, to study population level changes in firing, synchrony was measured using pair-wise cross-correlation analyses. For units recorded simultaneously, average cross-correlation was computed over short time windows (4 s long) and normalized to account for non-stationarities. The average pairwise cross-correlation across all units within individual groups depicted a peak at 0-lag in the dopamine depleted and 3K mice (Fig 2.4E). To determine the number of synchronous pairs, a threshold of units with zero-lag synchrony, computed for a 10 ms window, greater than 3-SDs from the mean of the 0.5 to 4s baseline period on either side, was used (Fig 2.4F). For the trajectories that included more than 2 simultaneously recorded units (N=8 control, N= 11 3K, and N=13 Depleted), the differences in the percent synchronous pairs was not significant between the groups (Fig 2.4F, p-value relative to controls 3K = 0.82, Depleted =0.63). This might suggest that the synchronous units that drive the peak seen in the average cross-correlation might make up the minority of units recorded in this cohort. Oscillatory analyses using a combination of the power spectral density and phase-shift criteria described previously (Whalen et al., 2020) was performed to identify whether the 3K mice showed any oscillations coherent with the tremor frequency. The tremor range chosen for this analysis was 5-12 Hz (Fig 2.4G). Oscillations in the delta-range 0.5-4Hz were also computed due to their relevance in the basal ganglia of dopamine depleted mice (Whalen et al., 2020). While the number of oscillatory units were limited, there were more oscillatory units in the tremor range in the 3K mice than either of the other groups (Fig 4H). Burstiness and irregularity seen at rest in the 3K and dopamine depleted mice, persisted during movement (Fig 2.4I-K), with the control animals also showing a shift towards more irregular firing during induced movement.

To directly probe the relationship between the aberrant pattern of firing and the symptoms seen in 3K mice, a correlation was computed between tremor frequency and the burstiness of VAL firing in three 3K mice that underwent both neural recordings and behavioral studies. There was a correlation between these features, but more data would be required to confirm this effect (Fig 2.4L, tremor peak frequency vs burst metric $R^2 = 0.47$). Similarly, a higher proportion of bursts predicted a lower velocity during movement (Fig 2.4M, velocity during movement vs burst metric, $R^2 = 0.37$).

Together the changes in firing pattern suggest that the 3K mice and dopamine depleted mice showed bursty firing relative to controls at rest that persists during movement. A small proportion of

units demonstrate high degree of synchronous firing in both pathological groups. Oscillations are limited, with some evidence for tremor-encoding cells in the 3K mice. Additionally, the frequency of

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the tremor and severity of bradykinesia might be related to the burstiness in the VAL.

Figure 2.4 VAL in 3K and dopamine depleted mice showed pathological firing patterns.

A, Rasters depicting spiking activity from 5 representative units from Control (black), 3K (orange), depleted (blue) mice. Red line depicts clusters of spikes qualifying as bursts. **B**, Bursts detected using the surprise burst method (S=3) for the bouts of rest during the baseline 5 minutes of recording. Rest bouts were computed using a threshold of <0.003 rad/s rotational velocity of the wheel during this period. Proportion of spikes within bursts were computed per unit only during bouts of rest longer than 20sec and compared between 3K/Depleted mice and Control mice. (Control = 0.43 ± 0.01, n=54, N=6; 3K= 0.48 ± 0.01, n=77, N=8; Depleted=0.51 ± 0.01, n=102, N=6, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K p=0.04, Control vs Depleted p =0.003). **C**, Irregularity of firing measured as the coefficient of variation of the interspike interval (CV of ISI) per group specifically for periods of rest >20s. (Control = 1.27 ± 0.04, 3K= 1.38 ± 0.03, Depleted=1.41 ± 0.02, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K p=0.01, Control vs Depleted p =0.0013). **D**, A simple linear regression was performed between the irregularity and the burst metric (proportion of spikes within bursts) from panels B-C. (Control = R² = 0.58, Slope of line = 0.25, slope significantly non-zero; 3K = R² = 0.77, Slope of line = 0.32, slope significantly non-zero; Depleted= R² = 0.6, Slope of line = 0.47, slope significantly non-zero). **E**, Average cross-correlation values computed for all pairs of simultaneously recorded units on a trajectory at rest (Control, n=8 trajectories; 3K n=11 trajectories; Depleted n=13 trajectories). Cross-

correlation was performed using 1ms bins for 12s long windows with a 4s overlap on each side. The direct cross-correlation values for all pairs were averaged per group and depicted in this mean composite synchrony graph. **F**, A criteria was set for each pair based on the mean correlation of the 0.5s-4s window on either side. If the zero-lag synchrony value (for a 10 ms window around 0), exceeded 3-SDs from this mean, the pair was considered synchronous. % Synchronous pairs per group are shown. (Control = 25 ± 5 % synchronous pairs, 3K = 19 ± 7 % synchronous pairs, Depleted = 33 ± 9 % synchronous pairs, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.82$, Control vs Depleted $p=0.63$). **G**, Representative examples of detection of oscillatory units using a combination of the power-spectral density and phase-shift to account for contributions from low-frequency noise. Example oscillating unit shown for the delta oscillation range (0.5-4Hz) and the range relevant to the tremor frequency (5-12Hz). **H**, Number of oscillating units identified for a subset of recorded neurons from the VAL at rest. Within the delta range, oscillating units were seen in all groups (Control = 2/18, 3K = 4/18, Depleted = 2/18). Within the tremor range, most units were seen in the 3K group (Control = 1/18, 3K = 5/26, Depleted = 2/18). **I-K**, Analyses like those shown in panels B-D performed during periods of induced movement as described before. Proportion of spikes in bursts during movement: (Con_Mov = 0.46 ± 0.02 , $n=40$, $N=4$; 3K_Mov = 0.50 ± 0.01 , $n=65$, $N=6$; Dep_Mov = 0.52 ± 0.01 , $n=90$, $N=4$, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.07$, Control vs Depleted $p=0.018$). Irregularity during movement: (Con_Mov = 1.38 ± 0.08 , 3K_Mov = 1.43 ± 0.04 , Dep_Mov = 1.51 ± 0.08 , Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.22$, Control vs Depleted $p=0.47$). Regression of irregularity and burstiness: Control = $R^2 = 0.79$, Slope of line = 0.23, slope significantly non-zero; 3K = $R^2 = 0.76$, Slope of line = 0.27, slope significantly non-zero; Depleted = $R^2 = 0.27$, Slope of line = 0.09, slope significantly non-zero). **L**, For 3 animals that underwent both behavioral and recording experiments in the 3K group, the proportion of spikes in bursts during movement are correlated with the peak tremor frequency in the tremor box ($R^2 = 0.47$, Slope of line = 0.34, slope non-significant $p=0.52$, Simple linear regression). **M**, Similarly, the burstiness was correlated with the velocity during movement in the open field using a simple linear regression. ($R^2 = 0.37$, Slope of line = -0.02, slope non-significant $p=0.59$, Simple linear regression).

2.4.5 Basal ganglia recipient-thalamus in the tremor mice was normal, and showed suppressed movement modulation in dopamine depleted mice

The basal ganglia-recipient thalamus (VM) which receives inputs from the SNr, is thought to have altered activity in a parkinsonian state (Albin et al., 1989). The study of VM in the 3K mice is motivated by the presence of pathological aggregation in the SNr, and the modest bradykinesia seen in these mice.

With a similar approach to the one described previously (Fig 2.6A-B), VM neuronal activity was studied in 3K mice ($n=108$ units, $N=8$ animals, age = 16 ± 4 months), age-matched wild-type control animals ($n=130$ neurons, $N=6$ animals, age = 11 ± 5 months), and dopamine depleted animals ($n=85$ neurons, $N=7$ animals, age = 9 ± 6 months). At rest, the 3K mice showed no differences in the average firing rates 5.57 ± 0.48 Hz, whereas the dopamine depleted mice showed a suppressed firing rate 3.38 ± 0.41 Hz relative to control animals 5.03 ± 0.32 Hz (mean $\pm$ SEM). (Fig 2.6D-E, p -value 3K > 0.999 , Depleted $p<0.0001$). Notably the firing activity of the VAL (5.99 ± 0.8 Hz) and VM was comparable in the control group, as has been previously shown (Anderson and Turner, 1991).

To test movement-related activity, a similar set of comparisons like in the VAL, were repeated for VM neurons recorded in the three groups. The movement-related activity of the control VM, while

lower than the VAL in control animals, was comparable to the 3K VM activity (Fig 2.6G-H Control_VAL = 4.9 ± 1 Hz, Control = 3 ± 0.47 Hz, 3K = 3.85 ± 0.54 Hz, Hz, n=90 units, N=4, Control vs 3K p=0.62, Fig 2.6I, p=0.79, K-S test). The dopamine depleted mice, like the changes observed in the VAL, showed a dampened response to movement related activity (Fig 2.6G) The change in firing rate from the pre-movement to movement state was reduced in dopamine depleted mice depicted by a leftward shift in the delta FR distribution (Fig 2.6I, Depleted 1.7 ± 0.34 . Control vs Depleted p = 0.046 Fig 2.6I, p=0.02, K-S test).

In control mice, 68% of VM neurons showed a significant change in firing rate during movement which was lower than the 77% in the VAL, with 58% (45 of 78 units) showing an increase in firing rate and 10% (8 of 78 units) showing a decrease. In 3K and 6-OHDA groups, this percentage was shifted to 75% (55 of 73 units) and 76% (68 of 90 units), respectively (Fig. 2.6J, p<0.052, Chi-square comparing all three groups for the 3 categories). The number of excited units (the most common movement-related response) in the 6-OHDA group was lower (50%) and the number of inhibited units was higher (26%) compared to the control group (p=0.036 Ctl. vs 6-OHDA, Fisher's exact test comparing positively modulated vs non-positively modulated units between two groups). Within the population of excited units, deltaFR was significantly attenuated relative to the 3K group but not the control group (Fig. 2.6I, Ctl. vs 6-OHDA p=0.37, 6-OHDA vs 3K p=0.005). Hence the reduced movement-related activity of the VM in the dopamine depleted mice, is likely due to a shift in the number of responsive units only.

Figure 5. Nanivadekar, 2025

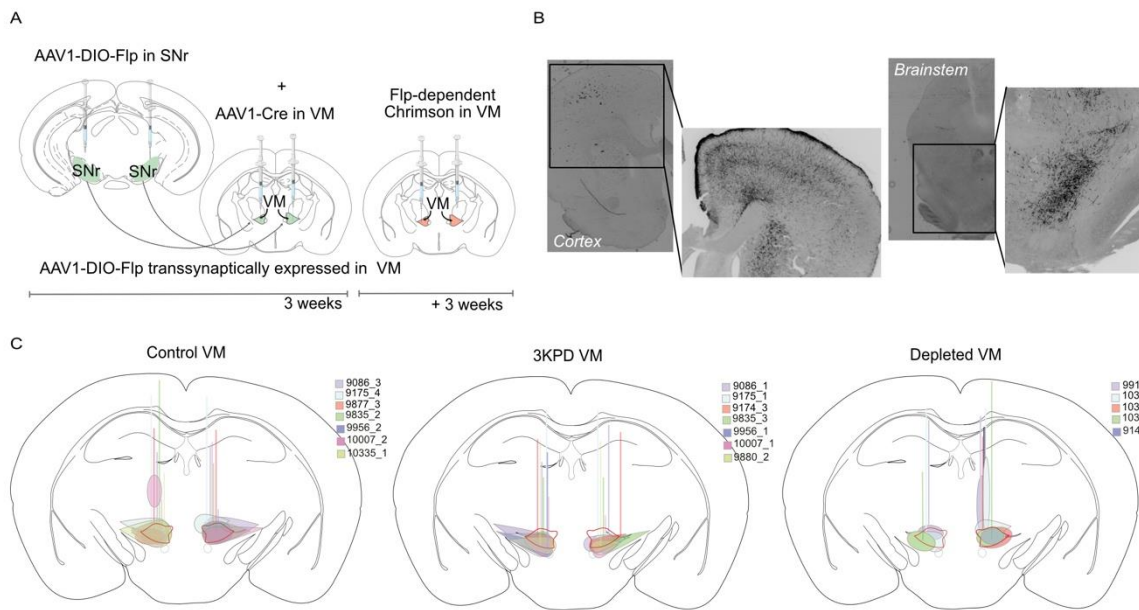


Figure 2.5 Transsynaptic approach to target the basal ganglia-recipient thalamus (VM)

A, Schematic describing the viral approach using an AAV1-DIO-Flp injected in the SNr that is transsynaptically expressed in the VM. In the same surgery, an AAV1-Cre viral construct is injected in the VM thalamus. After 3 weeks, the Flp-dependent Chrimson opsin is injected in the VM to visualize the SNr-recipient thalamic region allowing for post-hoc validation of recording site. Viral expression after the second surgery was ~3 weeks. **B**, Representative example of the projection patterns of the VM to the layer 1 of the motor cortex, striatum, and the pontine reticular nucleus of the brainstem in an animal (10335_3, see panel C) that underwent the viral approach described in A. **C**, Schematic overview of the spread of viral labeling around the VM and recording trajectories reconstructed from histological slices for animals used in Figures 6 and 7.

Figure 5. Nanivadekar, 2025

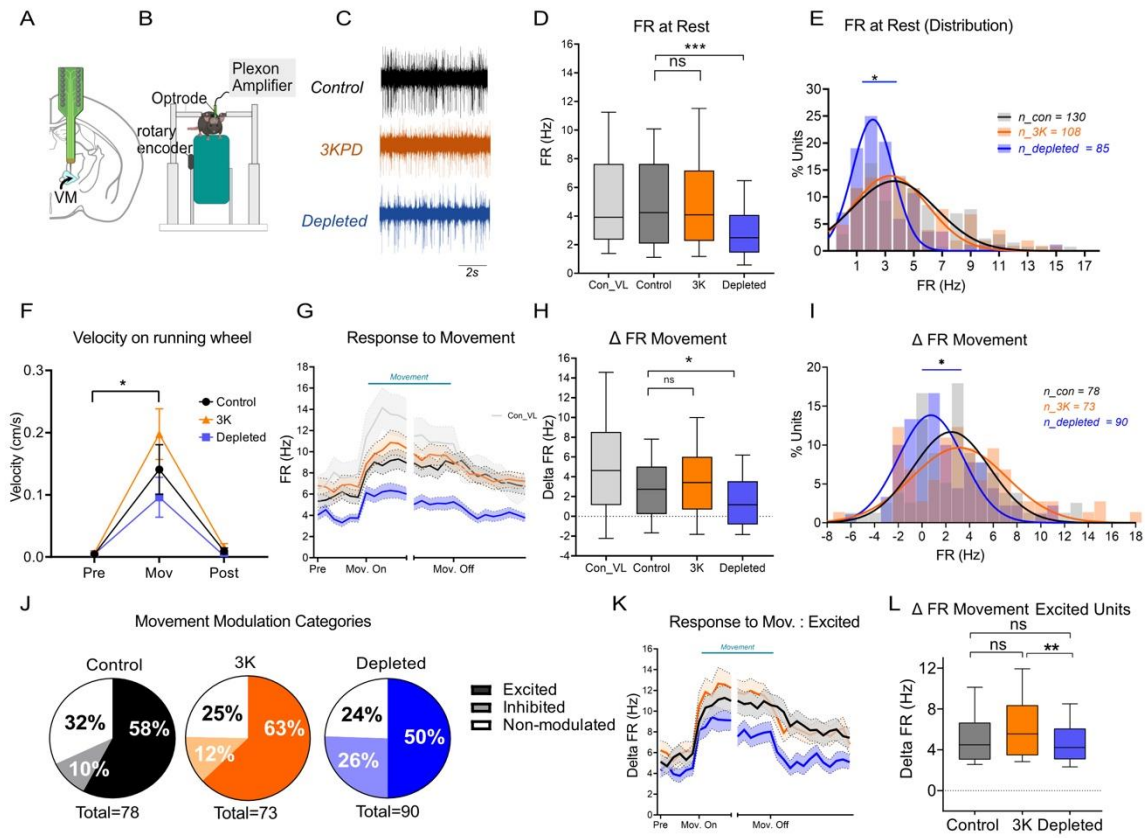


Figure 2.6 VM showed normal movement modulation in 3K, and a suppressed firing rate and dampened response to movement in dopamine depleted mice

A-B, Schematic showing recording electrode targeting the VM for awake, head-fixed recordings. **C**, Representative examples of units from a Control (black), 3K (orange), and Depleted (blue) mouse VM over a 10-second period. **D**, Firing rates of VM neurons compared between the 3K/Depleted groups vs Control animals. Firing rate of VAL control units also depicted (light grey) (Control_VAL = 5.99 ± 0.8 Hz, $n=54$ units, $N=6$; Control = 5.03 ± 0.32 Hz, $n=130$ units, $N=8$; 3K = 5.57 ± 0.48 Hz, $n=108$ units, $N=8$; Depleted = 3.38 ± 0.41 Hz, $n=85$ units, $N=7$; Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p>0.999$, Control vs Depleted $p<0.001$). **E**, Distribution of firing rates of VAL neurons from panel D plotted as a histogram (bin-width = 2Hz, range = 0-32 Hz, relative frequencies depicted) with a Gaussian non-linear curve fit per group. Cumulative distributions compared using the K-S test. (Control vs 3K $p=0.98$, Control vs Depleted $p<0.0001$). **F**, Velocity induced during forced locomotion by a manually applied rotational force to the wheel quantified as angular velocity (radius of wheel = 7.5cm). Raw rotary encoder data smoothed with a Gaussian kernel smoothing and converted to velocity traces that are computed for the pre-movement, movement, and post-movement periods. Mean velocity during these periods computed and pre-to-movement comparisons performed for each group. (Control pre = $7.8 \times 10^{-5} \pm 5.2 \times 10^{-5}$ rad/s, mov = $2.5 \times 10^{-3} \pm 7.1 \times 10^{-4}$ rad/s post = $1.7 \times 10^{-4} \pm 8.7 \times 10^{-5}$ rad/s; 3K pre = $1.2 \times 10^{-4} \pm 5 \times 10^{-5}$ rad/s, mov = $3.5 \times 10^{-3} \pm 7.2 \times 10^{-4}$ rad/s post = $2.7 \times 10^{-4} \pm 1.1 \times 10^{-4}$ rad/s; Depleted pre = $8.2 \times 10^{-5} \pm 2.9 \times 10^{-5}$ rad/s, mov = $1.7 \times 10^{-3} \pm 5.7 \times 10^{-4}$ rad/s post = $3.2 \times 10^{-5} \pm 1.6 \times 10^{-5}$ rad/s, Control pre-vs-mov $p=0.0038$, 3K pre-vs-mov $p<0.0001$, Depleted pre-vs-mov $p=0.016$, Dunnett's multiple comparison test, 2-way ANOVA. **G**, Peristimulus time-histogram (PSTH) plotted binned firing rates (bin size = 5s) for 30 seconds pre-movement-onset, for the variable duration of the movement bout (> 20s) and 60 seconds after movement offset for the population of VM units (grey line = Control VAL – same as figure 3, black line = Control, orange line = 3K, blue line = depleted, shaded area = SEM). **H**, Changes in firing rate from pre-movement period (30s) to during movement bout was computed for each

recorded unit and mean delta FR was depicted for the 3 groups. (Control_VAL = 4.9 ± 1 Hz, n=42 units, Control = 3 ± 0.47 Hz, n=78 units N=5; 3K = 3.85 ± 0.54 Hz, n=73 units, N=6, Depleted = 1.7 ± 0.34 Hz, n=90 units, N=4, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.62$, Control vs Depleted $p = 0.046$). Movement analysis could be performed only on a subset of total neurons due to missing long duration (>20s) movement bouts. **I**, Cumulative distributions for the delta firing rates during movement depicted for the 3 groups, with a Gaussian non-linear curve, distributions were compared using K-S test. (Control vs 3K $p=0.79$, Control vs Depleted $p = 0.02$). **J**, Ratios of positively, negatively, and non-modulated units depicted as pie-charts (Control = 45/78 Excited, 8/78 Inhibited, 25/78 Non-modulated; 3K = 46/73 Excited, 9/73 Inhibited, 18/73 Non-modulated; Depleted = 45/90 Excited, 23/90 Inhibited, 22/90 Non-modulated). **K-L**, Similar analyses to panel G-H, performed only for the positively modulated units. PSTH plotted for the subset of neurons that showed excitation in response to movement. Changes in FR from pre to movement periods compared for the Control, 3K, and Depleted units (Control = 5.31 ± 0.53 Hz, n=45 units N=5; 3K = 6.37 ± 0.55 Hz, n=46 units, N=6, Depleted = 4.16 ± 0.39 Hz, n=45 units, N=4, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.338$, Control vs Depleted $p = 0.368$, 3K vs Depleted $p = 0.005$). ns, not significant; * $p = 0.05-0.01$; ** $p = 0.01-0.001$; *** $p=0.001-0.0001$; **** $p<0.0001$

2.4.6 Basal ganglia-recipient thalamus showed normal firing patterns in 3K mice and increased synchrony and delta oscillations in dopamine depleted mice

Firing pattern changes like pathological oscillations and burst firing in the basal ganglia have been shown to be associated with bradykinesia (Hutchison et al., 2004). In a dopamine depleted state, the SNr shows pathological changes that include alterations in firing rate as well as an increase in pathologically synchronous activity seen as bursts of spikes with intermittent pauses (Mastro et al., 2017). The VM consequently is hypothesized to reflect these alterations in the dopamine depleted mouse model. In the context of pathological aggregates in the SNr of 3K mice, it is of interest to study activity of the VM neurons to see whether the firing pattern changes seen in the VAL generalize to the VM or demonstrate circuit-specificity.

The VM neurons in control animals showed more bursty and irregular firing relative to the VAL neurons (Fig 2.7B, C) with a strong correlation between these metrics suggesting a common underlying mechanism. Unlike the VAL, the 3K and dopamine depleted mice did not abnormalities in individual neuronal firing patterns (Fig 2.7B-C, bursts relative to control: p-value 3K=0.2, Depleted=0.86; irregularity relative to control: p-value 3K=0.52, Depleted>0.99). When looking at population level metrics of neuronal firing, like synchrony, both the 3K and dopamine depleted mice depicted a strong peak at 0-lag synchrony (Fig 2.7E). Specifically, the dopamine depleted mice showed a significantly higher number of synchronously firing units relative to control (Fig 2.7F, p-value 3K = 0.8, Depleted = 0.004). The power-spectral density and phase-shift analysis for oscillatory signals in the VM revealed a greater proportion of units in the dopamine depleted condition showing oscillations

in the delta-frequency range (4/16 units). (Fig 2.7G-H). No oscillatory units were detected in the tremor range among the 3K VM neurons.

Burstiness was worsened among the dopamine depleted mice during movement relative to controls (Fig 2.7I, p-value 3K>0.99, Depleted = 0.045). Additionally, the correlation between the irregularity and burstiness was abolished in the dopamine depleted condition during movement, suggesting some potential compensation for the firing pattern abnormalities (Fig 2.7K, Control $R^2 = 0.68$, Depleted $R^2 = 0.27$). Since the effects in the VM were restricted to the dopamine depleted animals, a behavioral correlation was performed between the velocity of movement and the burstiness during movement. The proportion of spikes in bursts was inversely correlated with the velocity of movement in the open field for a cohort of depleted animals, suggesting a possible mechanism for the bradykinesia (Fig 2.7L, $R^2 = 0.92$).

As predicted by the pathological activity in the basal ganglia of dopamine depleted mice, there is increased synchrony and burstiness among dopamine depleted mice relative to controls. This suggests that the suppression of movement encoding, and the overall reduction of movement velocity could be explained by the aberrant firing patterns seen in the VM of dopamine depleted mice. In the VM, the 3K mice showed no differences in firing rate or individual/population level metrics of firing patterns compared to controls.

Figure 6. Nanivadekar, 2025

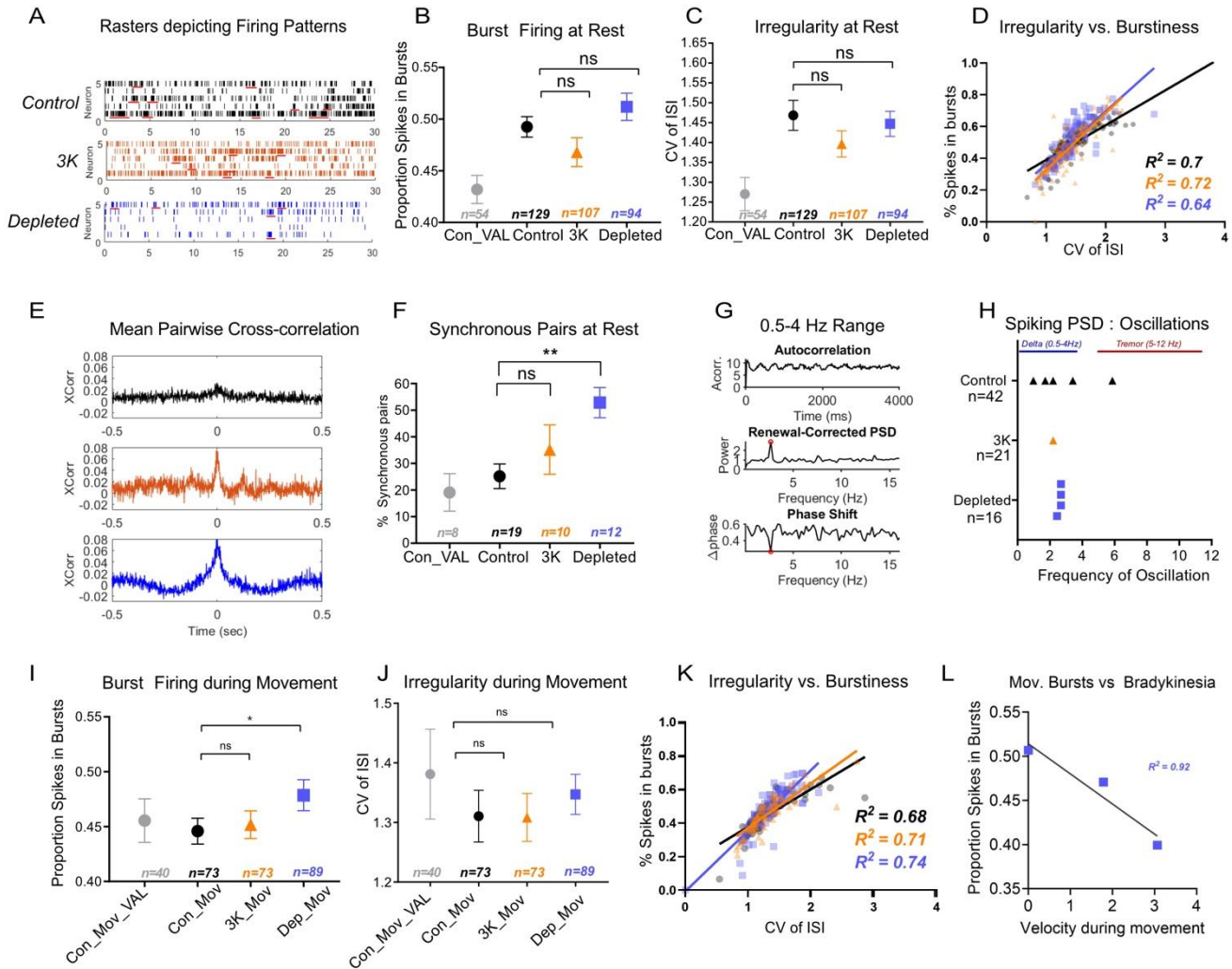


Figure 2.7 VM showed normal firing patterns in 3K, and increased synchrony in dopamine depleted mice.

A, Rasters depicting spiking activity from 5 representative units from Control (black), 3K (orange), depleted (blue) mice. Red line depicts clusters of spikes qualifying as bursts. **B**, Bursts detected using the surprise burst method ($S=3$) for the bouts of rest during the baseline 5 minutes of recording. (Control_VAL = 0.43 ± 0.01 , $n=54$, $N=6$, Control = 0.49 ± 0.01 , $n=129$, $N=8$; 3K = 0.47 ± 0.01 , $n=107$, $N=8$; Depleted = 0.51 ± 0.01 , $n=94$, $N=7$, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.2$, Control vs Depleted $p=0.86$). **C**, Irregularity of firing measured as the coefficient of variation of the interspike interval (CV of ISI) per group specifically for periods of rest $>20s$. (Control_VAL = 1.27 ± 0.04 , 3K = 1.38 ± 0.03 , Control = 1.47 ± 0.04 , 3K = 1.4 ± 0.03 , Depleted = 1.45 ± 0.03 , Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.51$, Control vs Depleted $p > 0.99$). **D**, A simple linear regression was performed between the irregularity and the burst metric (proportion of spikes within bursts) from panels B-C. (Control = $R^2 = 0.7$, Slope of line = 0.22, slope significantly non-zero; 3K = $R^2 = 0.72$, Slope of line = 0.36, slope significantly non-zero; Depleted = $R^2 = 0.64$, Slope of line = 0.33, slope significantly non-zero). **E**, Average cross-correlation values computed for all pairs of simultaneously recorded units on a trajectory at rest (Control_VAL, $n=8$ trajectories; Control, $n=19$ trajectories, 3K $n=10$ trajectories; Depleted $n=12$ trajectories). The direct cross-correlation values for all pairs were averaged per group and depicted in this mean composite synchrony graph. **F**, A criteria was set for each pair based on the mean correlation of the 0.5s-4s window on either side. If the zero-lag synchrony value (for a 10 ms window around 0), exceeded 3-SDs from this mean, the pair was considered synchronous. % Synchronous pairs per group are shown. (Control_VAL = 25 ± 5 % synchronous pairs, Control = 25 ± 5 % synchronous pairs, 3K = 35 ± 9 % synchronous pairs, Depleted = 53 ± 6 % synchronous pairs, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p=0.8$,

Control vs Depleted $p = 0.004$). **G**, Representative examples of detection of oscillatory units using a combination of the power-spectral density and phase-shift to account for contributions from low-frequency noise. Example oscillating unit shown for the delta oscillation range (0.5-4Hz) **H**, Number of oscillating units identified for a subset of recorded neurons from the VM at rest. Within the delta range, oscillating units were seen in all groups with the highest proportion in the depleted mice (Control = 4/42, 3K = 1/21, Depleted = 4/16). **I-K**, Analyses like those shown in panels B-D performed during periods of induced movement as described before. Proportion of spikes in bursts during movement: (Con_Mov_VAL = 0.46 ± 0.02 , $n=40$, $N=4$; Con_Mov = 0.45 ± 0.01 , $n=73$, $N=5$; 3K_Mov = 0.45 ± 0.01 , $n=73$, $N=6$; Dep_Mov = 0.48 ± 0.01 , $n=89$, $N=4$, Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p > 0.99$, Control vs Depleted $p = 0.045$). Irregularity during movement: (Con_Mov_VAL = 1.38 ± 0.08 , Con_Mov = 1.31 ± 0.04 , 3K_Mov = 1.31 ± 0.04 , Dep_Mov = 1.35 ± 0.03 , Kruskal-Wallis with Dunn's multiple comparisons test, Control vs 3K $p > 0.99$, Control vs Depleted $p = 0.47$). Regression of irregularity and burstiness: Control = $R^2 = 0.68$, Slope of line = 0.22, slope significantly non-zero; 3K = $R^2 = 0.71$, Slope of line = 0.27, slope significantly non-zero; Depleted = $R^2 = 0.74$, Slope of line = 0.36, slope significantly non-zero **L**, For 3 animals that underwent both behavioral and recording experiments in the depleted group, the proportion of spikes in bursts during movement are correlated with the velocity during movement in the open-field using a simple linear regression. ($R^2 = 0.92$, Slope of line = -0.03, slope non-significant $p = 0.18$, Simple linear regression).

2.4.7 Seizure-like periodic multi-channel discharges observed in both thalamic sub-regions of the 3K mice

In addition to the changes observed in the firing of the VAL neurons in the 3K mice, a novel phenomenon was observed in the 3K mice where periodically all channels demonstrated a signal that was highly regular within the episode and lasted for a variable duration of time and occurred sporadically. When studied for the 300s baseline period of rest, among the 3K animals ($N=12$) for both the VM and VAL, a total of 383 events were manually labelled. The average duration of these events was 1.45 ± 1 second, ranging from 0.2 to 6 seconds long. These events were not correlated with any observable behavioral feature (based on video recording observations) and were abolished during movement. It might hence reflect a drifting to a different arousal state involving sleep or an absence-seizure like phenomenon. Simultaneous EEG recordings from the cortex might provide more insights on the phenomenology of these events.

A representative example of such an event from the VM of a 3K mouse is depicted (Fig 2.8A, inset: zoomed in on one channel during the 3s long event). Quantification of the number of spikes within the event reveals 24 spikes in 3 s suggesting an 8 Hz intra-event frequency. The inter-event frequency is highly variable. Number of multi-channel episodes and the average duration of episodes (Fig 2.8B-C) is quantified for multiple trajectories per animal in the VAL and depicted for the 3K mice ($N=8$) and littermate controls ($N=3$). All 3K mice showed at least one occurrence of this multi-channel event in the VAL, with some showing up to 23 events in a single recording session. Some, not all, littermate control animals showed shorter and infrequent sporadic occurrences of these events

in the VAL. Similar observations are made in recordings from the VM (Fig 2.8D-E) showing similar numbers of events as the VAL, with a maximum of 18 events in a session. Littermate control VM animals match the duration of 3K-VM events but are less frequent.

Given the presence of these events in both sub-regions of the thalamus, combined with the presence of synucleinopathy throughout the cortex, the mechanism underlying these seizure-like episodes could be explained by cortical pathophysiology. Other measurements of cortical activity could shed light on the time-course of these events. Additionally, to determine the role of arousal states in the occurrence of these events, level of arousal can be measured using assays like pupil diameter to identify the behavioral relevance of these episodes.

Figure 8. Nanivadekar, 2025

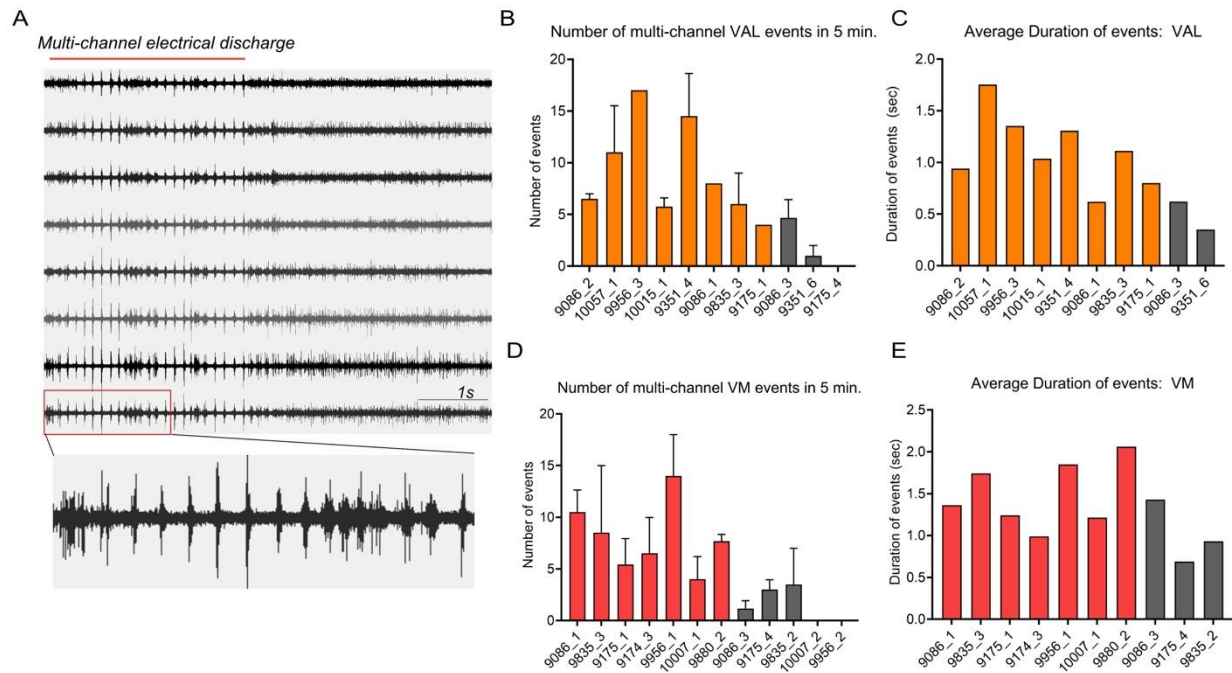


Figure 2.8 Periodic irregular discharges across channels representing seizure-like activity observed in the 3K mice.

A, Representative example of seizure-like event with multi-channel electrical discharge lasting for ~ 3seconds, inset zoomed in version of the event on a single channel. **B**, Number of these episodes manually counted for the 300s baseline rest period for multiple trajectories in individual animals with recordings performed in the VAL (3K, Range = 1-17 events in 5 minutes, N=8, Littermate control, Range = 0-5 events in 5 minutes, N=3). **C**, Average duration of all events per animal reported in panel B (3K range = 0.5-1.75 seconds, Control = 0.5-0.75 seconds). **D**, Number of episodes counted for recordings in the VM (3K, Range = 4-14 events/5 minutes, N=7, Littermate control, Range = 0-4 events/100 seconds, N=5). **E**, Average duration of VM events, (3K range = 1-2 seconds, Control = 0.6-1.4 seconds).

2.5 Discussion

Here we investigated the role of the cerebellar and basal ganglia input to the thalamus in distinct pathological models of PD, to establish neural pathways implicated in tremor and bradykinesia. In the 3K synuclein-model of PD where the predominant symptom is an action tremor at around 10 Hz, we showed selective dysfunction in the VAL with normal firing rate and abnormal firing patterns, and a dampened response to movement, suggesting afferent cerebellar dysfunction. In the dopamine depletion model of PD where bradykinesia is the primary symptom, we showed dysfunction in the VM involving changes in firing rate and pattern and a dampened response to movement, secondary to afferent basal ganglia dysfunction. Surprisingly, we also showed changes in firing patterns and movement responsivity in the VAL of dopamine depleted mice that are indistinguishable from the 3K mice, in addition to a suppressed baseline firing rate. These results suggest the existence of partially overlapping circuits passing through the motor thalamus that show alterations to produce distinct motor phenotypes.

While the aSyn pathology and dopamine depletion pathology produce divergent symptoms in the two models, namely tremor and bradykinesia, we showed that pathophysiology of the VAL in these models is similar (Fig 2.9A). Understanding this requires a two-fold approach – firstly we must identify whether the cerebellum is involved in both pathological models of PD, and secondly, we must explore whether there is a relationship between the specific cerebellar neural patterns projecting to the thalamus and the symptoms seen in these models.

Role of the cerebellum in pathological models of PD

The involvement of the cerebellum in the aSyn tremor model comes from our evidence showing a pattern of aggregation in the cerebellum, which has been reported in patients with PD (Seidel et al., 2017), as well as the prominent tremor which is often thought to arise from cerebellothalamic circuits (Hellwig et al., 2001; Marsden, 2000; Pedrosa et al., n.d.; Schnitzler et al., 2009; Y.-M. Wang et al., 2024; Wong et al., 2022). In the dopamine depleted condition, while we do not see direct evidence of cerebellar pathology, growing evidence in the literature suggests a role of the cerebellar connectivity with the basal ganglia in the manifestation of parkinsonian pathophysiology (Li et al., 2023). Specifically, dopamine loss in the context of PD has been shown to involve atrophy in the cerebellum

(Erro et al., 2020; Gellersen et al., 2017), hyperexcitability ((Herrera-Meza et al., 2014; Rolland et al., 2006) and a loss functional connectivity between the cerebellum and basal ganglia (Guan et al., 2017).

Role of cerebellar thalamus in tremor and bradykinesia

Having established cerebellar involvement in both the synuclein and depletion models of PD demonstrating divergent phenotypes, we might then ask whether these symptoms are encoded by the cerebellar or cerebello-thalamic activity in these models. Recent evidence from analysis of cerebellar physiology suggests that the cerebellar outputs might be critical in the manifestation of different motor phenotypes in movement disorders like PD, dystonia, and ET (Van Der Heijden et al., 2024). Previous studies in other mouse models of tremor have established patterns of activity in the thalamus coherent with the tremor frequency and shown a suppression of the tremor with thalamic interventions (Wang et al., 2024). The patterns of activity in the cerebellum involved in producing a tremor seem to be distinct from other cerebellar disorders establishing a direct relationship between the neural activity patterns and the phenotype (Van Der Heijden et al., 2024). In our study, we showed a similar alteration that is representative of changes in the cerebello-thalamic circuit. We showed specific changes in the firing pattern of the VAL, and not the VM, with increased burstiness, and irregularity at rest that persisted during movement. Additionally, the neural response to induced locomotion was dampened in the 3K-VAL suggesting that the shifts in oscillatory activity might have effects on other aspects of movement encoding. Gamma oscillatory activity in the VIM, human equivalent of the VAL, has been shown to be associated with reaction time in patients with ET performing a motor task (Brücke et al., 2013). The mild bradykinesia, suppression in novelty-seeking (Fig. 1) and the previously characterized deficits in gait and posture, might be secondary to the altered VAL activity projecting to motor and premotor cortices in the 3K mice. For bradykinesia, apart from functional connectivity and metabolic changes in the cerebellum of rats undergoing dopamine depletion, little is known about the specific cerebellar patterns that reflect a bradykinetic phenotype (Herrera-Meza et al., 2014; Menardy et al., 2019; Rolland et al., 2006).

Hyperactivity of the cerebellum-basal ganglia connectivity (Wu & Hallett, 2013), secondary to either Purkinje cell dysfunction or dopamine loss, might be common to both these phenotypes and might provide an explanation for the common physiological changes observed in the VAL of our 3K and dopamine depleted mice. This mechanism provides a unified framework to understand the heterogeneity of PD symptoms. Alternatively, the cerebellum might encode phenotypic features distinctly (Van Der Heijden et al., 2024), but the thalamus, which is thought to act as a gating resistor,

might produce similar pathological signatures in response to tremor-related cerebellar inputs and bradykinesia-cerebellar inputs in the context of bradykinesia due to a common top-down motor cortical influence. The VAL pathophysiology in the dopamine depleted condition might also be reflective of the potential confound of recording from the overlapping population that receives inputs from both the SNr and the DCN (Roth et al., 2024).

Alternate sources of tremor in the synuclein mouse model

While the physiological changes in the VAL are reflective of cerebello-thalamic dysfunction in the 3K mice, we cannot rule out the involvement of alternate mechanisms of tremor generation due to a lack of coherence with the tremor frequency seen in the 3K-VAL (Fig 2.9B). The cerebellar-IO closed-loop is thought to be involved in harmaline-induced mouse tremor models and patients with ET (Wang et al., 2024). This circuit, involving Purkinje cell dysfunction (Brown et al., 2020) might be self-sufficient in generating oscillations that either do not make it to the VAL or are abolished by compensatory mechanisms in the VAL due to the slow progression of symptoms in the 3K model. Tremor might also be generated through the cerebellar inputs from PV neurons in the dentate and fastigial nuclei to the gigantocellular reticular nucleus like has been previously demonstrated in a Synaptotagmin-2 (Syn2) deletion model (Zhuo et al., 2020). This is particularly likely given the aSyn pathology in 3K mice that is also involved in synaptic transport functions, like the defects observed with Syn2 deletion. Finally, cerebellar-outflow tremors manifesting as ataxic, irregular tremors or peripheral mechanisms for tremor generation either at the level of motor neurons like in the case of physiological tremors, or spinal-cord mediated aberrant sensory inputs might all be alternate pathways for tremor generation in this model. There has been growing interest in whether the thalamic tremor encoding is a cause or a consequence of tremor manifestation (Pedrosa et al., 2014). In this context, while the head-fixed setup for recordings allowed us to eliminate other behavioral confounds in the interpretation of neural activity, this setup limited the manifestation of the tremor activity in these mice. If the tremor-related activity in the VAL is an effect of the physiological manifestation, as opposed to the cause, it might alter the thalamic neural activity we observe in this setup where the tremor is not fully expressed.

Pathophysiology of bradykinesia in the motor thalamus of PD mouse models

The classical view of PD pathophysiology predicts a hyperactivity in the basal-ganglia output (Albin et al 1989, DeLong 1990). The evidence from basal-ganglia recipient motor thalamic activity has been limited and inconclusive. Mild changes including a reduction in firing rate (Schneider et al.,

1996), a loss of receptive field specificity (Pessiglione et al., 2005), and dysrhythmia in the form of inappropriate synchrony with respect to cortical beta oscillations (Nakamura et al., 2021) have been shown in the motor thalamus of dopamine depleted rodents. Using slice electrophysiology, an increase in intrinsic excitability of the basal-ganglia recipient thalamus due to a decrease in the M-type potassium currents has been demonstrated in 6-OHDA lesioned mice (Bichler et al., 2021). The loss of movement speed and issues related to movement initiation in the dopamine depletion model are thought to be mediated by the influence of the basal-ganglia on the thalamo-cortical circuit. In patients with PD, the pallidal-receiving area of the thalamus shows a suppression of firing rate and a lower number of regularly firing neurons (Molnar et al., 2005). Consistent with the predictions from the classical model, we see a suppression in the firing rate of the VM in the dopamine depletion model demonstrating bradykinesia (Fig 2.9C). Additionally, we also observe a higher proportion of synchronously firing and delta oscillatory units. These findings suggest the involvement of the basal-ganglia-thalamo-cortical circuit in the pathophysiology of depletion-mediated bradykinesia. Manipulations that suppress the pathological basal-ganglia projections to the VM are needed to confirm the causal role of this circuit. The normal activity of the VM in the 3K mice further supports the notion that the tremor mice demonstrate a different pathological circuit that likely has minimal basal-ganglia involvement.

Circuit-specific targeting of thalamic sub-regions

Given the divergent movement deficits seen in these two mouse models, with the dopamine depletion mice showing mostly bradykinesia, and the 3KPD mice showing tremor, the fact that there are differences and commonalities in the pathological changes in neural activity in different parts of the thalamus for these two phenotypes is impactful. Using the transsynaptic approach, we can specifically target basal-ganglia recipient and cerebellar-recipient thalamic sub-regions to ask whether the alterations in the 3K mice are specific to the cerebellar-input as might be predicted based on the pathology.

The 3K mice showed specificity in their pathological changes in the VAL, providing a substrate for therapeutic intervention in these mice that is circuit specific. The common pathophysiological changes in the VAL of dopamine depleted mice and 3K mice warrants further investigation of the cerebellar contributions to heterogeneous phenotypes seen in PD.

The dopamine depleted mice showed a global suppression of firing, and movement-related activity in the thalamus. Both VM and VAL tend to show these changes, leading to the idea that non-

specific thalamic interventions in dopamine depleted conditions might produce motor benefits. In both the 3K, and the dopamine depleted mice, the patterns of neural firing seem to play a larger role than the firing rate, suggesting that interventions that can alter the pattern of activity, and not just the level of activity, might lead to specific therapeutic benefit.

These findings provide the unique opportunity to update our understanding of the parallel pathological loops for tremor or bradykinesia within the thalamus. These results provide the foundation for therapeutic interventions that target specific sub-circuits in the thalamus to attain symptomatic relief generalizable to multiple symptoms with minimal off-target effects.

Figure 9. Nanivadekar, 2025

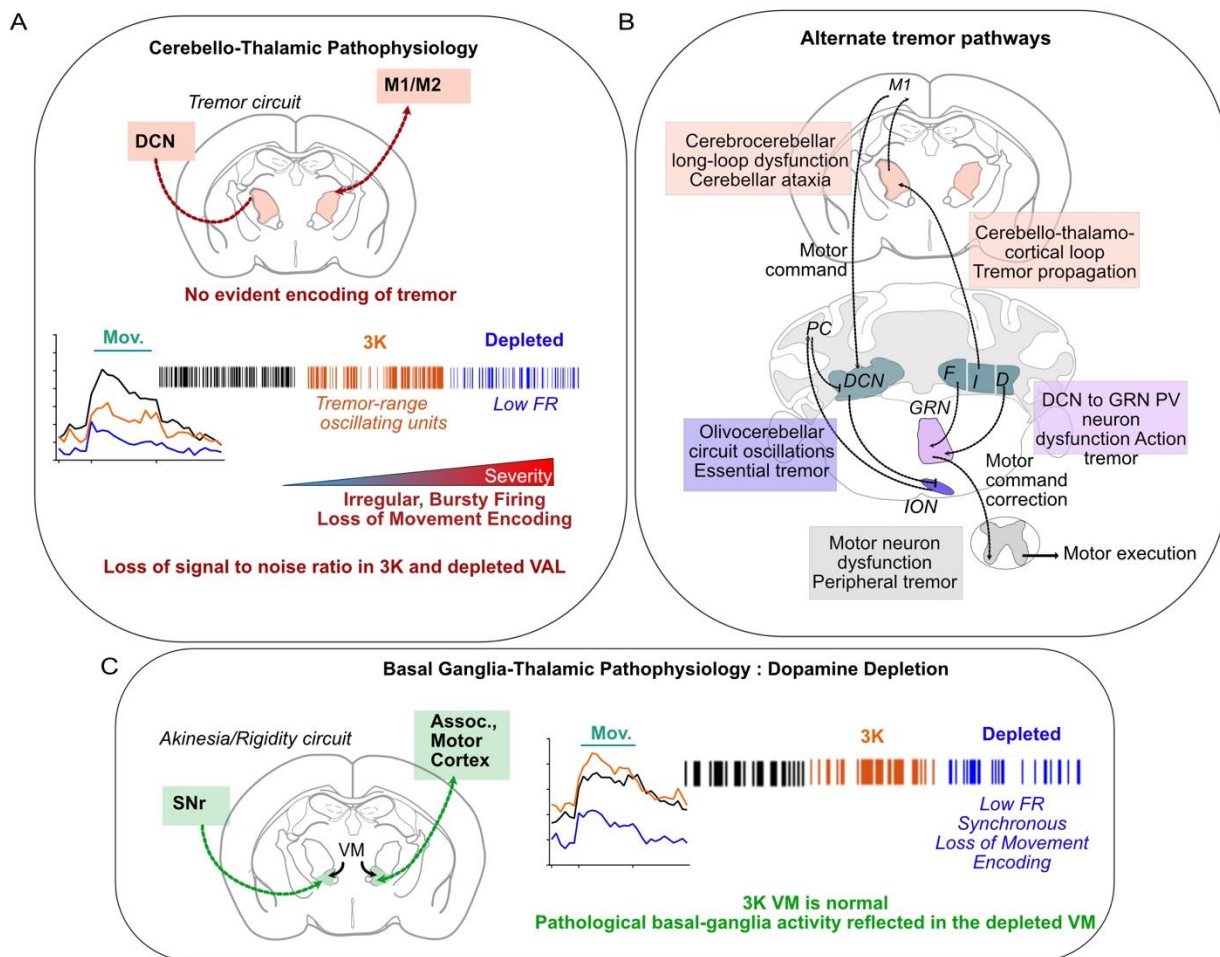


Figure 2.9 Pathophysiological mechanisms in the motor thalamus governing motor symptoms in 3K and dopamine depleted mice.

A, Cerebello-thalamic pathophysiology in the motor thalamus. In the VAL, 3K and Depleted mice showed similar changes, with greater severity in the depleted mice. There are some oscillating units at the tremor frequency in the 3K mice only, and a suppressed FR in the depleted VAL. VAL in both groups demonstrates bursty, irregular firing at rest, that persists during movement, and a suppressed response to movement-related modulation. **B**, Alternate mechanisms for tremor

pathophysiology in the 3K mice depicted based on existing evidence in the literature. The cerebrocerebellar long-loop is thought to be involved in irregular tremors generated as a consequence of the malfunctioning of the cerebellar forward model wherein lack of coordination of the efference copy of a motor command and the sensory feedback delay produces an irregular tremor-like activity called ataxia (Kakei et al., 2021). The cerebellothalamic loop has been implicated in multiple clinical studies showing the role it plays in tremor propagation and suppression, but not tremor generation. The Olivocerebellar short-loop is involved in pathological oscillatory tremor models like the harmaline-induced tremor and in ET patients. (Y.-M. Wang et al., 2024). Finally, the projections from PV neurons in the dentate and fastigial nuclei of the cerebellum to the gigantocellular reticular nucleus is thought to produce an ET- like action tremor (Zhou et al., 2020). Peripheral tremors are thought to arise from motor neuron dysfunction. **C,** The VM showed no changes in the 3K mice, but a suppressed firing rate, synchronous firing and dampened movement response in the VM of depleted mice that might contribute to the dopamine depletion-mediated bradykinesia.

Chapter 3

3 Further characterization of the mechanism underlying motor and non-motor symptoms in 3K mice with aSyn histology and an SNr-lesion study

3.1 Abstract

The 3K model allows us to study the neural mechanisms of tremor, mobility, and gait disturbances in the context of a synucleinopathy, like in Parkinson's disease. In addition to the aSyn pathology characterized in the original study, we sought to broaden the scope of our histological analyses to include the brainstem, spinal cord, and cerebellum to explore other extra-cortical mechanisms underlying symptoms demonstrated by this mouse. We demonstrate the presence of aggregates in the cortex, SNr, DCN, and sensory brainstem nuclei, and dorsal sensory regions of the spinal cord. In addition to the effects of cerebellar pathology characterized in Chapter 2, the presence of dense aggregates in sensory regions outside the brain, suggests a possible role of pathological sensory feedback in the generation of the 3K tremor. Given the lack of pathophysiology in the VM of 3K mice (demonstrated in Chapter 2), and the limited loss of TH in the striatum, we hypothesize that the 3K symptoms are likely basal ganglia-independent.

While the behavioral phenotype has been established in the paper (Nuber et al., 2018), in this study we developed motor and non-motor assays to compare deficits in the 3K mice at various stages of disease progression, as well as to demonstrate their dependency on the basal ganglia in an SNr lesion study. Therapeutic approaches were also tested to causally establish the role of dopamine in the pathophysiology. We showed that the tremor, gait, and novelty-seeking features in 3K mice are worsened with age. By performing diphtheria toxin-induced lesions in the SNr, we showed that a pathological SNr is not involved in the production of tremor or non-motor symptoms in the 3K mice but might contribute to their hindlimb postural instability. DBS delivered in the EPN of a severe, older 3K mice led to increased mobility, whereas L-DOPA had a minimal effect on symptoms. In

summary, this study establishes the age-effect of various motor and non-motor behavioral deficits, establishes a comprehensive histological map of the pathological aggregates, and suggests a basal ganglia-independent mechanism for the tremor in this genetic synuclein-induced PD model.

3.2 Introduction

In comparison to acute toxin models of PD, the 3K mice show more similarities to clinical idiopathic PD in terms of their time-course and symptoms. In addition to showing subtle deficits in mobility, these mice show a progressively worsening tremor and imbalance secondary to limb dysfunction. The Nuber paper establishes this model as a PD model due to the degeneration of striatal DA neurons, reduced DA levels and some L-DOPA responsive symptoms like gait dysfunction. Additionally, the pathological aggregates they show in the cortex, striatum, SNc, and cerebellar cortex, corroborate the motor deficiencies demonstrated. While this establishes a rationale for the involvement of the basal ganglia in the manifestation of motor symptoms in the 3K mice, our findings in Chapter 2 establish that there is little to no change in striatal dopamine levels relative to littermate controls, and that the VM is normal in the 3K mice. This suggests that the symptoms in the 3K mice might be basal ganglia-independent. To more directly probe how the basal ganglia might influence behavior in these mice, we performed an experiment where we lesioned the SNr in the 3K and control mice.

In a previously performed study in our lab, an SNr lesion in bilaterally dopamine depleted mice prevented the akinesia caused by dopamine loss, while retaining their postural dysfunction (unpublished). This result suggested that the aberrant activity in the SNr might be responsible for producing bradykinesia, and not postural deficits in the depletion model. To test whether the 3K symptoms were also similarly SNr-dependent, we non-specifically lesioned the SNr in 3K and control mice and compared their performance on various motor and non-motor tasks. We showed that the SNr lesion did not reduce the tremor or non-motor deficits, led to a reduction in ambulation, and produced changes in postural paw angles in the 3K mice. L-DOPA treatment also produced no evident effect on motor symptoms further eliminating the possibility of a dopamine-mediated pathophysiology. DBS delivered in the EPN showed an increase in open-field mobility suggesting that other nodes in the basal ganglia could be leveraged despite the lack of SNr contribution to the 3K motor syndrome.

With the goal of establishing the regions in and outside of the basal ganglia that might be implicated in the pathophysiology, we performed whole-brain aSyn labeling assays, extending the analysis to the brainstem and spinal cord as well. In addition to the cortical aggregates previously reported, we saw aggregates in the sensory brainstem nuclei and the sensory domains of the spinal cord, which provides interesting avenues to be explored regarding potential sensory contributions to the motor deficits seen in these mice. An abundance of aggregates in the DCN provided further pathological basis for the cerebello-thalamo-cortical dysfunction demonstrated in Chapter 2.

3.3 Material and Methods

3.3.1 Animals

Experimental procedures were approved by the Carnegie Mellon University Institutional Animal Care and Use Committee and in accordance with the guidelines set by the NIH and SfN Use of Animals in Neuroscience Research. Males and female mice on a C57BL/6J background were used for all experiments. Males and female heterozygotes from 4 months to 16 months of age were used from the 3K mouse line. Animals were group housed (2-5 per group) in a 12h/12h light dark cycle until used for experimentation.

3.3.2 Behavior Analysis

Open-field Behavior for Movement and Hole-board analysis

Mice were placed in the center of a square open-field chamber with video monitoring using a bottom-up camera at a rate of 29.97 frames per second. 10-minute recordings were performed. Positions of the nose, tail, and body center were tracked using Noldus Ethovision XT. Movement was quantified by directly measuring the distance moved divided by the total trial time. Additionally, the type of movement was categorized as ambulation (when the start velocity was >1.2 cm/s and stop velocity > 1 cm/s), immobility when the pixel change was $<0.01\%$ averaged over 60 samples, and fine movements when neither ambulation nor immobility were detected.

Additional open-field experiments involved a top-down camera at a rate of 29.97 frames per second performed with a hole-board base. This allowed for quantification of exploratory behaviors by counting the number of nose-pokes performed by the mouse within 2 minutes of the trial time. The nose-pokes were manually counted by an observer post-hoc. The number of nose-pokes was normalized for mobility differences during the task.

Corridor Setup for Postural analyses

A plexiglass corridor was created to restrict locomotion in a straight-line. This allowed for bottom-up video recordings which were then processed using DeepLabCut to establish paw-positions. Custom code in MATLAB was written to compute the center of mass of the animal based on the average x-y coordinates of the nose and the tail markers. The paw-positions were then normalized relative to this reference point. Locomotor bouts were manually identified by visual video labeling. The normalized paw-positions were plotted as a function of time during these bouts of locomotion across the corridor, to represent the gait cycle. The relative angle between a reference paw and all other paws was computed as a metric of postural dynamics. Additionally, the distance between the hind-paws was computed as a measure of the stability of gait during locomotion (Appendix A).

Tremor Quantification

A video was recorded from the side of a small plastic box where the mouse was stationary. This allowed for focused tracking of the nose-point of the mouse using a trained model in DeepLabCut. The position of the nose-point was plotted over time to represent the head-tremor in the mice.

Tail-suspension task for Affect

The tail-suspension task involved suspending the mouse by a tape (~6 inches long) and measuring the mobility of the mouse over a 5-minute period. This was used as a metric of learned helplessness and therefore a measure of non-motor symptomatology in the pathological groups compared to control animals.

3.3.3 Therapeutic Approaches

L-DOPA Administration

Levodopa from Sigma Aldrich (D-9628) was administered intraperitoneally (IP) with benserazide to reduce peripheral conversion to dopamine and reduce side effects. 5 mg/kg L-DOPA and 2.5 mg/kg benserazide was administered IP. Behavior was recorded in the open field from 10 minutes before LDOPA administration and for an hour post-administration.

DBS Paradigm

Fabricated DBS electrodes were implanted in the entopeduncular nucleus and biphasic, bipolar DBS was delivered to the EPN in a severe 16-month-old 3K mouse (See DBS methods in Chapter 4).

3.3.4 Histology

Plaque pathology

At the end of recording experiments, mice were euthanized and perfused with 4% paraformaldehyde (PFA). The brain was extracted and stored in PFA overnight. It was then switched to a 30% sucrose solution for another day. Tissue was sectioned using a freezing microtome (Microm HM 430; Thermo Scientific). A 1 in 5 series was performed for the entire brain up to the brainstem. Spinal cord extraction was performed on some 3K mice to collect slices in the cervical, thoracic, lumbar and sacral spinal cord. Plaque pathology in aSyn mice was assessed with phospho-S129-a-Syn staining (rabbit anti-phospho-S129-a-synuclein EP1536Y 1:100; AbCam). Primary antibody incubations were left in a 4 degree C refrigerator for 24 hours. Secondary antibodies to the rabbit primary, included Alexa Fluor 647-conjugated donkey anti-rabbit (1:500, Life Technologies) that were incubated for 3 hr. at room temperature. Spinal cord sections labeled with Alexa Fluor 488-conjugated donkey anti-rabbit (1:500) incubated for 3 hrs. at room temperature. Epifluorescent images were taken at 4x magnification (Keyence BZ-X) to identify aSyn expression.

3.3.5 Surgical procedures for lesion experiment:

All surgical procedures were performed under general anesthesia. Cellular ablation of the SNr was achieved using a Cre-targeted diphtheria toxin strategy. An AAV carrying Cre - pENN-AAV-CMV_s-PI-Cre-rBG (AAV1) and an AAV carrying a Cre-inducible diphtheria toxin receptor (AAV2-DIO-DTR-GFP) were stereotactically injected 3 weeks apart into the bilateral SNr of 3K mice. Anesthesia

was induced using 3% isoflurane in an induction chamber and maintained throughout the surgery with 1.5% isoflurane. Mice were placed in a stereotaxic frame, and bilateral holes were drilled in the skull. (SNr: -2.8 posterior, 1.5 lateral). 150 nl of virus was injected with a Nanoject (Drummond Scientific) through a pulled glass pipet (30 um tip diameter) whose tip was placed on the top of the dura (SNr: 4.65 mm). The pipet was left in the brain for 5 minutes after the viral injection to prevent backflow. Post-surgery, animals received 3mg/kg Ketofen and 0.2 ml saline and were placed on a heating pad for recovery, before returning to their home cage. Two weeks later, an injection of 1 ug diphtheria-toxin (DTX) was administered IP. Behavioral experimentation was performed 2 weeks after the DTX administration.

3.4 Results

3.4.1 3K tremor mice show aSyn aggregates throughout the cortex, in the basolateral amygdala, DCN, brainstem nuclei, and the spinal cord

To expand the search for pathological aggregation beyond the cortex and striatum and demonstrated in ((Nuber et al., 2018), immunological staining for the pathological form of the protein was performed in a whole brain series, in addition to spinal cord sections in a 3K and age-matched wild-type control. Aggregates were seen throughout the cortex and in the basolateral amygdala (BLA) (red box, Fig 3.1A). No evident aggregates were visible in the striatum, SN, or other regions of the basal ganglia and thalamus (black box, Fig 3.1B). Additionally, in the brainstem, multiple cranial nerve nuclei like the trigeminal nucleus (red box, Fig 3.1B) the facial nerve nucleus and the cuneate showed dense aggregates. The spinal cord sections also showed predominant aggregates in the dorsal horn, with some in larger ventral motor neurons. (GFP labeling, Fig 3.1D). The control animals in comparison showed no aggregates in any of these outlined regions (black boxes, Fig 3.1A-D). A follow-up analysis designed to look at whether the aggregates colocalized with neurons vs. non-neuronal cell types, involved co-staining with aSyn and a neuronal marker (NeuN). Most aSyn labelling colocalized with the neuronal marker supporting a PD-like pathology (Appendix B) as opposed to multiple systems atrophy (MSA) which involves aSyn aggregates in the oligodendrocytes. Further testing with a glial marker to label oligodendrocytes in conjunction with the aSyn could help rule out an MSA comorbidity.

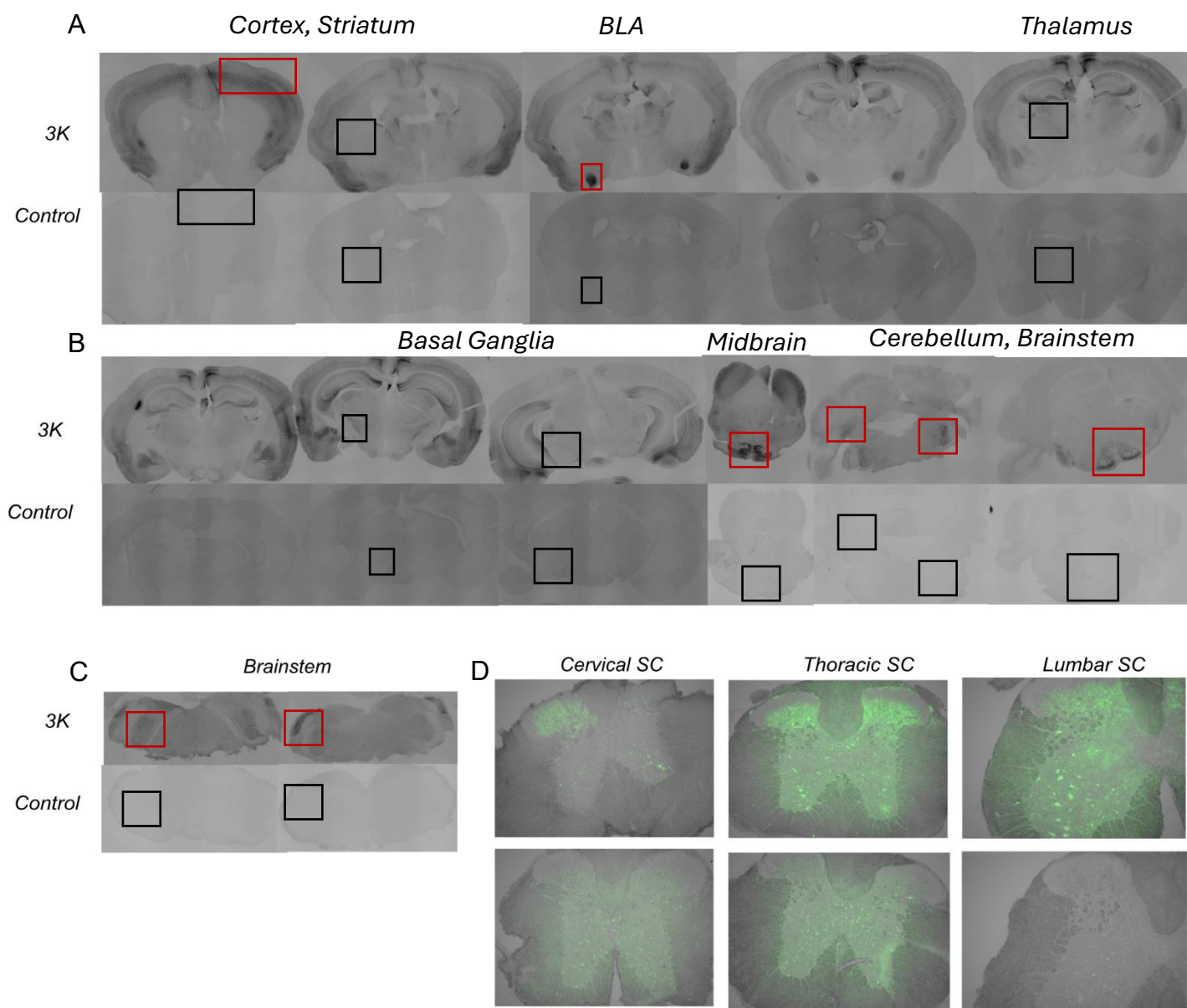


Figure 3.1 3K tremor mice showed aSyn aggregates throughout the cortex, in the DCN, and the spinal cord.

A, Brain series in cortical sections run in parallel from a 3K and wild-type mouse with far-red labeling for pathological aSyn. Positively labeled aggregates only seen in the 3K cortex and BLA (red box). Other regions depicted with no aggregates – 3K striatum, thalamus, Control cortex, striatum, BLA, thalamus (black box). **B**, Aggregates seen in the pontine grey, deep cerebellar nucleus, spinal trigeminal nucleus, and the magnocellular reticular nucleus in the 3K (red box). **C**, Other cranial nerve nuclei showed aggregates in 3K. **D**, Dorsal spinal cord showed GFP labeling for pathological aSyn in 3K and not controls.

3.4.2 3K mice showed progressive worsening of gait difficulties, postural imbalance, and novelty-seeking behaviors with age

To better develop strategies to probe mechanisms underlying the symptoms seen in this model, we sought to establish a more detailed analysis of the behavioral deficits seen in this mouse model. Using more sophisticated tools for mouse tracking like DeepLabCut (DLC), we were able to visually depict the head-tremor observed in these mice (Fig 3.2B). A comparison was made between four 3K mice and four control mice at age 4 months and 7 months old. Mice were allowed to run freely in an open-field arena for 10 minutes and their locomotion was assessed using Noldus Ethovision tracking to measure the total distance moved for the duration of the task. Open-field analyses showed a reduction in the velocity of movement in the 3K mice that worsened with age (Fig 3.2A, 3K vs Control 4month p-value = 0.2, 7-month p-value = 0.03) suggesting some bradykinesia. Additionally, mice were allowed to run freely in a corridor setup where a camera filmed their locomotion from the bottom-up. Tracking methods using DLC allowed us to measure gait-metrics like interpaw-distance and paw-position during a gait cycle (Fig 3.2C-D). The interpaw-distance showed a wide-based gait stance in the 3K mice as compared to controls evidenced by an increase in their hindlimb distance (Fig 3.2C, 3.2D red arrow) suggesting postural instability. The position of every paw was computed in reference to the center of mass (mid-point between tail and nose points) and the angle and distance between the paws relative to a reference paw (Hind-Right paw) was measured to estimate changes in limb-coordination during a gait cycle. Specifically, during the footfall in the gait cycle, the position of the other paws relative to the hind-right paw is shown. The 3K mice showed smaller angles between the hind-right and all other paws, and an increase in the distance between the hind-right and the contralateral front-left paw (Fig 3.2D, 3K – purple lines, Control – grey lines). To assess non-motor symptoms, the 3K mice were placed in an open-field with a hole-board base that allowed for nose-poking or novelty-seeking behavior. The 3K mice showed deficits in their novelty seeking behavior with an age-dependent reduction in the number of nose-pokes on the task (Fig 3.2E, 4month p-value = 0.2, 7month p-value = 0.06). Nose pokes were normalized to the % mobility of the mice in the open field to account for confounds from mobility deficits in the 3K mice. In the tail-suspension task that measures affect, the 3K's were comparable to control animals at 4 and 7 months of age (Fig 3.2F, 4month p-value = 0.11, 7month p-value = 0.5). In summary, we established that at baseline, the 3K mice showed

progressively worsening bradykinesia, postural changes, a tremor, and some deficits in novelty-seeking behaviors.

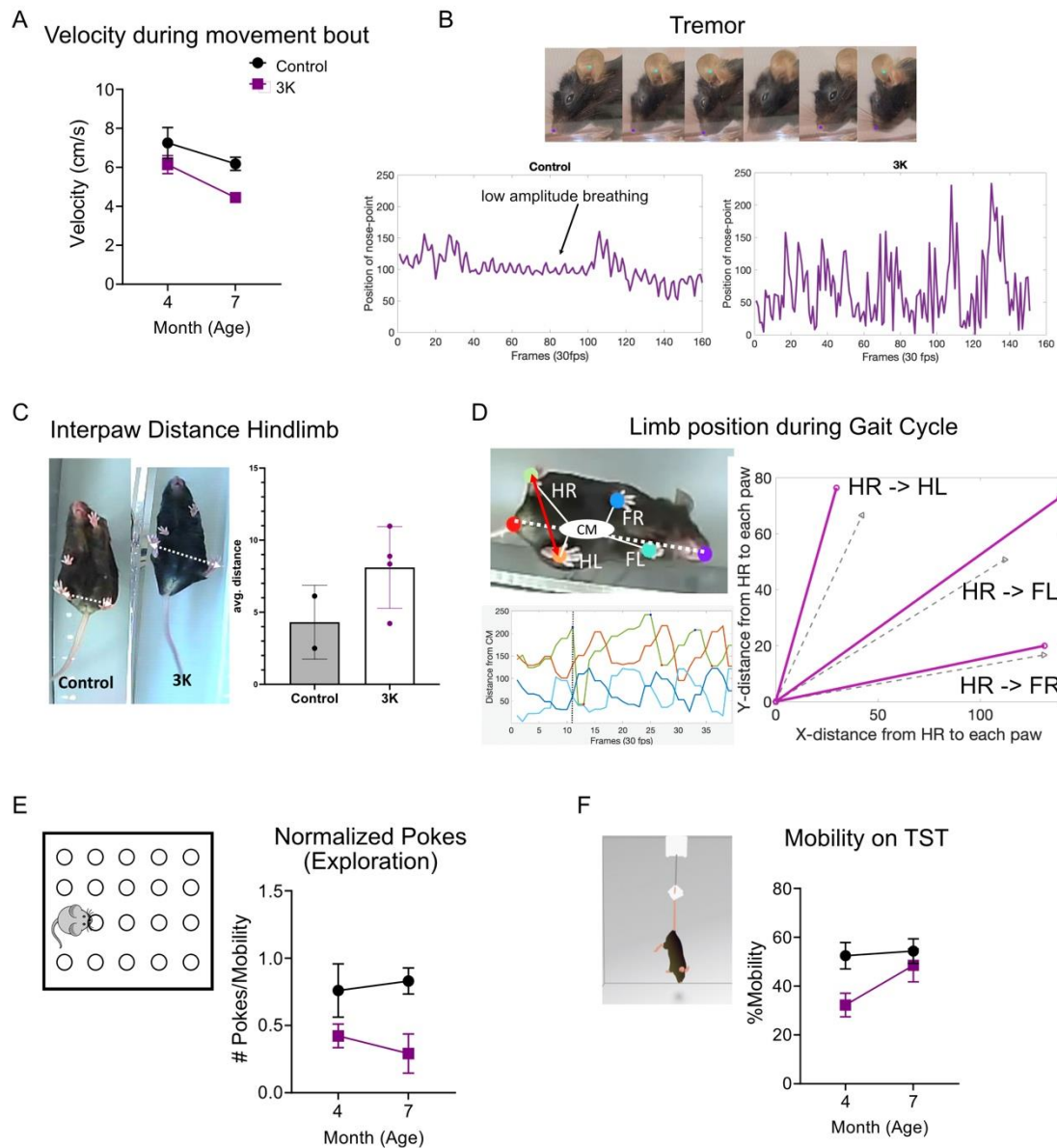


Figure 3.2 3K mice showed motor deficits including tremor, gait difficulties, postural imbalance, and changes in novelty-seeking behaviors.

A, Velocity during movement compared between control and 3K mice (Control 4m 7.26 ± 0.79 , Mean $\pm$ SEM, 7m 6.18 ± 0.34 ; 3K 4m 6.14 ± 0.46 , 7m 4.45 ± 0.27 , 4m p-value = 0.2, 7m p-value = 0.03, Mann Whitney U test) **B**, Tremor measured using nose-point tracking in DLC. Low amplitude breathing observed in Control mice, with evident oscillatory activity in 3K nose-point trace. **C**, Hind-limb dysfunction depicted for Control vs 3K mice as a measure of distance between hind-paws. **D**, Gait analysis using bottom-up video with DLC tracking, vectors drawn from hind-right paw to all other paws during a footfall in the gait cycle. Purple arrows – 3K limb vectors, Dashed grey lines – Control Limb vectors. **E**, Number of nose pokes normalized to mobility (Control 4m 0.76 ± 0.2 , Mean $\pm$ SEM, 7m 0.83 ± 0.1 ; 3K 4m 0.42 ± 0.09 , 7m 0.29 ± 0.15 , 4m p-value = 0.2, 7m p-value = 0.06, Mann Whitney U test). **F**, % Mobility measured on tail-suspension task, compared for 4 and 7 months between 3K and control mice. (Control 4m $53.45 \pm$

5.4, Mean $\pm$ SEM, 7m 54.36 $\pm$ 5; 3K 4m 32.24 $\pm$ 4.81, 7m 48.52 $\pm$ 6., 4m p-value =0.11, 7m p-value=0.5, Mann Whitney U test).

3.4.3 Lesioning the SNr does not suppress tremor, but reduces postural instability

Inspired by previous findings from our lab suggesting that a pathological SNr output could be responsible for the movement deficits but not the changes in posture brought about by a dopamine depleted state (unpublished), we predicted that if the SNr was responsible for the tremor or gait dysfunction in this mouse model, ablating it would alleviate these symptoms. This result could provide both therapeutic and mechanistic insights for future studies in this model. In our SNr lesion studies performed in four 3K and four Control mice, we injected AAV1-Cre and the Diphtheria toxin receptor (DTR) in the SNr 3 weeks apart. Diphtheria toxin was then injected intraperitoneally 2 weeks after the DTR injection. Behavioral assays were performed 2 weeks after the toxin injection. The expected outcome of this study would be a near complete lesion of the SNr mediated by the DTX binding to the Cre-dependent DTR injected in the SNr expressing AAV1-Cre (Fig 3.3A). Our histological analysis remained inconclusive regarding the degree of lesion we successfully achieved in the SNr. Further investigation of a direct measure of the lesion or the injection site using a microglial stain would be warranted.

In this cohort, we observed a reduction in ambulation and no changes in movement velocity only in the 3K mice (Fig 3.3B). This pattern suggests that unlike the role played by the SNr in the dopamine depleted condition, the SNr might not be responsible for the bradykinesia seen in the 3K mice, and in fact might be assisting in some compensatory locomotion that is lost in the lesioned condition. While the mobility-type remains unchanged in the controls post-lesion, the mice showed a higher speed of movement which aligns with the classical view of the SNr sending an inhibitory output to downstream targets and hence a likely disinhibition of the movement circuit in the context of the lesion (Fig 3.3C). Our DLC tracking for tremor revealed no differences in the qualitative aspects of the tremor as visualized by the nose-point jitter in the context of the SNr lesion (Fig 3.3D). The hindlimb paw distance reduced in all 3K mice with the SNr lesion suggesting that the SNr might be responsible for postural instability in 3K mice (Fig 3.3E, Appendix A). SNr lesion in control animals seemed to induce an anxiogenic effect with increases in mobility on the TST, as well as increased nose-pokes. The increased mobility on the TST and increased nose-pokes seen in controls after the SNr

lesion were not seen in the 3K mice, secondary to possible comorbid dysfunction in other non-motor regions like the BLA. (Fig 3.3F-G).

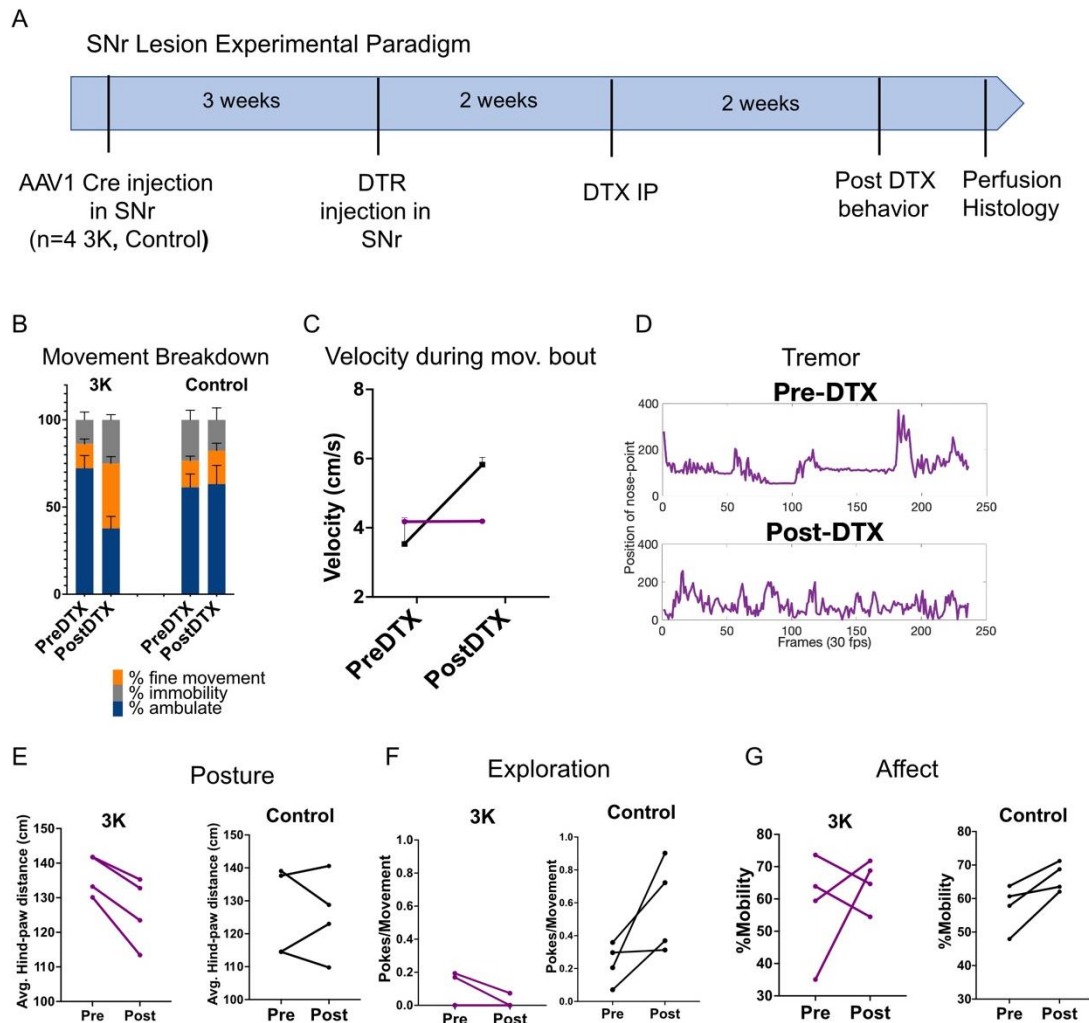


Figure 3.3 Lesioning the SNr does not suppress tremor but reduces postural instability in 3K mice.

A, Experimental paradigm for DTR-mediated SNr lesion in four 3K and four Control mice. **B**, SNr lesion in the 3K mice showed a suppression of ambulation, unlike controls. **C**, Velocity during movement bouts measured using Noldus tracking. Control animals showed an increase in velocity while the 3K velocity remains unchanged after the lesion. **D**, Tremor showed worsening in some mice, as shown in the representative example from pre-to-post DTX lesion. **E**, Hind-limb distance for the four 3K and four Control mice depicting a reduction in the pathological limb position in the 3K mice post-lesion. **F-G**, Novelty-seeking and learned helplessness tasks in 3K and Control mice depicted pre- vs post-lesion. Controls showed increase in nose-pokes and TST mobility, whereas 3K mice showed variable effects.

3.4.4 Effect of L-DOPA administration and EPN DBS on mobility and tremor

L-DOPA was administered with benserazide to a 16-month-old severe 3K mouse. The mobility was measured as the mouse moved freely in the open field an hour post-administration. There was an acute increase in mobility in the first few minutes, with a return to baseline mobility soon after (Fig 3.4A). L-DOPA administration did not change the tremor activity as measured by the % time spent in the tremor state based on the body-point jitter tracked by the Noldus Ethovision tracking. (Fig 3.4B). EPN DBS was delivered to the 16-month 3K mouse both in the conventional continuous pattern, and the burst pattern with 175Hz frequency delivered for 200ms/second. There was an increase in the distance traveled by the mouse acutely when the stimulation was on with both types of stimulation. Burst DBS produced more ambulation with longer movement bouts. Post-stimulation the mouse returned to baseline mobility with both kinds of stimulation. (Fig 3.4C). Acutely, tremor activity seemed suppressed with both kinds of stimulation, with a slightly longer lasting effect with burst DBS (Fig 3.4D). The confound of increased locomotion in the depicted suppression of tremor needs to be investigated further.

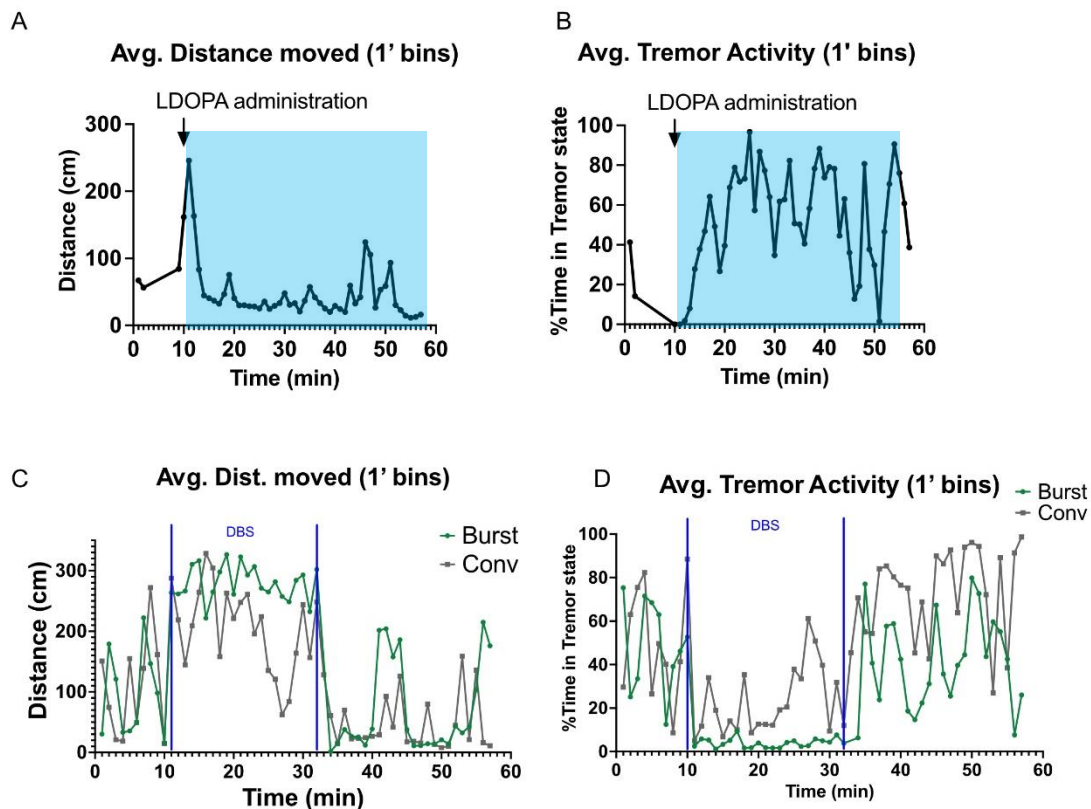


Figure 3.4 Therapeutic effect of L-DOPA and EPN DBS limited to an increase in mobility acutely.

A, Distance moved during L-DOPA administration measured in a severe 16-month 3K mouse. Acute sharp increase in movement lasting for less than 10 minutes followed by a return to baseline. **B**, Tremor assessment showing no change after L-DOPA administration. **C**, Distance moved before, during, and after EPN DBS delivered for 30 minutes. Acute

increase in distance traveled during stimulation, with a return to baseline post-stim. **D**, Suppression in the tremor metric during stimulation measured by jitter in body point.

3.5 Discussion

In this study, we established the presence of pathological aSyn aggregation in the cortex, DCN, cranial nerve nuclei, and sensory spinal cord regions. In specifically assaying the role of the basal ganglia in the deficits seen in these mice, we see that the SNr does not seem to be involved in either the tremor or the bradykinesia seen in these mice but might play a role in some of the postural changes observed. The intact SNr seems to be involved in non-motor aspects like novelty seeking and affect as evidenced by the increased anxiogenic features seen in the control group when the SNr is lesioned. These functions could be mediated through non-motor targets of the SNr. Interestingly, the effects of the SNr lesion on the non-motor aspects of behavior are not seen in the 3K mice, possibly due to the dysfunction in other nodes of the pathological circuit responsible for non-motor features. Additionally, the reduced ambulation in the 3K mice in the absence of the SNr, suggests that the intact SNr might play a compensatory role to maintain ambulation disrupted by other mechanisms.

Progression of behavioral effects and its consequences on therapy

The clear progression of symptoms with age provides a unique opportunity and challenge in studying pathophysiology in this mouse model. Identifying the age at which symptoms like tremor can be reliably quantified is possible based on the analyses performed at different age ranges. The 7–9-month range seems like the minimum age to be able to reliably quantify motor deficits differentially from control animals. However, to develop predictors of behavioral outcomes, animals in the prodromal stages at ~ 4 months could be studied. Behavioral or pathological markers at this age can be correlated to long-term outcomes to identify early predictors of prognosis. The non-motor behaviors studied here including novelty seeking/exploration and affect, do not seem to be more prevalent at earlier stages like has been shown in human PD patients. To test whether this model demonstrates the non-motor prodromal features seen in PD, future studies could focus on evaluating features like circadian rhythm disturbances using 24-hr activity monitoring or anxiety-related performance measures on the elevated-plus maze task.

Pathology in sensory regions and behavioral implications

In addition to the pathological aggregates seen in cortical and cerebellar regions, the brainstem showed dense pathological aggregates in multiple cranial nuclei including trigeminal nuclei, facial nerve nuclei, as well as pontine nuclei. Additionally, the spinal cord also showed aSyn aggregates in the dorsal horn region of the spinal cord predominantly, along with some larger ventral motor neurons showing aSyn labelling. A paper studying the patterns of aSyn expression in the brain and spinal cord established increased levels of aSyn in the dorsal region of the spinal cord (Geertsma et al., 2024). Another study suggests that the spinal cord pathology in Lewy body disorders spreads retrogradely from the axons of the dorsal horn and is associated with lower regions of the brainstem like the medullary reticular formation and the locus coeruleus, but not with the amygdala or olfactory bulb (Sumikura et al., 2015). The functional implications of these aggregates in the sensory regions of the brainstem and spinal cord remain unclear. Hind-limb clasping has been previously shown to be associated with the spread of aSyn to sensorimotor areas including the DCN, vestibular nuclei, and primary motor cortex (Ferreira et al., 2021). In the context of the tremor seen in these mice, there could be contributions from these sensory pathologies to the amplitude modulation of tremor. A study looking at the interaction between audiovisual sensory feedback and action tremor (Welzel et al., 2024) suggests a shift in the perception and arousal state based on sensory feedback which can be communicated to the tremor-related cerebellothalamic circuit via the locus coeruleus. In the 3K mice, sensory feedback-based interventions could be studied as modulators of tremor.

Limitations and Caveats of SNr lesion study

While the diphtheria-toxin mediated lesion of the SNr allows for a targeted approach to study its role in the pathophysiology of the 3K mice, there are certain methodological considerations that make the interpretation complicated. Primarily, validating the extent and efficacy of the DTX-mediated lesion in the SNr presents a challenge. One approach taken to confirm the lesion, was the co-injection of a color-changing reported Nuc-flox(mCherry)-EGFP which would label cells containing Cre green and non-Cre cells, red. In the DTX-lesioned conditioned, we expected to see few cells remaining that are only labeled red. In our experiments, we did not see many green or red cells in the SNr, making it hard to discern whether this was due to a failure of this construct, or due to the failure of the DTX-mediated lesion. Additionally, we used a PV-antibody to stain the SNr PV neurons to identify if there was any reduction in the number of neurons in the DTX-lesioned condition. While there were qualitative differences in the PV-staining across animals that were used to determine the degree of ablation, it is

important to note that the PV-antibody shows considerable variability in staining in normal conditions as well, further complicating its interpretability.

With these methodological considerations in mind, to confirm that the SNr is not involved in the production of the tremor or the mild bradykinesia in these mice, we would need to run more detailed SNr lesion studies with better quantification of the extent of the lesion. Since the aSyn aggregation does not show a pattern consistent with a basal ganglia-thalamic pathology, and the behavioral deficits are not altered by the SNr lesion, there is further support of the idea that the tremor and other motor deficits seen in the 3K mice are likely a consequence of a circuit outside the basal ganglia, potentially involving the cerebellum.

Chapter 4

4 Circuit-specific neuromodulation can produce long-lasting motor recovery in a dopamine depleted mouse model

4.1 Abstract

Dysfunction of basal ganglia neural circuitry leads to neurological disorders like PD. However, the complexity within these circuits in terms of cell-type and circuit-specificity pose serious challenges for delivering therapies. DBS improves PD symptoms acutely but does not differentiate between neuronal circuits, and its effects decay rapidly if stimulation is discontinued. Recent findings suggest that optogenetic manipulation of distinct neuronal subpopulations in the GPe provides long-lasting therapeutic effects in dopamine depleted mice. In vitro electrophysiology revealed that brief bursts of electrical stimulation can lead to excitation of parvalbumin-expressing GPe neurons and inhibition of lim-homeobox-6-expressing GPe neurons simultaneously due to synaptic differences. In this chapter we transform this knowledge of circuit architecture into a translatable therapeutic approach by 1) testing the circuit-inspired electrical DBS protocol in vivo, 2) establishing the pattern and anatomical targets of stimulation that produce long lasting motor recovery, and 3) investigating the neural correlates of these therapeutic effects using in vivo electrophysiology in the SNr. We showed that in dopamine depleted mice, the circuit-inspired DBS provided long-lasting therapeutic benefits that far exceeded those induced by conventional DBS, extending several hours after stimulation. We showed that the persistent rescue effect is specific to burst stimulation delivered at a frequency of 175Hz for 200ms/1 sec to the EPN/anterior STN territory, likely due to a coactivation of the D1 and STN inputs to the GPe as suggested by in vitro studies. A persistent response irrespective of the pattern of stimulation is also observed when the DBS electrode is implanted in the posteromedial regions relative to the STN, in the zona incerta (ZI), or the border between posterior hypothalamus (PH) and midbrain

reticular nucleus (MRN). In the small cohort of animals that received DBS in the STN, we showed a predominantly acute response. Finally, we outline the challenges and approaches to optimize simultaneous electrophysiological studies in animals undergoing DBS.

4.2 Introduction

DBS is an effective treatment used in patients with PD who demonstrate symptoms that are inadequately controlled by medications. This treatment involves the delivery of continuous high frequency stimulation to either the STN or GPi. DBS improves motor symptoms acutely, and its effects decay rapidly when stimulation is turned off. The need for constant stimulation increases the risk of side effects and the frequency of battery replacement. Hence the investigation of alternative patterns of stimulation that produce long-lasting recovery is critical. Such stimulation paradigms could minimize adverse outcomes caused by constant current delivery while also inducing therapeutic plasticity by reversing the aberrant synchronous activity of the basal ganglia seen in PD patients. Since the cellular mechanism of action of DBS is unknown, the clinical advances in identifying these patterns have been limited.

Recent findings in the Gittis lab suggest that optogenetically manipulating distinct neuronal subpopulations, specifically, activating PV neurons and inhibiting Lhx6 neurons in the GPe, provides long-lasting reduction in immobility in dopamine depleted mice. In vitro electrophysiology studies probing the synaptic characteristics of these cell subtypes suggest that the activity of the PV- GPe and Lhx6-GPe neurons might be dissociable using electrical stimulation as well. Using insights from the synaptic features of these cell-types, in vitro, we identified that electrical stimulation delivered in the entopeduncular nucleus (EPN, rodent homolog of the GPi) as bursts can produce the same cell-type modulation described above. In vivo,{Citation} this pattern of stimulation is predicted to produce long-lasting recovery in a dopamine depleted mouse model. Indeed, the electrical stimulation that led to maximal separation of the two cell-types, produced motor recovery that lasted for hours after stimulation was stopped when delivered in the EPN.

While the finding that burst DBS delivered in the EPN shows long-lasting improvements in motor symptoms, is exciting, it opens many doors to ask questions related to the circuits that might be engaged in this behavioral response. Would stimulating at any node in this circuit produce a similar behavioral outcome? Or is there something unique about the location of stimulation that would not translate to other conventional sites of stimulation like the STN? Answering these questions is valuable

from a translational standpoint, since patients who undergo neurosurgery to get DBS implants continue to keep these implants for an average of 20 years (Bove et al., 2021). Hence identifying the impact of this alternate stimulation protocol in patients with already implanted STN DBS electrodes, would be impactful. Additionally, mapping the circuits that produce acute and persistent responses based on stimulation site can provide further mechanistic insights into the functional connectivity of pathological circuits that lead to motor symptoms seen in PD. We hypothesize that the pathways implicated in generating an acute stimulation-induced response vs. a long-lasting post-stimulation recovery are distinct. By using an atlas-based reconstruction of the DBS electrode trajectories, we showed that the DBS delivered anterior to the STN produces a persistent response only to burst stimulation, DBS delivered within the STN produces limited acute-only recovery, and DBS delivered posteromedial to the STN produces a persistent response to either kind of stimulation.

Understanding the mechanism by which burst DBS produces long-lasting effects is key to develop a better understanding of the potential reversal of pathology that might be brought about by this stimulation. While the circuit-mapping provides a connectivity-based framework to understand this effect, a more direct investigation of the electrophysiological effects of the stimulation can be performed by recording activity in the SNr before and after the delivery of stimulation. Patients with PD show profound oscillatory activity in the beta frequency range throughout the basal ganglia circuit (Dostrovsky, 2004; Hutchison et al., 2004; Levy et al., 2000). While the presence of these beta oscillations has been limited in mice, following dopamine depletion, the SNr in mice shows increase in burst firing and synchronous activity. This pathological activity is thought to be involved in the mechanism underlying parkinsonian symptoms, and reducing this pathophysiology is implicated in the motor rescue seen with the optogenetic manipulations of specific GPe cell-types (Mastro et al., 2017). Here we probe the SNr firing patterns in the context of electrical stimulation to test whether the persistent rescue seen after delivering burst DBS to the EPN, is correlated with a suppression of bursty neurons in the SNr like seen in the optogenetic studies. This also presents the opportunity to identify patterns of activity in the SNr that might be associated with optimal therapeutic benefit, which could serve as a biomarker to drive adaptive or closed-loop DBS.

In this chapter we demonstrate a clinically translatable optimized burst DBS protocol which can produce long-lasting motor recovery. We anatomically map the regions that produce acute vs. persistent responses and outline other behavioral and pathological predictors of the therapeutic duration in a dopamine depletion mouse model to serve as a guide map for optimization of clinical DBS targeting and programming.

4.3 Methods

4.3.1 Animals:

Experimental procedures were approved by the Carnegie Mellon University Committee for the Use and Care of Animals and in accordance with the guidelines set forth by the National Institute of Health and Society for Neuroscience Use of Animals in Neuroscience Research. Male and female heterozygous mice on a C57BL/6J background were used for all experiments.

4.3.2 Surgical procedures:

All animals underwent a bilateral dopamine depletion and electrode implantation procedure in the same surgery. Mice weighing greater than 20g were chosen. Anesthesia was induced using 3% isoflurane in an induction chamber and maintained with 1.5% isoflurane during the surgery. Mice were placed in a stereotaxic frame and bilateral holes were drilled in the skull over the medial forebrain bundle and the implant target site. (MFB: -1.0 posterior, 1.0 lateral from bregma, EPN: -0.88 posterior, 1.8 lateral from theoretical bregma). 1 uL of 5 ug/uL 6-OHDA (TOCRIS) was injected in the MFB (4.9 mm from top of the dura) over the course of 5 minutes using a Nanoject (Drummond) with a pulled glass pipet (30 um tip diameter). The pipet was left in the brain for 5 minutes after the injection. The prefabricated electrical stimulation implant was attached to a holder and then slowly lowered at the EPN (4.6 mm) from the top of the dura. The electrode was affixed to the skull with Metabond and Orthojet dental cement.

After dopamine depletions, the weight of the mice was closely monitored, with daily post-operative saline injections (0.9% NaCl, administered IP). Mice were placed in a new recovery cage, partially placed on a heating pad, and consisting of soft food, chocolate pellets, hydrogel. 2 days of recovery was allowed after the surgery before any experiments were performed.

STN Targeting

Instead of targeting the EPN, bilateral holes were drilled in the skull over the STN (-1.4 posterior, 1.65 lateral from bregma) and the prefabricated stimulation implant was lowered at the target site (depth: 4.5mm) from the top of the dura. For the newly identified posteromedial site of stimulation, the following coordinates were used: -1.7 posterior, 1.2 lateral from bregma, 4.45mm depth).

Headbar implantation for SNr Recordings

In addition to the MFB and the DBS implant, holes were drilled in the skull to create craniotomies above the SNr (-2.8 posterior, 1.5 lateral from bregma). A stainless-steel head bar was affixed to the skull with dental cement (Metabond, Orthojet). This would allow head-fixation during recordings. A well of dental cement was built around the skull to create a saline pool for grounding, this was filled with a silicon elastomer to secure until the day of recording. Additionally, a female gold-pin was implanted over the posterior region of the brain (around the cerebellum), touching the dura. This would connect to the male gold-pin connector attached to the recording electrode for grounding.

4.3.3 Electrode fabrication and implantation:

Electrodes for in vivo DBS were fabricated as previously described (Schor & Nelson, 2019), with the following modifications. Briefly, each of the two six-lead bipolar stimulating electrodes was constructed by twisting together 76.2 μ m coated wires, heating the wires to slightly fuse the coating, and then cutting the fused end at an angle. The six twisted ends therefore formed three electrode pairs at various depths. The six untwisted ends were then stripped using a razor blade and fit into female Millimax connectors. Prior to surgical implantation, electrodes were tested for short circuits and the bipolar pulse trains of desired current amplitude were tested at each contact on both electrodes.

4.3.4a Behavior experiments and analysis:

(Methods reported in (Spix et al., 2021))

Electrode pair selection

An isolated constant current bipolar stimulator (WPI) was used to deliver electrical stimuli. The timing of stimulus delivery was controlled by TTL input from an Arduino. Mice were tethered to an electrical

input cable, and optimal stimulation (electrode pair and amplitude) was determined based on the animal's behavioral response to 100 Hz constant stimulation in the open field. As higher stimulation amplitudes are known to lead to seizures, amplitudes beginning at 65 μA were tested and then increased incrementally by 10 μA until the animal increased movement in the open field (2,500 cm^2 transparent open field with an opaque, smooth base), indicating a therapeutic effect.

Conventional and burst DBS parameters

Our DBS settings were modeled after conventional DBS settings in humans and rat models (Brocker et al., 2017). During 'burst stimulation', animals received stimulation at 175 Hz (200 ms duration), with an average amplitude of 110 μA and pulse width of 60 μs . Stimulation for 175 Hz bursts was delivered as 35 biphasic, bipolar pulses in a 200 ms period per second of stimulation (200 ms of stimulation followed by an 800 ms pause). (Schor & Nelson, 2019) previously showed the therapeutic efficacy of DBS is related to the total electronic energy delivered during stimulation (19). Therefore during 'conventional DBS', we kept this value constant by adjusting the frequency and current amplitude values during stimulation such that animals received continuous bipolar, biphasic stimulation at 100 Hz, with an average amplitude of 109 μA and pulse width of 60 μs . However, to ensure that the different therapeutic duration observed following 'burst' vs. 'conventional' DBS were not due to difference in stimulation frequency, in 2/8 mice, conventional DBS was delivered at 175 Hz instead of 100 Hz. In these mice, 175 Hz 'conventional DBS' produced similar velocity, immobility, and therapeutic duration results as mice who received 100 Hz 'conventional DBS'. In order to test the validity of the machine-learning model in predicting the degree of cell-type modulation and the consequent motor recovery caused by other stimulation settings, we tested 3 additional burst DBS protocols including a stimulation at 175 Hz delivered either as 1s stimulation followed by a 4s pause or 10s stimulation followed by a 40s pause; and a stimulation at 50 Hz delivered for 200 ms with an 800 ms pause.

Open field behavior

Following optimal electrode pair and amplitude selection, efficacy of each DBS stimulation protocol was assessed as follows. In a crossover design, mice were pseudorandomly assigned to either receive conventional (constant) stimulation or burst stimulation on the first day. Mice were placed in the open field and 10 minutes of baseline data were collected. Afterward, the assigned stimulation

protocol was delivered for 30 minutes. Post-stimulation data was then collected for an additional 2-3 hours to monitor potential persistent motor effects. Mice were tested on alternate stimulation protocols on successive days, and all mice were tested on both stimulation protocols for comparative analysis.

Movement analysis

Following each behavior experiment, video-tracking software (Noldus Ethovision) was used to calculate immobility and average velocity. Behavioral metrics were binned in 30 s intervals. Average velocity was computed by dividing the total distance moved (in cm) by the bin duration. Percent immobility was then calculated by taking the fraction of time that the animal was ‘immobile’ within the bin. Animals were defined as ‘immobile’ when center-point velocity was less than 0.75 cm/s for 15 samples, sampled at 29.97 frames per second. Therapeutic duration was defined as the amount of time after DBS was turned off before mice returned to pre-stimulation immobility, or >80% immobility, across 3 consecutive 10-minute intervals. Statistical analysis regarding immobility and average velocity was performed using a Wilcoxon signed-rank pairwise comparison between pre-, stim-, and 90’ post periods.

4.3.4b Updates to Behavior Setup and Mobility characterization (Results 4.4.4-4.4.6)

Initial current amplitude tested was 100 uA and was gradually increased up to 300 uA to elicit a movement response. The open field consisted of a cylindrical enclosure with a diameter of 10 inches, and a smooth clear base. A camera filmed the activity of the mouse in this open-field from the bottom-up. A crossover design like that described previously was used, with mice pseudorandomly assigned to receive either conventional DBS or burst DBS on day 1 of behavior (day 3 post-implantation). In the open field, a 10-minute baseline period video was collected, followed by 30 minutes of stimulation with the assigned protocol. Finally post-stimulation data was collected for 90 minutes to study persistent motor effects. Mice were tested on alternate protocols on successive days. In animals that underwent SNr recording, burst DBS was delivered on day 1 in the original square (2,500 cm²) open-field, and SNr recording was performed on day 2 with the delivery of burst DBS in an awake, head-fixed state.

For further characterization of the form of mobility induced by stimulation, mobile periods were categorized as including ambulation (defined by the center point velocity greater than 1.2 cm/s) or fine movements (center point velocity above 0.75 cm/s but below 1.2 cm/s). Further analysis of stimulation induced mobility involved immobile, ambulation, and fine-movement periods. Furthermore, velocity was analyzed specifically during bouts of ambulation.

4.3.5 Histological analyses:

At the end of 2 days of behavioral experiments, mice were euthanized and perfused with 4% paraformaldehyde (PFA). The brain was extracted and stored in PFA overnight and switched to a 30% sucrose solution the next day. Tissue was sectioned using a freezing microtome (Microm HM 430; Thermo Scientific) and blocked with 10% normal donkey serum for 1 hr. Primary antibodies against TH (rabbit anti-TH, 1:250) and Iba (rabbit anti-Iba, 1:500) were incubated for 24 hours in a 4° C refrigerator. Degree of dopamine denervation was assessed using immunofluorescence against tyrosine hydroxylase in the striatum. A littermate control was processed in parallel to provide reference TH intensity ranges. The Iba staining was used to label microglial activation depicting the placement of the stimulating electrode in the EPN. Primary antibodies were detected with Alexa Fluor 647-conjugated donkey anti-rabbit (1:500, Thermo Fisher Life Technologies), incubated for room temperature (90 mins for TH, 3h for Iba). Tissue was mounted, and cover-slipped with Vectashield before imaging. Epifluorescent images were taken at 4x magnification (Keyence BZ-X) to identify electrode placement using a combination of the dye, and microglial activation with Iba. 10 magnification windows of striatal slices were quantified using the pixel intensity tool in ImageJ. 3 100 x 100 um squares from each hemisphere were measured, normalized to the intensities of the cortex, and compared to the healthy control striatal measurements. A threshold of less than 30% remaining TH in the striatum was required for the mouse to be considered depleted compared to controls.

4.3.6 Electrode reconstruction using AP-histology: (Results 4.4.4-4.4.5)

CM-DiI was applied to the tip of electrode prior to implantation to allow better visualization of the electrode post-hoc. Approximately 25 slices from throughout the basal ganglia and midbrain regions were selected per animal for mapping of the electrode trajectory. These images were analyzed using

the AP histology software in MATLAB. Once each slice image was mapped and aligned with a corresponding Allen-CCF mouse brain atlas image, the electrode locations were drawn. The software then compiled the electrode locations from each marked image and created a 3D representation of the electrode trajectory (Appendix C).

4.3.7 In-vivo recording: (Results 4.4.6-4.4.7)

On the day of recording, mice were placed on top of the running wheel and allowed ~10 minutes to acclimate to the head position. The silicon elastomer was removed, the craniotomy was cleared, and a 64-channel silicone probe with sites 50um apart (Cambridge Neurotech) was attached to the micromanipulator (Sutter Instruments). The 4-shank probe consisted of 16 channels per shank. CM-DiI was applied to the tip of the probe with a paintbrush to allow for better post-hoc visualization of the recording site. The middle two shanks were centered on the midline marker labelled during surgery at the appropriate distance for the SNr posterior to bregma. The probe was advanced slowly until the appropriate depth for the SNr was reached. High frequency firing that showed occasional bursts in the depleted condition confirmed SNr recording placement. Post-hoc tissue analysis for dye expression and implant location was used to confirm targeting. The probe was left at the desired depth for about 15 minutes before beginning to record. The male gold-pin connector soldered onto the ground-wire of the probe was then attached to the female grounding pin implanted over the dura. Extracellular recordings for spiking were acquired using the Intan RHD USB Interface board recording system connected to a TDT processor. This interface also allowed for a direct readout of movement using the digital input signal from a rotary encoder attached to the wheel-axle.

4.3.8 Neural Analysis: (Results 4.4.6-4.4.7)

The processing of neural data acquired from the Intan system would be performed using a custom MATLAB code and Kilosort which is an automated spike sorting platform (processing pipeline described in Appendix D). Preprocessing included converting rhd files to .bin files, high/low pass filtering from 250 Hz to 5 KHz and median subtraction to reduce noise. Kilosort was used for semi-automated spike sorting, followed up by manual curation in Phy. Units were graded for quality based on ISI-violations, auto correlogram, stability, waveform shape and movement artifact-related

modulation of firing. When there was an unstable unit over time, a good period of recording was identified to perform further spiking analyses. Analysis included computation of burst firing using the surprise burst method, measuring the firing rate and coefficient of variation of the inter-spike interval (CV of ISI) as well as movement analyses to compute movement-related firing modulations. Analysis of oscillatory activity and synchronous firing using the methods described previously (Whalen et al., 2020; Willard et al., 2019) was also performed.

4.4 Results

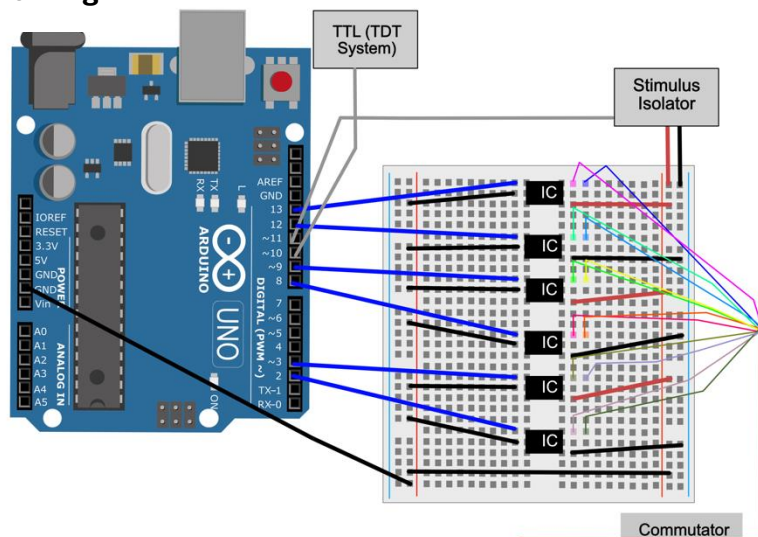
4.4.1 In-vivo mouse model of electrical DBS was developed to probe various circuit-inspired stimulation patterns and targets

We modeled our stimulation setup on the system developed by Jonathan Schor in Alexandra Nelson's group at UCSF. Electrodes were fabricated (see methods, Fig 4.1B) to allow for 3 pairs of contacts for stimulation, to improve the chances of being in the small target area (EPN). The stimulator circuit was built in 3 compartments, the stimulation rig, the testing rig, and the behavior rig (4.1A-D) The stimulation rig was designed to allow delivery of pulses of stimulation with varying frequency, amplitude, and duty cycles to test various burst DBS paradigms. An Arduino-driven controller was used to provide a paired pulse output to one pair of contacts from the 3 pairs on each electrode. This paired pulse was then transmitted to a stimulus isolator that allowed modulation of current amplitude, with a fixed voltage configuration. In addition to expanding this setup for a bilateral configuration, I was able to code for various burst stimulation patterns to be delivered across these contacts using customized Arduino scripts.

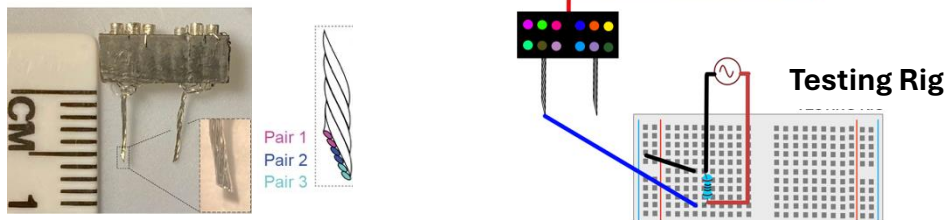
An important aspect of reliably generating pulses, was to ensure that the circuit was delivering the current amplitude and biphasic waveform we intended for it to deliver. I devised a testing setup that would deliver continuous current across a 100-ohm resistor using each contact on the electrode tip as one of the poles. The voltage waveform across this resistor corresponding to the chosen current was then visualized on an oscilloscope. All electrodes were tested before implantation using this setup (Circuit diagram, Fig 4.1C). The final step of this setup involved the integration of the stimulator with the open-field behavior setup. Both the stimulator, and the top and side camera recording behavior, were connected to a TDT recording system. This allowed the integration of the TTLs from the

stimulation pulses with the corresponding behavior of the animal. Each of the connection from the stimulator circuit led to individual points on a connector that, through the commutator, connected to the electrode implanted in the animal's brain. Hence the animal was tethered but freely moving in the open-field setup (Fig 4.1D).

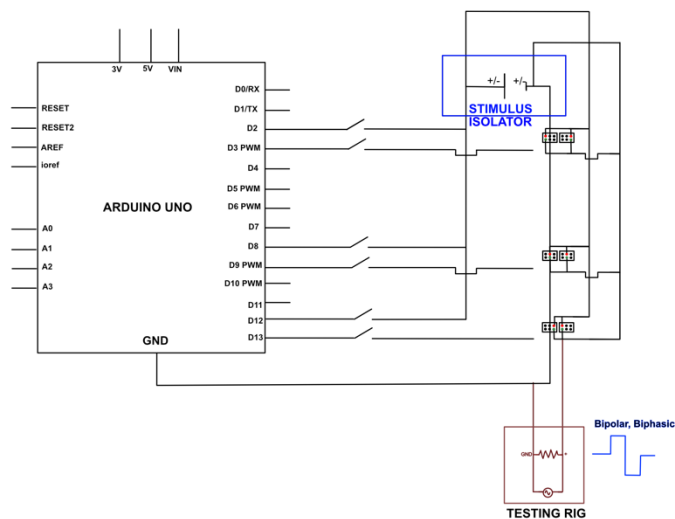
A Stimulation Rig



B



C



D Behavior Rig

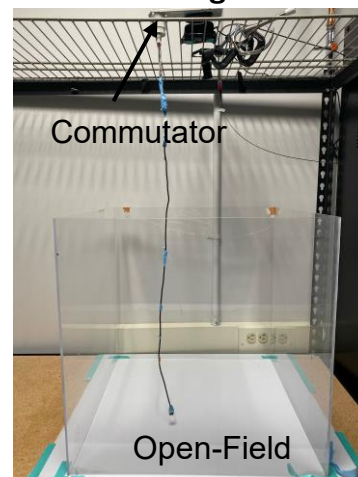


Figure 4.1 Stimulator for in-vivo DBS using various stimulation paradigms.

A, Schematic depicting the stimulation and testing rig connected to deliver biphasic, bipolar stimulation on 3 pairs of contacts on either side. **B**, Photo and schematic of fabricated electrode. **C**, Circuit diagram for stimulator circuit. **D**, Integration with behavior rig for open-field behavior with animal tethered via a commutator undergoing stimulation

4.4.2 Burst DBS produces long-lasting motor recovery in bilaterally depleted parkinsonian mice

(Findings reported in (Spix et al., 2021))

To test whether an electrical DBS protocol tailored to drive population-specific neuromodulation would mimic the persistent therapeutic effects induced by cell-type specific optogenetic manipulations (Mastro et al., 2017) we compared the therapeutic efficacy of conventional vs. burst DBS in mice rendered parkinsonian by bilateral injections of the toxin 6-hydroxydopamine (6-OHDA) into the MFB (Fig. 4.2A). Burst DBS consisted of short bursts delivered once a second with a duration of 200 ms and an intraburst frequency of 175 Hz.

The therapeutic effects of burst versus conventional DBS were then compared using a cross-over study design with a 1-day washout period between treatments in dopamine depleted mice (Fig. 4.2C), using an approach shown to recapitulate several features of DBS in humans (Schor & Nelson, 2019). Six-lead DBS electrodes were implanted bilaterally near the entopeduncular nucleus (EPN) (Fig. 4.2B, Fig 4.3), an electrode placement often used in human patients receiving DBS in the GPi (Au et al., 2021; Vitek et al., 2004) and an area where fibers from the STN and D1- SPNs can be coactivated.

Before stimulation, dopamine depleted mice were highly immobile. Both conventional and burst DBS reduced this immobility, enabling mice to move around the arena during stimulation (Pre and DBS in Fig. 4.2, D to F). After receiving conventional DBS, mice quickly returned to the immobile state (Fig. 4.2, D to H). Conversely, after receiving burst DBS, mice continued to move around the arena (Fig. 4.2, D to H), with five out of the eight mice still moving at the end of the trial, ~2.5 hours after stimulation (Fig. 4.2, I and J). On average, the therapeutic effects of burst DBS persisted >4.5 times as long as those of conventional DBS (Fig. 4.2I), a conservative estimate as only 37% of mice had returned to the immobile state at the end of the trial (Fig. 4.2J).

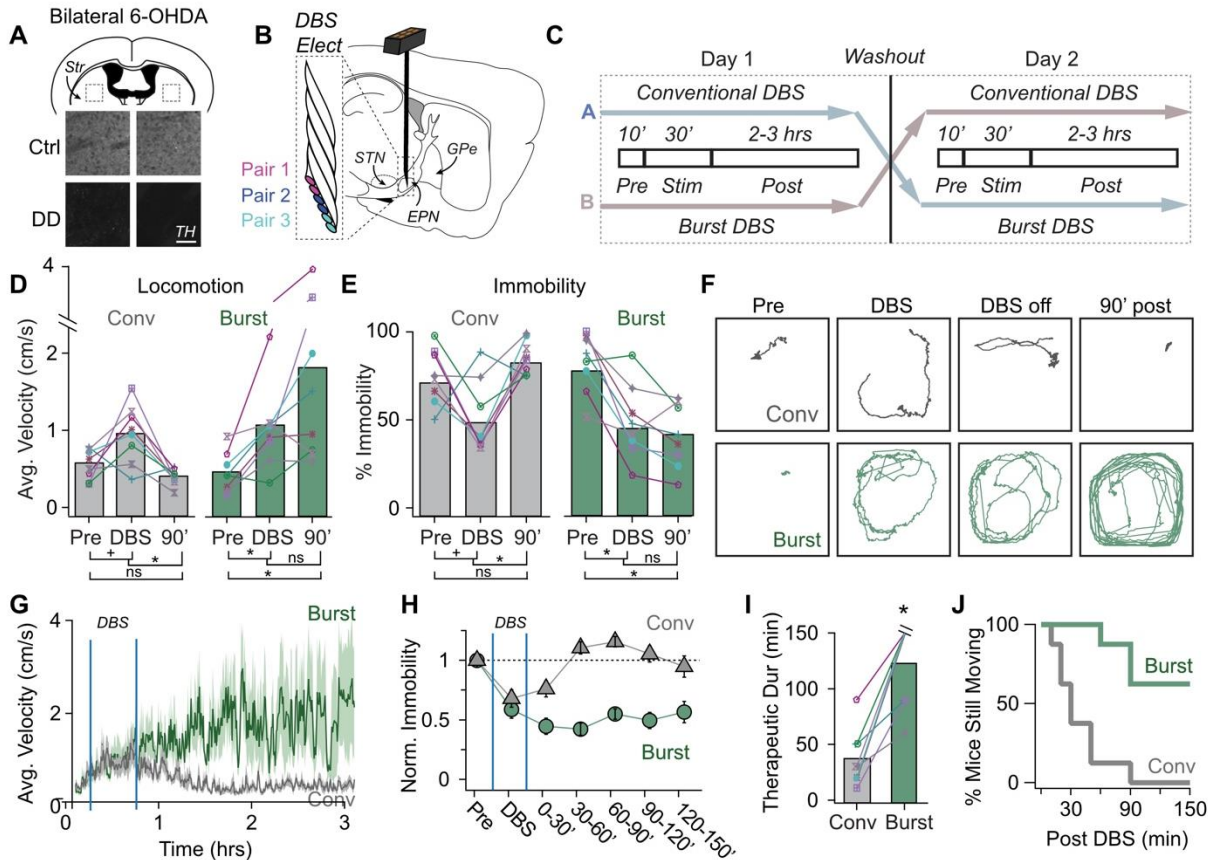


Figure 4.2 Burst stimulation restores movement persistently in parkinsonian mice.

A, Schematic showing bilateral dopamine depletion (DD, $6.43 \pm 0.09\%$ TH remaining compared to littermate controls). **B**, Schematic showing DBS electrode placement near the EPN. **C**, Crossover design illustrating pseudorandom experimental protocol where dopamine depleted mice receive conventional or burst DBS on alternate days. **D**, Average velocity of dopamine depleted mice increased during conventional DBS (gray) but did not persist after stimulation (Pre: 0.54 ± 0.07 , Stim: 0.94 ± 0.13 , Post: 0.40 ± 0.06 cm/s) (Wilcoxon Signed Rank WSR: pre vs stim, $^+p = 0.055$; stim vs 90' $^*p = 0.008$; pre vs. 90' $p = 0.195$, $n = 8$ mice). Average velocity increased during Burst DBS (green) and was still elevated 90 min after stimulation (Pre: 0.47 ± 0.09 , Stim: 1.03 ± 0.19 , Post: 1.77 ± 0.54 cm/s) (WSR: pre vs stim $^*p = 0.016$; stim vs 90' $p = 0.195$; pre vs. 90' $^*p = 0.008$). Data from individual mice are shown and colors and symbols are consistent throughout. **E**, Immobility of dopamine depleted mice decreased during conventional DBS (gray), but did not persist (Pre: 75 ± 6 Stim: 51 ± 7 , Post: 87 ± 4 % time immobile) (WSR: pre vs stim, $^+p = 0.056$ stim vs 90' $^*p = 0.008$, pre vs 90' = 0.195). Immobility decreased during burst DBS (green) and was still low 90 min after stimulation (Pre: 82 ± 6 , Stim: 48 ± 7 , Post: 45 ± 9 % time immobile) (WSR: pre vs stim, $^*p = 0.016$; stim vs 90'; $p = 0.641$; pre vs. 90' $^*p = 0.016$). **F**, Movement paths over 8 min intervals throughout a trial. Data are from the same mouse treated with conventional DBS (grey) or burst DBS (green). **G-H**, Average movement velocities (G) and immobility (H) plotted for the duration of behavioral trials. Movement velocities were averaged over 30 s time bins, % immobility was averaged over 30 min time bins. Grey, Conventional DBS; Green, Burst DBS ($n = 8$ mice). **I-J**, Amount of time after DBS was turned off before mice returned to pre-stim levels of immobility ('therapeutic duration'). Therapeutic duration was significantly longer following burst DBS (123.8 ± 37.4 min, green) than following conventional DBS (37.5 ± 25.5 min, grey) (WSR, $^*p = 0.012$, $n = 8$ mice), shown for individual mice (I) and cumulatively across the population (J).

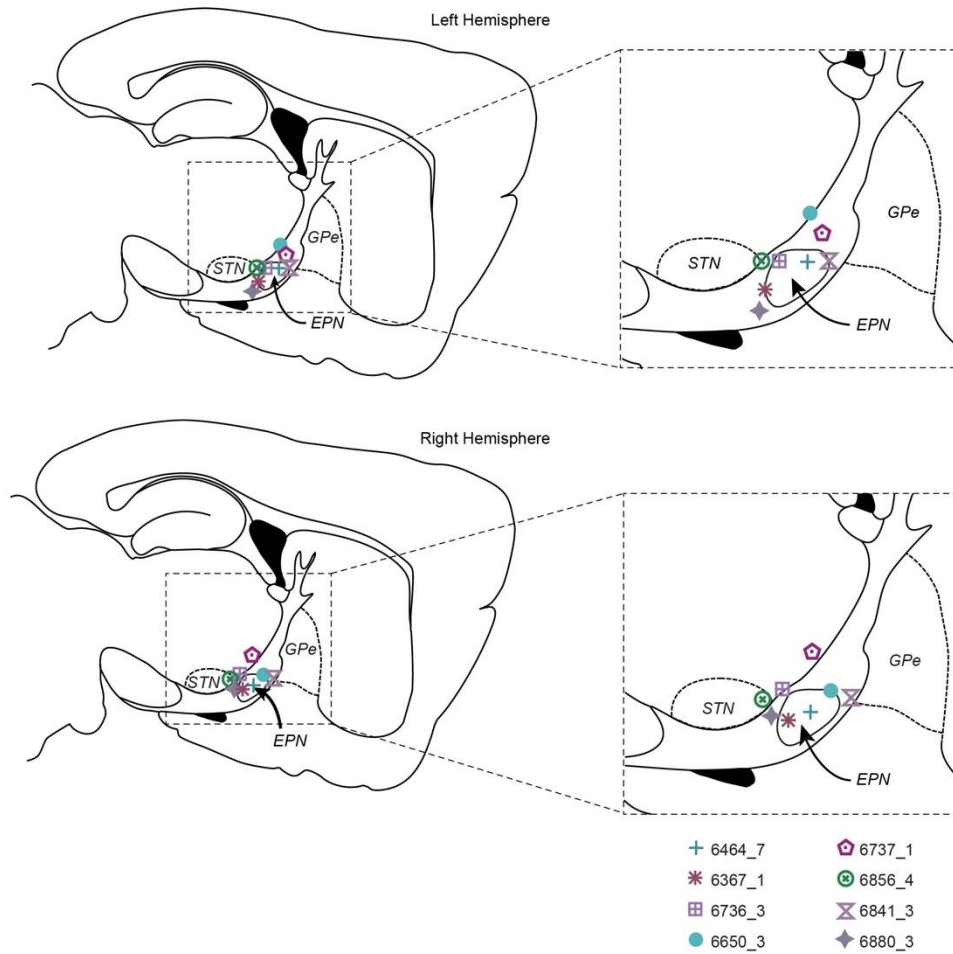


Figure 4.3 Histologically verified electrode placements.

Markers represent electrode placements near the EPN for all mice included in figure 4.2 for the left hemisphere (*top*) and right hemisphere (*bottom*). Symbols for placement in each mouse correspond with symbols used in Fig. 4.2

4.4.3 Stimulation parameters that predict maximal cell-type specific separation in-vitro, yield longest motor recovery in-vivo

(Findings reported in (Spix et al., 2021))

Given our demonstration of persistent motor recovery produced by one combination of burst stimulation parameters (175Hz bursts delivered during a 200ms stim. period every second) as compared to conventional DBS, it remains unclear whether these parameters are more effective relative to other combinations. To identify the range of parameter combinations that separate the responses of PV-GPe and Lhx6-GPe neurons (modulation factor), we used an iterative machine

learning approach. Experimental data were used to train models to predict potential nonlinear relationships between parameter combinations and firing responses of PV-GPe and Lhx6-GPe populations. The Gaussian stimulus optimization model sampled the parameter space comprising of stimulation duration, amplitude, and frequency and generated modulation factor predictions across many artificially generated combinations.

While next-generation DBS devices will likely be able to deliver bursts as short as 200 ms, most currently implanted devices can only deliver bursts in 1s increments. This translational need, combined with our Gaussian model predictions regarding the localized nature of the effective parameter combination, motivated us to identify how quickly the effectiveness of the persistent recovery drops as the parameters deviate from the previously tested effective combination. We hypothesized that DBS protocols predicted to be less effective at dissociating the responses of PV-GPe, and Lhx6-GPe neurons will be less effective at inducing persistent rescue.

To test this, we compared the therapeutic effects of three other stimulation combinations (175Hz On for 1s every 5s, 175 Hz On for 10s every 50s, and 50 Hz On for 200 ms every sec) in addition to the conventional continuous DBS, and the 175 Hz 200ms-800ms protocol (optimal burst) (Fig 4.4A-G). The behavioral outcomes for these protocols aligned with the cell-type modulation factor predictions from the Gaussian model (Fig 4.4A-D). Specifically, the 1s, 175 Hz protocol led to an intermediate response to stimulation, relative to the acute response from conventional stimulation and persistent response from the optimal burst protocol (Fig. 4.4B). The 50Hz -200ms protocol and the 175Hz – 10s led to acute response much like conventional DBS (Fig 4.4C, D). This is in line with the modest modulation of cell-types seen with these protocols in the parameter space surface plots showing predictions from the model (Fig 4.4A).

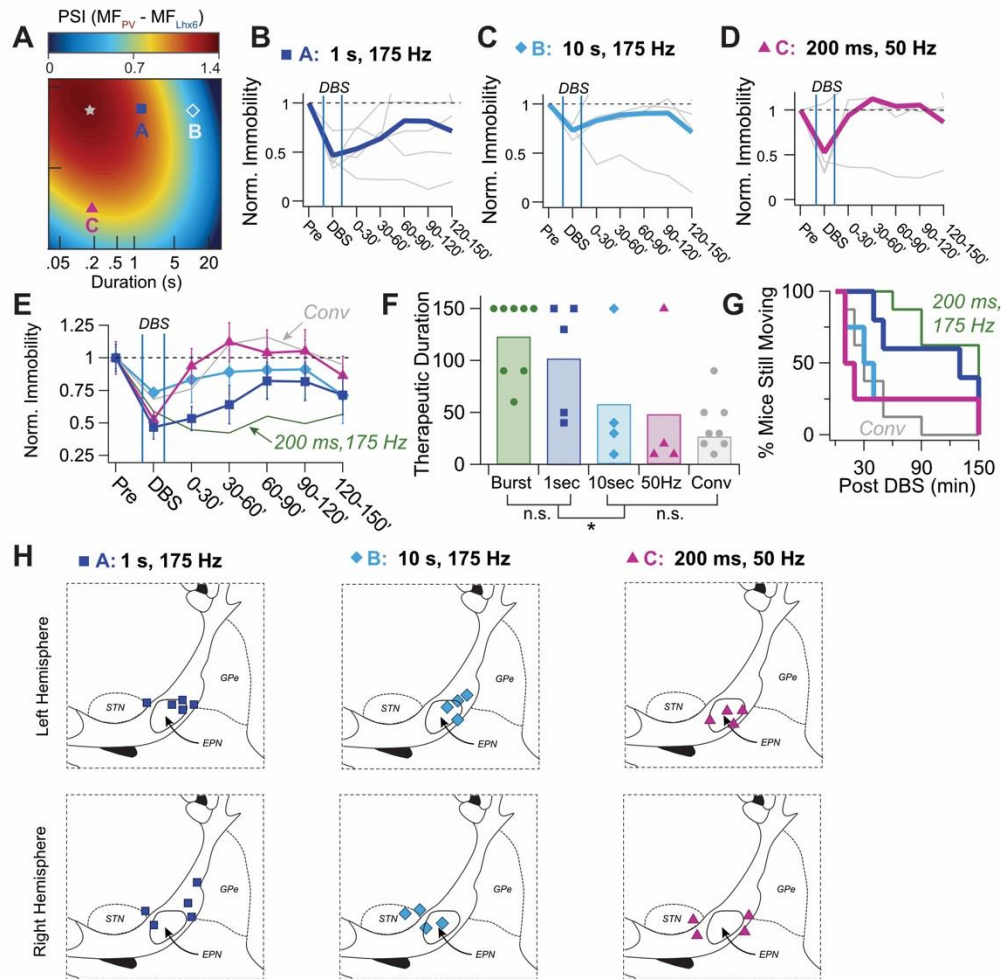


Figure 4.4 Projected population-specificity of burst protocols predicts therapeutic efficacy in vivo.

A, Surface plot from Fig. 2B, showing location of additional burst parameters tested (*blue square* = 175 Hz/1 s every 5 s; *teal diamond* = 175 Hz/10 s every 50 s; *pink triangle* = 50 Hz/200 ms, every 1 s). The *star* marks the location of the ‘optimal’ burst protocol, tested in Fig. 4 (175 Hz/ 200 ms, every 1 s). **B – D**, Normalized immobility plotted for the duration of behavioral trials. *Thin gray lines*, individual animals; *thick colored lines*, population averages, (B) 175 Hz/1 s: N = 5 mice; (C) 175 Hz/10 s: N = 4 mice; (D) 50 Hz/200 ms: N = 4 mice. **E**, Overlay of averaged normalized immobilities for the 3 burst protocols, binned in 30 min intervals. *Error bars, sem*. For comparison, average normalized immobilities from conventional and burst DBS experiments (Fig. 4) are re-plotted in *thin green* (burst) or *thin gray* (conventional) lines. **F**, Therapeutic durations for individual animals following the offset of stimulation for each DBS protocol. Symbols denote individual animals and filled bars are group averages. Compared to conventional DBS, therapeutic duration was significantly longer following 175 Hz/1 s DBS (104 ± 54.6 min, *blue*) (Mann-Whitney U, MWU, vs conv * $p = 0.023$), but not following 175 Hz/10 s DBS (57.5 ± 62.9 min, *teal*) (MWU, vs. conv, $p = 0.804$) or 50 Hz /200 ms DBS (47.5 ± 68.5 min, *pink*) (MWU, vs conv, $p = 0.436$). There was no significant difference in the therapeutic duration between 175 Hz/1 s and 175 Hz /200 ms (MWU, 1 s vs. 200 ms, $p = 0.368$). **G**, Plot of the decay of therapeutic duration within groups of mice. **H**, Electrode placements near the EPN for all mice included in figure S6 for the left hemisphere (*top*) and right hemisphere (*bottom*).

4.4.4 DBS delivered to the STN/EPN territory showed 3 response types based on electrode placement: Acute only (STN), Persistent to burst DBS (anterior to STN), and Persistent to both kinds of stimulation (posteromedial to STN)

(unpublished)

Based on the in vitro electrophysiology experiments, delivery of burst stimulation in the EPN was thought to drive GPe cell-type activity differentially, primarily because of the coactivation of the STN inputs and the D1-SPN collateral inputs to the GPe. Delivery of continuous high frequency stimulation in the STN is thought to produce symptomatic relief in patients with PD in symptoms including akinesia and rigidity, as well as tremor. Recent tractography work has demonstrated that targeting specific sub-domains within the STN is critical to get symptom-specific improvements (Horn, 2024). Conversely, off-target side-effects including dyskinesias and cognitive defects are common in patients receiving STN DBS. Given the translational impact of delivering burst-DBS in terms of reduction of the amount of current delivered, potentially minimizing side-effects, we sought to test whether similar patterns of stimulation in the STN led to acute or persistent motor recovery. STN DBS would either be sufficient to elicit similar effects as the EPN DBS, due to the proximity of the nuclei and potentially the coactivation of D1-fibers, or the effects seen with the EPN DBS would be highly specific to the more anterior location of stimulation. In a cohort of mice, we explored stimulation targeting in the antero-posterior and medio-lateral axis relative to the STN/EPN border to develop a map of the regions of placement that would elicit either acute or long-lasting effects. As summarized in Fig. 4.5, we saw 3 main response groups and a group of non-responders. The normalized immobility for these animals and their corresponding placements are summarized in Fig. 4.5, and 4.6.

Response group 1 involved animals that only produced an acute response to stimulation and returned to baseline immobility when the stimulation was stopped (n=3, Fig 4.5A). 2 of these animals were placed in the dorsomedial STN (Fig. 4.6M) and responded acutely to both conventional and burst DBS (Fig 4.6A-C) whereas one was within the EPN and showed some delayed persistent response to both stimulation types (Fig 4.6C, M). This suggests that conventional stimulation and the optimized burst DBS protocol (175Hz, 200ms on/sec) delivered in the STN would produce only an acute response. More animals need to be studied to validate this effect.

Response group 2 included animals that showed a persistent response to burst stimulation and only an acute response to conventional DBS (n=6, Fig 4.5B, Individual animals – Fig 4.6 D-F). Most

of these animals were clustered at the border of the STN and EPN, with 2 placed more anteriorly at the GPe/EPN border, and 1 placed more medially in the ZI (Fig 4.6M). This effect is consistent with our previous finding that burst stimulation placement closer to the EPN would likely have a higher chance of activating the neural pathways that produced persistent motor effects (therapeutic duration for burst DBS, higher than conventional DBS, Fig 4.6F).

Response group 3 included animals that showed a persistent response to both kinds of stimulation ($n=5$, Fig 4.5C, Individual animals – Fig 4.6G-I). If the pathway of persistent motor recovery diverges from the acute response pathway at the level of the GPe or the SNr, this response would suggest that downstream nodes in this network could be leveraged in a way that do not depend on the pattern of stimulation. Most of the animals in this response group were placed in the medial, posterior regions relative to the STN, in the ZI, PH, and anteromedial parts of the MRN (Fig. 4.6N-O).

Finally, there were a non-trivial number of non-responders ($n=6$) that showed minimal response to stimulation with either kind of DBS which was defined as less than 10 minutes of drop in immobility below 80% of baseline immobility during the stimulation period. These animals spanned the spatial map ranging from the EPN to the STN, reflective of either the biological variability in responses to stimulation of these regions, or other factors like electrode malfunction or tissue impedance that might limit the amount of current delivery, and subsequently limit the behavioral response.

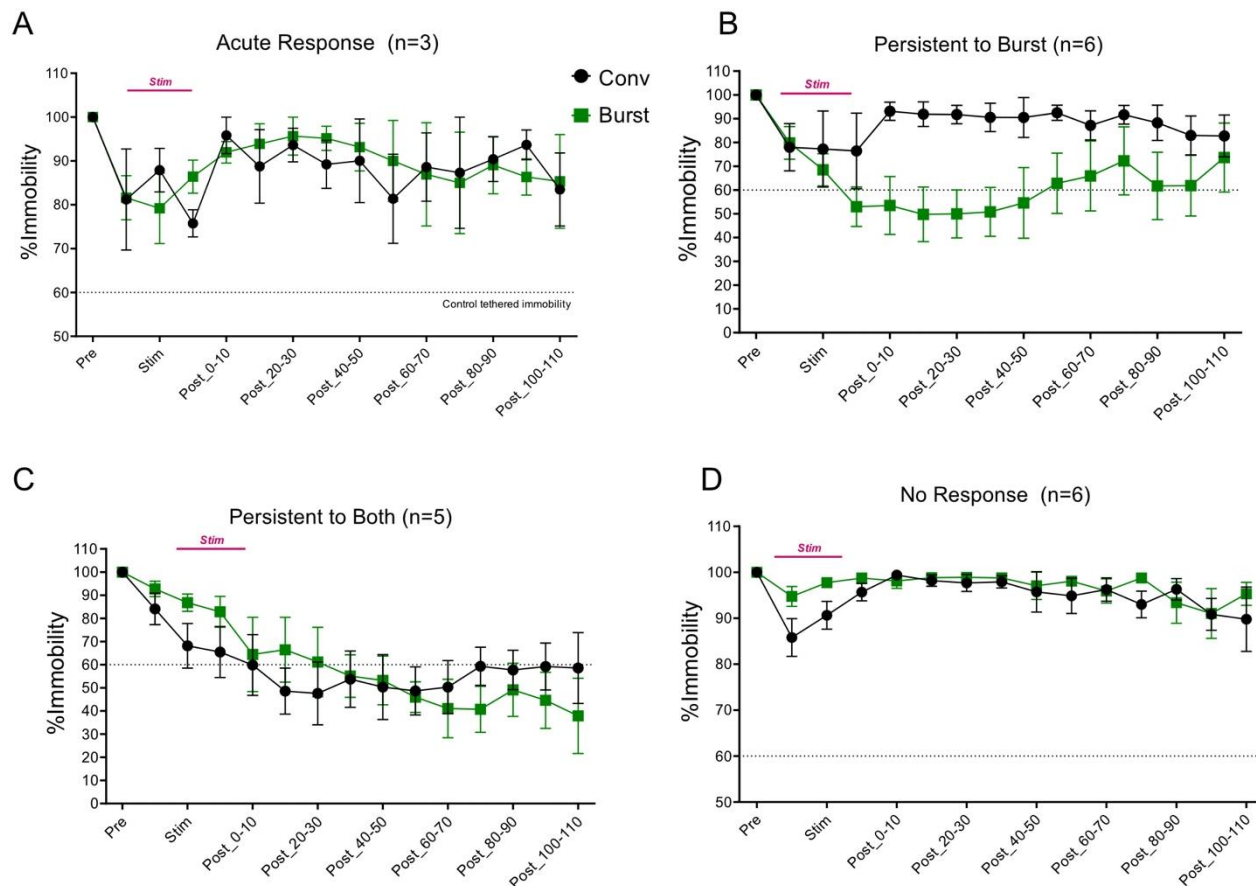


Figure 4.5 Stimulation delivered near the STN/EPN territory produces 3 response types.

Normalized Immobility measured pre-stimulation (10'), during stimulation (30') and post-stimulation (110') from a cohort of mice during conventional (grey) and burst DBS (green). Dotted line – average %immobility of 2 controls tethered in a similar setup. **A**, Response group 1 depicting acute responders only (N=3) (range depicted 50-110%) **B**, Response group 2 depicting acute response with conventional DBS, followed by a return to baseline immobility, and a persistently recovered mobility in response to burst DBS (N=6, range depicted 0-110%) **C**, Response group 3 depicting persistently recovered mobility in response to both conventional and burst DBS (N=5, range depicted 0-110%) **D**, Group of non-responders including animals that showed no or minimal (>80%) response to stimulation (N=6, range depicted 50-110%).

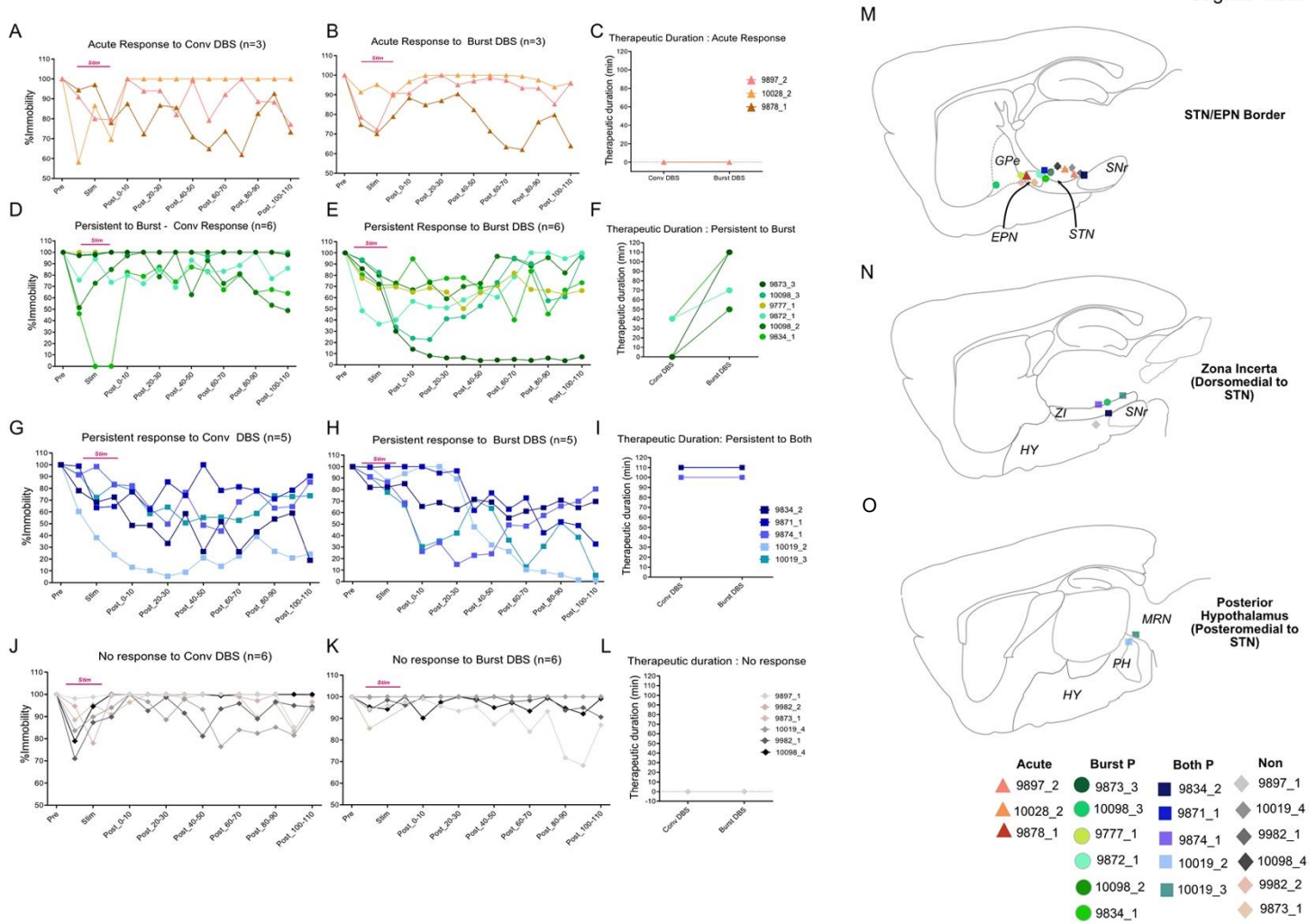


Figure 4.6 Response types map onto anatomical hot spots for engaging either persistent or acute response pathways.

A-C, Response to conventional and burst DBS for individual animals in response group 1 (Acute, red-orange triangles) depicted as the % immobility, as well as the therapeutic duration compared between the two types of stimulation. **D-F**, Similar response graphs for individual animals in response group 2 (Burst P, green dots). **G-I**, Response graphs for individual animals in response group 3 (Both P, blue squares). **J-L**, Response graphs for individual animals that showed no-response (Non-responders, grey diamonds). **M**, Lateral sagittal section including the STN and the EPN overlaid with individual markers for the animals depicted in panels A-L. **N**, Sagittal section medial to the STN including the dorsomedial zona incerta. **O**, Medial sagittal section including the posterior hypothalamic nucleus and midbrain reticular nucleus, overlaid with markers for animals targeting more posteromedial regions.

4.4.5 Acute response to stimulation and depletion status can serve as predictors of persistent rescue in response to burst DBS

In addition to the spatial mapping of response types to the stimulation target and the potential pathways that might be activated, other features that might serve as predictors of the degree of persistent responsivity, include the amount of dopamine remaining, as well as the dynamics of the acute response. The animals that showed a progressively increasing response to stimulation during the 30 minutes of active stimulation, showed long-lasting responses after stimulation likely by engaging the pathways that produce persistent effects. Alternatively, the animals that showed a sudden drop in immobility, that did not progressively lower during stimulation, tended to return to baseline mobility post-stimulation. To quantify this observation, regression analyses were performed between the change in the immobility during stimulation (by subtracting the % normalized immobility in the first 10 minutes of stim. from the last 10 minutes of stim.) and the therapeutic duration in response to either conventional or burst DBS (Fig 4.7A-B). We showed a correlation between the drop in immobility acutely, and the degree of persistence across animals. Animals that end up showing only an acute response (red/orange triangles), cluster at lower values of delta-immobility. Animals that showed persistent responses to burst stimulation are clustered at higher levels of delta immobility with burst stimulation but remain lower on the graph showing conventional stim. (green dots, Fig 4.7A-B). Animals that are persistently responsive to both stimulations cluster at a high level of delta immobility on both graphs (blue squares, Fig 4.7A-B). The association between these metrics might show a bimodal effect with a threshold of delta-immobility that must be reached to elicit the persistent motor response. Additionally, when we categorized the kinds of movements during and after stimulation as ambulation or fine movement, we showed a build-up of ambulatory activity during the acute response that was carried over after stimulation stopped in the case of response group 3- conventional and burst stimulation, and response group 2- burst stimulation (Fig 4.8B, C). For acute responses, the ambulation remained consistently high only during the period of stimulation, with no ramping effect (Fig 4.8A).

The level of depletion only showed a weak correlation with the therapeutic response to burst DBS, suggesting that the level of DA remaining might play a minor role in the translation of the burst DBS paradigm into the cell-type specific modulation predicted in the GPe (Fig 4.7C-D). Finally, the order of delivery of burst or conventional stim. did not affect the response duration. The variability in responses is likely predominantly due to the differences in stimulation placement (Fig 4.7E)

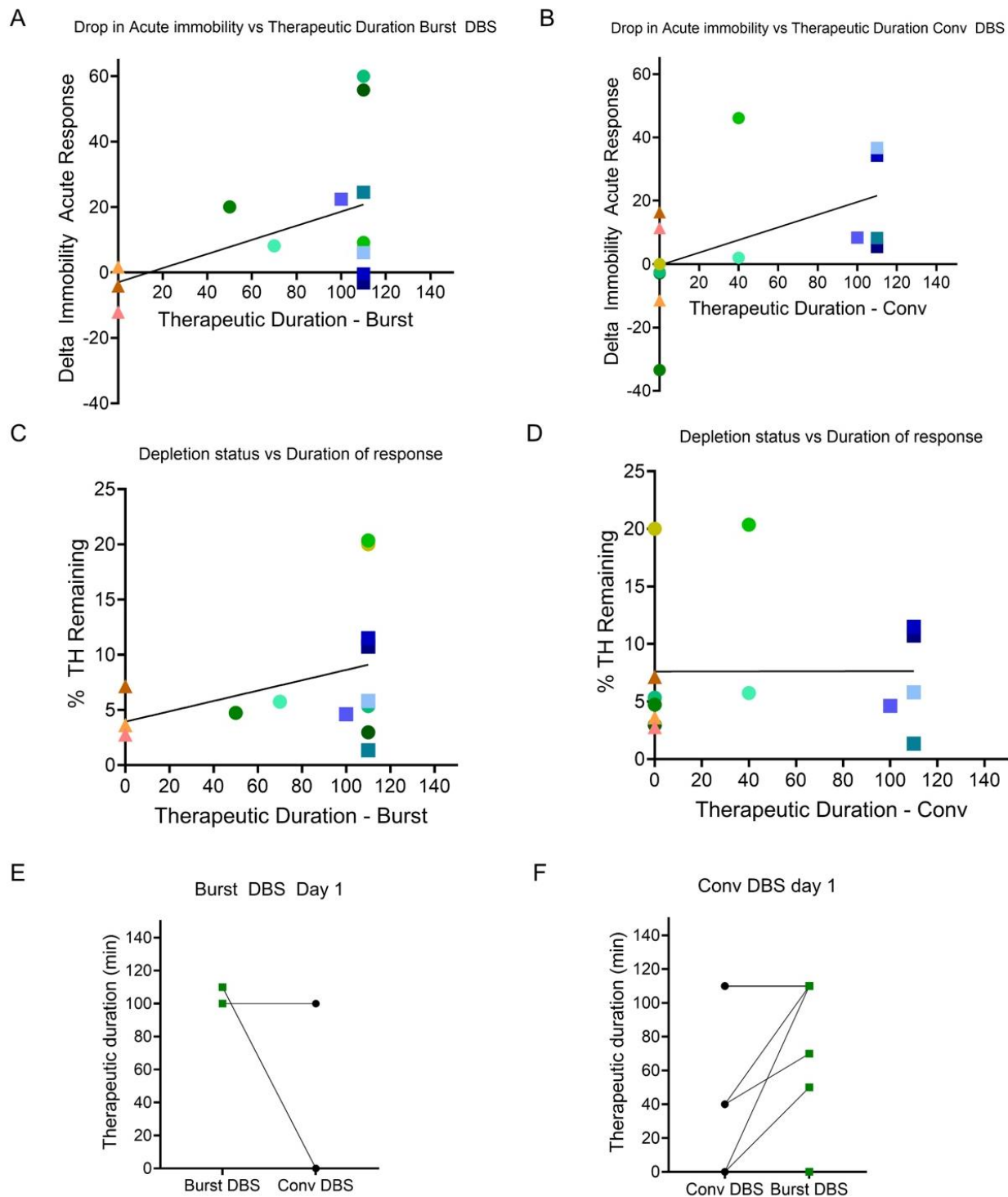


Figure 4.7 Other predictors of persistent motor response include acute responsivity, and level of depletion.

A-B, Regression between the change in immobility during active stimulation vs. therapeutic duration with burst and conventional DBS in all animals (non-responders excluded) (R^2 burst = 0.22, R^2 conv. = 0.25) **C-D**, Regression analysis for level of depletion based on the normalized % TH remaining in the striatum vs. therapeutic duration with burst and conventional DBS) (R^2 burst = 0.13, R^2 conv. = 8×10^{-6}) (Individual animal symbols same as Fig 4.6). **E**, Therapeutic

duration for animals that received burst DBS (green square) on day 1 (N=3), **F**, for animals that received conventional DBS (black dot) on day 1 (N=11).

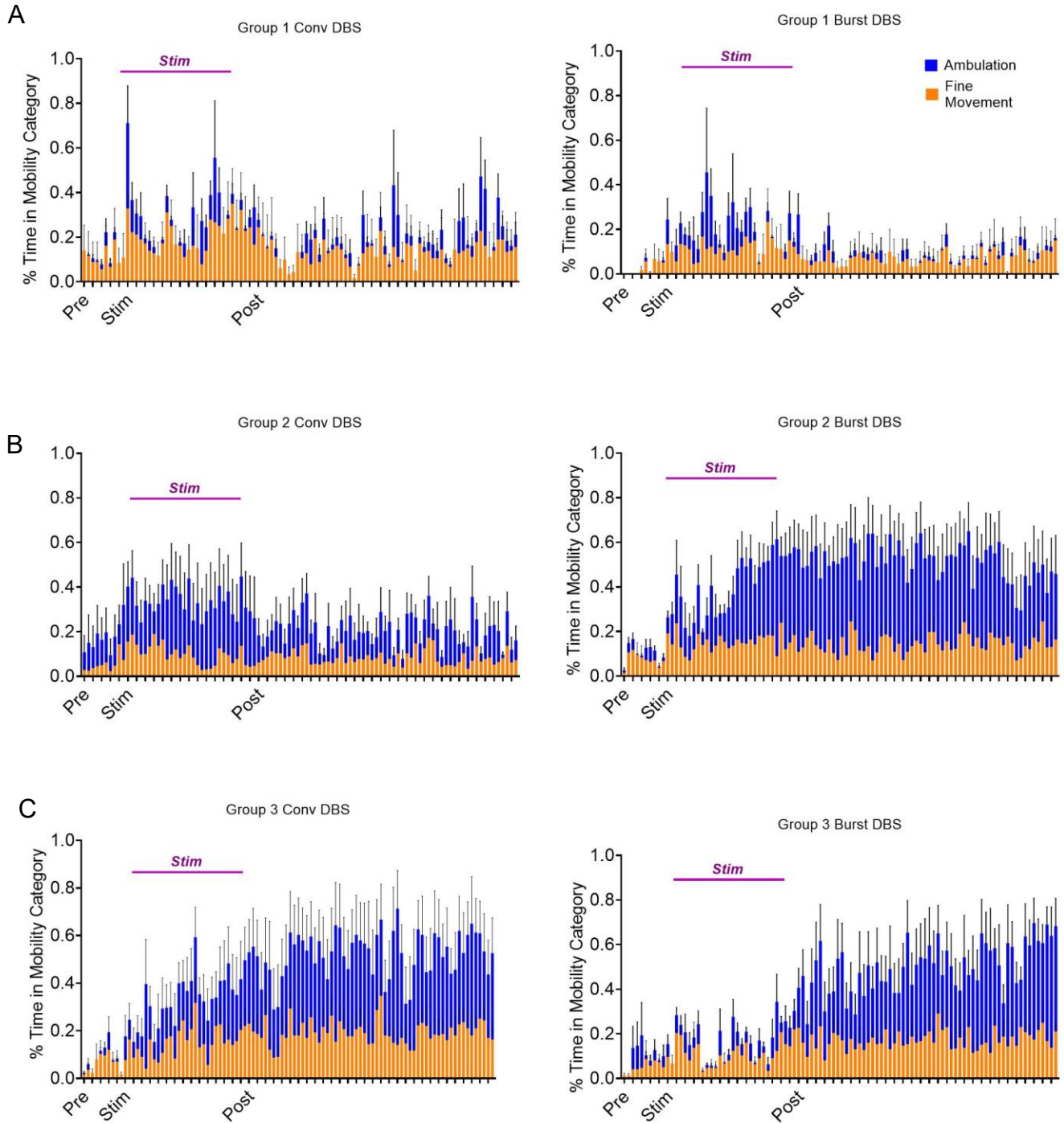


Figure 4.8 Changes in mobility type for acute and persistent responses.

A, Response group 1 (acute response), pre, stim, and post-stim. breakdown of % time spent in either mobility category for 1-minute bins is depicted – fine movement (orange), ambulation (blue), immobility not shown. Acute responders showed mild increase in ambulation during stim, with mostly fine movements contributing to the movement response. **B**, Response group 2 depicts the mobility responses when only burst DBS produces a persistent response. Ramping increase

in ambulatory activity seen only with burst DBS (right). **C**, Response group 3 demonstrating ramping increase in ambulation during and after stimulation with both conventional (left) and burst DBS (right).

4.4.6 Simultaneous recording in the SNr and stimulation in the EPN presents with some methodological challenges that remain unresolved

Like what was seen in the optogenetics study that led to differential cell-type modulation and persistent behavioral rescue (Mastro et al., 2017), we predicted that the neural correlate of the persistent behavioral response would lie in the attenuation of pathological bursting in the SNr. This study was designed to span the mediolateral and anteroposterior extent of the SNr, using multi-shank electrode arrays (64 channels), while simultaneously delivering electrical stimulation in the EPN (Experimental paradigm, Fig 4.9A-D). While this approach provided a promising direction for future studies, it is important to address various methodological challenges, some of which were resolved, while others remain unresolved in running these experiments. As shown in Fig. 4.9F the challenges involved: a lack of real-time feedback about recording site and limited anatomical space that led to recording from off-target areas; variability in responses to stimulation due to the black-box nature of stim-delivery; and limitations of the recording interface in minimizing movement-related artifact. To improve recording-site validation, electrophysiological signatures of SNr firing were identified that guided real-time targeting. Post-hoc tracking was improved by using dye applied at the tip of the shanks. The mismatch observed post-hoc between the desired and achieved recording locations was corrected by approaching the SNr at an angle. However, the margin of error was high which led to recording sites ranging from the STN anteriorly to brainstem regions. Reliably achieving SNr targeting was hence deemed unfeasible with the given space constraints and variability in angled trajectories. A potential solution would involve making the surface component of the DBS electrode smaller, which would allow more room for the recording electrode to penetrate the SNr craniotomy. Additionally, using some viral optotagging approaches to confirm targeting in the SNr, could help achieve more certainty in real-time.

Another critical challenge was that the response to stimulation was highly variable. Some of the variability could be explained with post-hoc histology based on an improper stimulation site, or insufficient dopamine depletion. However, like the lack of real-time recording feedback, there was no real-time measurement of the actual current delivered because the tissue impedance was unknown once the chronic stimulation electrode was implanted. Add-ons to the setup to allow for real-time impedance measurement could help minimize the potential lack of responsivity related to tissue

impedance or electrode failure. Additionally, like in patients, there is biological variability in responses to stimulation that cannot be explained despite precise targeting approaches. The numbers required to overcome the biological response variability and the multiple points of failure in dopamine depletion and targeting are high, necessitating a simplification of the experimental approach.

Additionally, the Intan amplifier was highly susceptible to movement artifact, that would affect the quality of spike-sorting and make the interpretability of movement-related activity, or continuous-time-series analyses like synchrony or burst-analyses challenging. An amplifier, like the Plexon Omniplex system that is robust to any movement-related noise, would allow for better characterization of the neural correlates of behavioral response to stimulation.

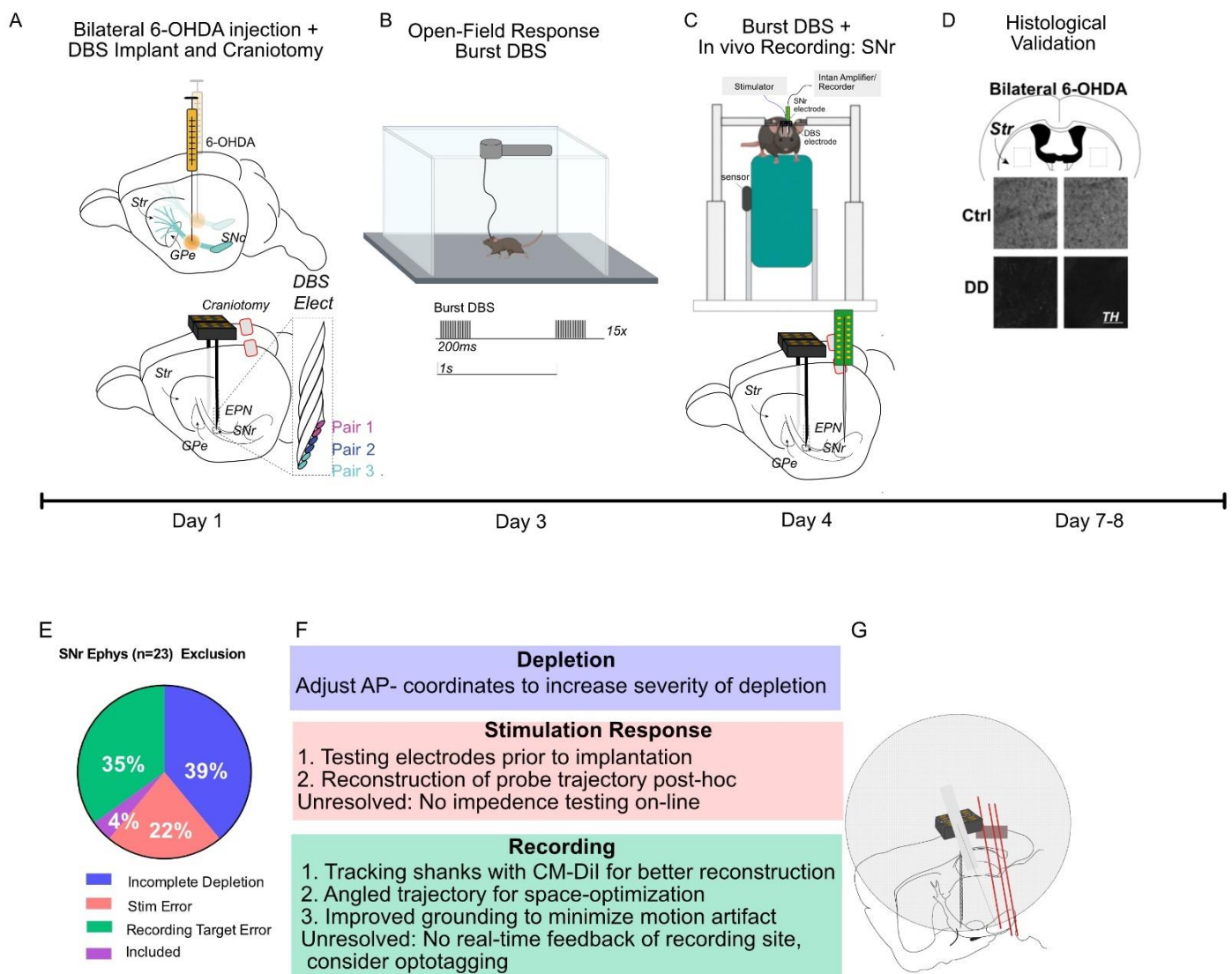


Figure 4.9 Experimental paradigm and challenges for simultaneous recording in the SNr and delivery of DBS in the EPN.

A, Surgical paradigm involving bilateral dopamine depletion, DBS implantation in the EPN, and craniotomies over the SNr. **B**, Assessment of behavioral response to burst DBS on day 3 after surgery. **C**, In vivo recordings in the SNr with delivery of burst DBS in the EPN. 64 channel electrodes used with the Intan amplifier. **D**, Histological validation of recording and stimulation probes. **E**, Exclusion criteria for cohort of 23 mice run on this setup including stimulation, and recording mistargeting, as well as issues with dopamine depletion. **F**, Resolved and unresolved factors for methodological challenges outlined. **G**, Example of an angled trajectory for SNr recording with a forward head-tilt on the head-fixed wheel setup.

4.4.7 Preliminary result showed suppressed pathological activity in the SNr of a rescued dopamine depleted mouse

Recordings were performed using 64-channel electrodes in a cohort of 23 mice. 5 control mice, and 3 dopamine depleted mice had valid recordings from the SNr that were compared at baseline. Dopamine depleted mice were confirmed to have a loss of striatal TH remaining relative to controls (<20% TH remaining, Fig 4.10A). The firing rate of dopamine depleted animals was slightly suppressed in the SNr (mean = 29.25 Hz), compared to controls (mean = 34.36 Hz) (Fig 5.2B). Irregularity measured as the CV of the interspike interval and the proportion of spikes in bursts computed using the surprise method, were higher in the depleted SNr compared to the control SNr units. (Mean CV of ISI Control = 0.8, Depleted = 1.35; Mean Bursts Control = 0.08, Depleted = 0.20) (Fig 4.10C-D).

After applying the exclusion criteria described in 4.4.6, one depleted animal that underwent EPN stimulation was analyzed for various firing metrics to identify the neural correlates of its day 3 behavioral response (Fig 4.10E-H). On day 3 post-operatively, the animal was placed in the open-field and burst DBS (175 Hz, 200ms/1s) was delivered in the EPN for 30 minutes. The animal showed an acute reduction in immobility, with limited (~30 minute) post-stimulation mobility improvement (Fig 4.10E). On day 4 post-op, the mouse was head-fixed on a running wheel and a 64-channel electrode was lowered into the SNr and units were recorded using the Intan amplifier. 10 minutes of pre-stimulation activity was recorded, followed by 30 minutes of stimulation. Post-stimulation activity was then recorded immediately after stimulation was stopped (0-10') as well as 100 minutes after stimulation was stopped (100-110'). The mouse showed an increase in firing rate, suppressed irregularity, and a suppressed ratio of bursts long after stimulation was stopped, unlike the effects seen in the acute post-stimulation period. Further analysis of the behavioral correlate of these neural effects could provide interesting insights regarding the role of the SNr in the rescue produced by EPN DBS.

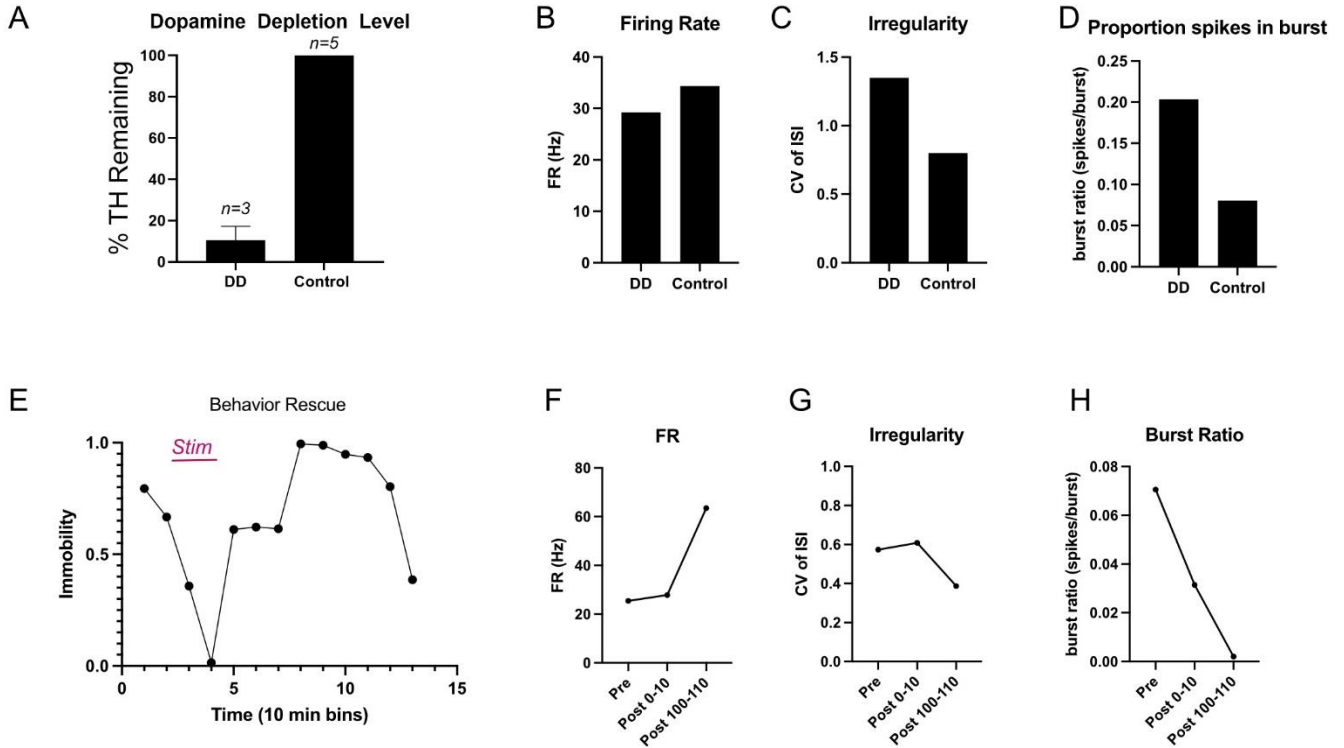


Figure 4.10 Preliminary finding (n=1) of SNr firing changes after burst DBS in the EPN.

A, Striatal TH quantification for a cohort of n=5 dopamine depleted and n=3 control mice. **B-D**, Measures of firing rate and pattern including CV and bursts at baseline in the SNr from the cohort described in **A**. **E**, Assessment of behavioral response to burst DBS on day 3 after surgery. **F-H**, In vivo recordings in the SNr with delivery of burst DBS in the EPN. Firing rate, and pattern (CV, bursts) measured pre-stimulation, acutely (0-10 minutes) post-stimulation, and longitudinally (100-110 minutes) post-stimulation.

4.5 Discussion

In this study we demonstrate how fundamental knowledge about the organization and function of basal ganglia circuitry can be used to refine the population specificity of electrical stimulation, ultimately prolonging the therapeutic benefits of DBS beyond those achieved with conventional methods. Using a burst stimulation protocol (175 Hz, 200 ms/1sec) delivered in the EPN and anterior STN, we were able to show long-lasting motor recovery in a dopamine depletion mouse model. In posteromedial regions closer to the ZI and PH, we could elicit this persistent response with both burst

DBS and conventional continuous DBS. These results provide some insights into the mechanisms that might underlie the long-lasting behavioral responses.

Translational relevance: Pilot study in humans

(Findings reported in (Kariv et al., 2024))

Other alternative forms of DBS, including adaptive and coordinated reset DBS can induce persistent therapeutic effects with the use of specialized implants, and it is plausible that the long-lasting effect observed in our study shares a common underlying mechanism with these approaches. However, our burst DBS protocol can be delivered through commonly used DBS implants and falls within US Food and Drug Administration– approved stimulus frequencies, enabling immediate testing in PD models across species, including human patients. In fact, a recent collaboration with Dr. Nader Pouratian’s team at UT Southwestern involved the study of the safety and efficacy of delivering our burst stimulation protocol in patients with PD that have DBS electrodes implanted in the GPi (Kariv et al., 2024). The delivery of acute burst DBS was tested in 6 patients to evaluate how well they tolerated this alternate pattern of stimulation relative to conventional DBS and whether their therapeutic effects continued to remain optimized. Burst DBS was equally well tolerated without significant side effects, and it produced equivalent reduction in the motor aspects of the UPDRS, while also increasing proactive inhibition relative to when the stimulation was off. This provides an exciting opportunity to perform further human testing to test the long-lasting effects of the burst stimulation protocol as demonstrated in our in vivo mouse model.

Circuits that produce acute vs long-lasting response to stimulation

Engaging various circuits in the STN/EPN regions of a parkinsonian mouse seems to produce distinct response to electrical stimulation. Some responses rely on the kind of stimulation, while some elicit the persistent motor response irrespective of the type of stimulation. Tractography-based methods have recently focused on functional effects of DBS on the white-matter connectivity in patients undergoing DBS in either the STN, or GPi. In this study, based on the response types and the anatomical localization of stimulation electrodes, we can speculate on circuits that might be engaged to produce the various response types.

Anterior STN targeting in the vicinity of the EPN, likely co-activates the D1 inputs to the GPe and the STN projections to the GPe producing the cell-type specific effect of PV excitation and Lhx6 suppression and long-lasting recovery with burst DBS. Targeting within the STN itself might engage

the STN projection to the GPe, that could excite both cell-types in the GPe, but might not include the coactivation of the D1 inputs to the GPe, critical for the *Lhx6* suppression. Downstream mechanisms of the GPe modulation have been studied further in our lab suggesting an anatomically segregated projection from *Lhx6* GPe neurons to the medial SNr and from PV GPe neurons to the lateral SNr (unpublished). This suggests that medial SNr activation might produce similar effects like the ones seen with our EPN DBS protocols. Similarly, the reciprocal connection between the SNr and the MLR is thought to play a role specifically in the persistent pathway of rescue, and not the acute response (unpublished). Stimulation anywhere downstream from the SNr that might engage these pathways could potentially produce the long-lasting improvement in mobility irrespective of the type of stimulation (Fig 4.11). Such a persistent response, independent of stimulation pattern, was observed when the DBS stimulating electrode localized medially in the ZI, PH and anteromedial regions of the MRN. Further investigation of the connectivity between the SNr and these areas might reveal circuit-inspired targets for long-lasting motor improvements in patients with DBS. ZI has been used as a site for DBS in PD patients and leads to symptomatic relief of bradykinesia, tremor and rigidity, likely due to its diverse connectivity to various motor structures including the basal ganglia, thalamus, and cerebellum (Ossowska, 2020). It is hence probable that in addition to the correction of pathological intrinsic firing, the fibers of passage through the ZI to specific parts of the SNr or MLR might be activated by this posterior stimulation, leading to the long-lasting motor effect. Hypothalamic DBS has been utilized in disorders outside of PD, for the treatment of aggressive behavior seen with intellectual disability (Benedetti-Isaac et al., 2021) or for the restoration of walking after spinal cord injury (Cho et al., 2024). Similarly, other hypothalamic and midbrain DBS approaches have leveraged the role of arousal networks in producing locomotor effects. (Garcia-Rill et al., 2015; Thevathasan et al., 2018). These studies suggest that DBS delivered to hypothalamic, or midbrain reticular regions activate upstream basal ganglia pathways and downstream brainstem locomotor pathways that are important for persistent motor effects and might not functionally involve the nuclei themselves. Mapping the anatomical localization of acute and long-lasting responses might allow us to identify novel targets for therapy that can potentially have neuroprotective effects.

Parkinson's Disease

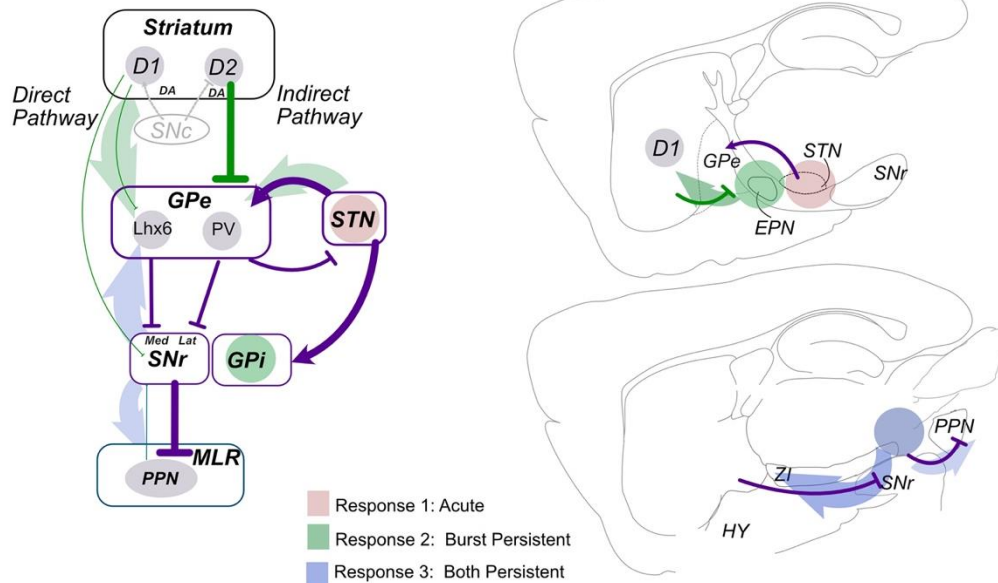


Figure 4.11 Pathways producing persistent vs acute motor recovery.

Stimulated areas depicted for the 3 response types (Acute – red, Burst Persistent – green, Both persistent – blue). Speculated pathways of activation with each response type depicted by shaded arrows of the same color. STN stimulation produces acute response with likely limited engagement of other pathways (red circle). STN/EPN border stimulation and EPN stimulation recruits the D1 inputs to the GPe and STN activation of GPe (green arrows) producing persistent responses to burst DBS. Zona incerta (ZI), posterior hypothalamus (PH) or midbrain reticular nucleus (MRN) stimulation might activate GPe inputs to medial SNr or SNr inputs to the MLR producing persistent responses irrespective of stimulation type (blue arrows).

Neural correlates of behavioral response

While the downstream mechanism of the differential modulation of GPe cell types that leads to the long-lasting behavioral rescue remains unknown, a key neural correlate observed in the original optogenetic study was the reduction in the proportion of pathologically bursty neurons in the SNr when the mice showed persistent motor recovery. Pathologically synchronous firing in the SNr and other areas within the basal ganglia has been shown in patients and animal models of PD (Abosch et al., 2002; P. Brown et al., 2001; Parr-Brownlie et al., 2007). Beta oscillations in the STN for example have long served as a biomarker to monitor therapeutic effectiveness via its suppression. Since mouse models of PD lack a robust beta frequency signal, the bursting and irregularity observed in the SNr of the dopamine depleted mice could serve as a biomarker for the degree of therapeutic rescue with DBS

interventions. The changes in firing patterns from pre-to post-stimulation serves as a readout for the effect of DBS on the neural circuits affected during dopamine depletion. The SNr recording study provides exciting future avenues for probing the pathological firing features in parkinsonian conditions that might be reversed in the context of the motor rescue. Such a discovery could lead to therapeutic interventions that go beyond symptomatic relief and provide correctional alterations to nodes in the pathological circuit.

Once the methodological challenges of the simultaneous recording in the SNr are addressed, through optotagging-based confirmation of recording site and space optimization as described before, large simultaneously recorded datasets from the SNr could facilitate analysis of dimensionality reduction based neural trajectories in relation to electrical stimulation. The prediction would be that the units recorded prior to burst and conventional DBS would cluster in a common neural principal-component (PC)-space. In the stimulation period, the units from animals that received conventional DBS would shift to a different PC space (black dots) representing the stimulation-induced activity, but soon return to the baseline space (black 'X') representing the pathological state. On the contrary, those that receive burst DBS would similarly switch to the stimulation-induced PC space like conventional DBS, but then enter a stable attractor-like state for longer and return to the baseline pathological state eventually (green lines and 'X') or shift to a new neural state (blue dashed lines, 'X') that could be representative of patterns of firing in a healthy mouse (Fig 4.12). Little is known about the presence of a state-change after burst-DBS. The behavioral outcomes being like the during-stimulation state would suggest that the extension of the stimulation-induced state is more likely than a switch to a new state after stimulation has ended. Neural recording data from the optogenetics study showed alterations in the burst firing of SNr neurons that correlated with the behavioral rescue. The progressive nature of these changes could point in the direction of some plasticity in the stimulation to post-stimulation period that could support such a state shift.

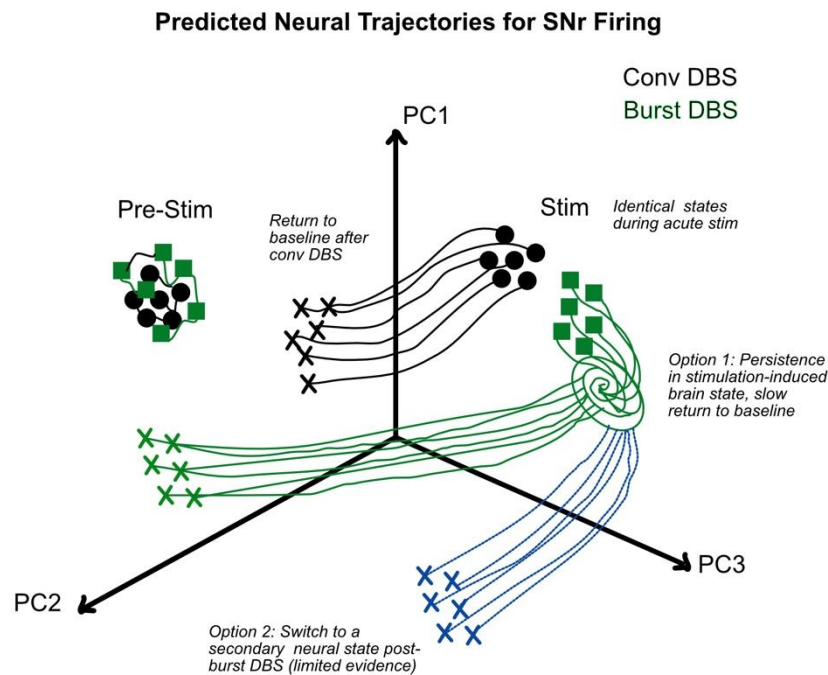


Figure 4.12 Prediction for change in SNr neural state with burst DBS.

Neural responses depicted in the pre-stim, stim, and post-stim. state for conventional (black dots) and burst DBS (green squares). In the burst DBS condition, neural state either enters a stable state in the stimulation-induced condition that lingers for longer before a return to baseline (green lines) or it switches from the post-stim state to a new healthy neural state (blue lines), whereas with conventional DBS, the neural state returns to the pre-stim. pathological brain state acutely.

Other measures of pathological firing in areas downstream from the SNr, like brainstem locomotor regions could also be tested to identify correlates of neural pathology and rescue. These might be engaged differentially when the mouse shows an acute response to stimulation, vs. a persistently induced state of locomotion. Arousal state has also been thought to play a role in the locomotion depicted by animals in the post-stimulation period. Recordings from sites relevant to arousal state like cortical areas or the hypothalamus in conjunction with pupil diameter measurements could enhance the interpretability of behaviorally relevant neural activity. Photometry approaches could alleviate the confound created by the artifact from electrical stimulation and provide more population level activity readouts for therapeutic effects in other parts of the striatum, basal ganglia, or brainstem locomotor regions. Additionally, a caveat of head-fixed recordings is that the behavioral response from a previously measured open-field session must be relied upon. However, switching to chronic neural recordings in a freely moving mouse could enhance the direct behavioral correlation of neural responses in the post-stimulation healthy motor state.

Future Directions: Updates to the stimulation setup to allow naturalistic behavior

The stimulation setup used in this study allowed us to study the behavioral responses to stimulation patterns that go beyond conventional stimulation. However, the combination of tethering, and the limited mobility with the dopamine depletion prevents the animals from being able to move naturalistically even when they are stimulated or rescued. We have previously shown that even control animals that are tethered show a non-trivial reduction in their mobility in the open-field due to the constraint and weight of the wire on their head (Fig 4.5 dotted line). While a wireless stimulator can allow for more naturalistic behavior, it presents its own challenges in terms of designing a stand-alone device that can be powered to run for a few hours without recharging. A collaboration with Dr. Jacob Robinson at Rice University has led to the development of a prototype of such a wireless bluetooth-powered stimulation-recording device that can interface with our implanted electrodes and allow for the delivery of current without the animal being tethered.

Chapter 5

5 Discussion and Final Remarks

5.1 Neural correlates of synuclein-induced tremor and dopamine depletion-induced bradykinesia

In **Chapter 1** we established that Parkinson's disease comprises of a constellation of symptoms including tremor, bradykinesia, gait dysfunction, and other non-motor deficits, all of which might not be represented in the parkinsonian animal models. While animal models like the toxin model produces a severe acute phenotype of bradykinesia and akinesia, it lacks features of tremor and gait difficulties. To investigate the neural basis of the heterogenous phenotypes produced by distinct parkinsonian pathologies, in **Chapter 2** we compared the electrophysiological features within the basal ganglia recipient (VM) and cerebellar recipient (VAL) sub-regions of the thalamus in two models – a tremor dominant model with aSyn dysfunction and a dopamine depletion model with bradykinesia. The thalamus which is an important integrator of information from the basal ganglia to the cortex and cerebellum to the cortex in parallel loops, has also been implicated in the pathophysiology of tremor and bradykinesia. The influence of the pathological basal ganglia circuit in the dopamine depleted condition and the aberrant cerebellar circuit in the tremor condition on the corresponding sub-regions of the thalamus make it an interesting mechanistic target for treatment. Appropriate tuning of treatment requires a comprehensive understanding of the pathophysiology of the circuits in the context of various symptoms.

We were able to show that in the tremor mice, the VAL was selectively affected, as predicted by the cerebellar basis of tremor and the aSyn pathology observed in the DCN. The VAL showed increases in burst firing and irregularity, along with a loss of motor encoding in the synuclein-induced tremor model. This validates the idea that the cerebello-thalamo-cortical circuit might be dysfunctional in these mice in the context of their tremor. To our surprise, we saw similar changes in the VAL of

the dopamine depleted mice, in addition to deficits seen in the VM with suppressed motor encoding and irregular firing and some evidence for low frequency delta oscillations. These results suggest a possible common mechanism at the level of the VAL thalamus in PD models and merits a deeper study of cerebellar contributions to motor symptoms in parkinsonian animal models. Since the VM showed deficits only in the dopamine depleted condition, the downstream effects of dopamine loss to the production of severe bradykinesia and akinesia in the dopamine depleted mice likely engaged the basal ganglia-thalamo-cortical circuit.

These observations suggest a convergence of pathophysiological mechanisms in the VAL in the tremor model and the dopamine depletion model (Fig 5.1) while maintaining a separate dopamine-mediated basal ganglia-dependent mechanism that is specific to the depletion models, and likely a correlate of the bradykinesia.

As a follow-up to our findings in Chapter 2, in **Chapter 3** we performed a detailed characterization of the patterns of aggregation of aSyn in a whole-brain and spinal cord series in the 3K mice. We established methods to study postural and gait deficits, along with a video-based tracking of tremor to then study the effect of lesioning the SNr on the motor and non-motor task performance of 3K and control mice. Consistent with our finding that the VM was normal in the 3K mice, we showed that the SNr-lesion did not influence the tremor in these mice. This further suggests that the pathophysiology of tremor in the 3K mice is independent of changes in the basal ganglia circuit.

5.2 Therapeutic interventions to produce long-lasting motor improvements

Previous work in the lab has led to the notion that cell-type specific modulation in the GPe can produce long-lasting motor improvements in mice with dopamine depletion. In vitro experiments studying synaptic features and input specific effects on cell-types in the GPe led to the discovery of patterns of electrical stimulation that can elicit the same cell-type modulation needed to produce this long-lasting rescue. In **Chapter 4**, we described the study of this pattern of stimulation in vivo to engage the pathway that shifts the behavioral state of an akinetic dopamine depleted mouse to a locomotor state that can last for hours after the stimulation has stopped. This can improve the therapeutic duration up to 4.5 times the existing conventional DBS treatments. In addition to

providing a clinically enhanced protocol for stimulation that can maximize battery power and minimize side effects, this study also raised the possibility of a plasticity-dependent correction of pathological activity that was sustained beyond the period of active intervention.

As a follow-up to our findings in the EPN, we sought to explore the distributed circuits underlying this behavioral response. In patients with PD, the STN is a commonly targeted region for symptomatic relief from rigidity, bradykinesia, and tremor. In our dopamine depleted mice, we tested whether the delivery of burst stimulation that led to persistent motor rescue in the EPN, would produce a similar behavioral outcome when delivered in the STN. In **Chapter 4** we describe animals that received both conventional and burst DBS in the STN. Targeting of DBS to regions adjacent to the STN produced other behavioral response types, including a posterior medial hotspot that led to persistent motor recovery irrespective of the pattern of stimulation. This circuit-mapping of the pathways that produce acute vs persistent motor effects allows us to propose an anatomical framework to guide DBS targeting that can leverage the persistent motor rescue pathway. Additionally, given the variability seen in patients undergoing DBS, there is a need for predictors of therapeutic response in the post-surgical setting. Programming performed in patients after the surgery often relies on a trial-and-error approach to optimize stimulation settings. In the same context, acute responsivity secondary to placement, can be used as a predictor for the degree of long-lasting response attainable in each patient. In our mouse model, we noticed that the ramping increase in responsivity to acute interventions, was best correlated with the entry into a state of long-lasting recovery. These insights could help guide clinical programming to maximize therapeutic effects.

Next-gen DBS devices have focused on improving the delivery of current to minimize side effects, either through better spatial targeting with directional leads, or through timing the delivery of current in response to the detection of a pathological signal or biomarker. In patients, the beta frequency oscillations in the basal ganglia are often associated with parkinsonian motor symptoms and are used as a biomarker to measure therapeutic impact. In rodents, an analogous synchronous bursty firing with the presence of low frequency delta oscillations has been observed in the SNr. In the second part of **Chapter 4**, we described the study attempting to use the bursting activity in the SNr of animals receiving burst DBS in the EPN as a measure for their therapeutic recovery in the post-stimulation period. This experimental approach presented with numerous challenges that deemed it unfeasible, but alternate approaches and resolvable solutions to these issues have been outlined for future studies. The closed-loop approach to DBS in conjunction with the DBS paradigms we discovered that lead to

long-lasting recovery could dramatically improve patient outcomes due to minimal current delivery, for maximal therapeutic benefit.

5.3 Unifying approach to tackle symptomatic variability in parkinsonian phenotypes

Personalized medicine has gained traction in a variety of clinical disorders. Similarly, in PD, the variability in patient symptoms and responses to therapeutic interventions has added complexity to our systems level understanding of the disease. Tractography based approaches have outlined the importance of circuit-specific targeting of DBS to optimize therapy for every individual. With that in mind, it is crucial to develop a framework to understand the pathophysiology of various parkinsonian phenotypes and the specific circuits that are responsible for producing them. Only then can we tailor our therapeutic interventions to target the relevant circuits in a patient-specific manner.

Existing thalamic interventions for tremor

The motor thalamus, specifically the cerebellar-recipient thalamus has been a therapeutic target in patients with essential tremor and other tremor-disorders like PD, either for thalamotomies which produce a structural lesion, or for DBS which can produce a functional lesion. The tremor-related signals originating in the cerebellum are thought to propagate through the cerebellothalamic tract to the ventral intermediate nucleus (VIM) of the thalamus which then sends the pathological signal to the motor cortex and downstream to the muscles that express the tremor. Hence lesions of the motor thalamus had been initially used to treat tremor in PD (Hassler & Riechert, 1955). DBS therapies have since been preferred over thalamotomies since they are reversible, and the stimulation parameters are adaptable according to the symptom and side effect profile (Okun & Vitek, 2004). These methods have primarily led to a reduction in tremor amplitude (Beuter & Titcombe, 2003; Duval et al., 2006; Mure et al., 2011).

Interventions in the cerebellar thalamus or the posterior ventrolateral thalamus (VLp) have shown clear improvements in tremor and some effects on rigidity, however its effects on bradykinesia, akinesia and gait in PD remains limited (Markham et al., 1966, Okun & Vitek, 2004). For a more comprehensive discussion of the history of thalamotomies and thalamic DBS see reviews- (Bosch-Bouju et al., 2013; Neudorfer et al., 2024). The effectiveness of these treatments combined with recent rodent work showing pathological tremor-related activity in the cerebello-thalamo-cortical loop (Pan, 2025) implicates the cerebellar thalamus in the tremor pathophysiology. While the cerebellar region

might be specific to tremor-related features of PD, other regions of the thalamus, that receive input from the basal ganglia and have more widespread cortical projection patterns have been implicated in motor learning (Canavan et al., 1989; Roth et al., 2024). The therapeutic opportunities for intervening in the thalamus for basal-ganglia dysfunction have been understudied. While there is evidence for pathological beta-frequency oscillations in the basal ganglia and cortex, and some evidence for oscillations in the cerebellar thalamus (Hanson et al., 2012; Holdefer et al., 2010; Kempf et al., 2009; Paradiso, 2004), there is a lack of similar reports in other parts of the thalamus due to restricted access to the thalamus in the context of DBS therapies. Rodent work has demonstrated selective synchronization in the basal-ganglia thalamus of anesthetized parkinsonian rats (Nakamura et al., 2014) which establishes a basis for therapeutic interventions in the basal ganglia-recipient thalamus to improve non-tremor motor symptoms like bradykinesia or rigidity, which have been conventionally treated by STN or GPi interventions.

Our insights from the investigation of the cerebellar-recipient and basal ganglia-recipient domains of the thalamus suggests that tremor is a symptom that is best tackled through a targeted intervention in the cerebello-thalamo-cortical pathway. Specifically, we outline pathological features of the thalamus in the synuclein-tremor model that can then serve as a biomarker in the context of effective tremor suppression (Fig. 5.1). On the contrary, bradykinesia and akinesia in dopamine depleted models, could benefit from a global thalamic intervention because of the pathological activity seen in both the VAL and the VM. Furthermore, within the basal ganglia, sub-circuits originating from distinct cell-types in the GPe could be leveraged differentially to produce acute or long-lasting motor responses. Symptom specificity can hence be attained in a projection-specific targeting approach, whereas the therapeutic efficacy including the duration and dynamics of response can be altered using insights from cell-type connectivity and synaptic properties (Fig 5.1).

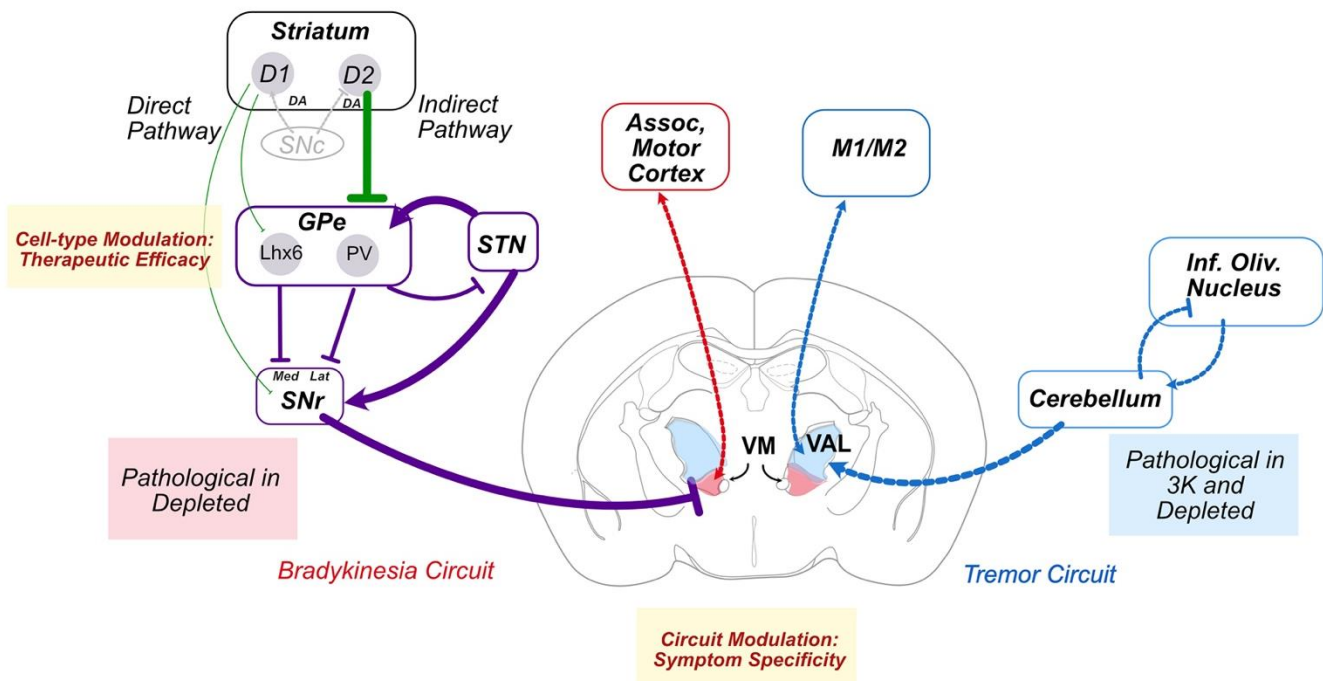


Figure 5.1 Symptom-specific circuit mapping.

Bradykinesia circuit represented by the aberrant basal ganglia output to the VM thalamus in the context of dopamine depletion (red). Pathological activity in the VM seen in depleted mice make it a good therapeutic target for bradykinesia. Tremor circuit represented by oscillatory activity from the cerebello-olivary circuit transmitted to the VAL (blue dashed arrows). Pathological activity seen in the VAL in 3K mice make it a good therapeutic target in tremor mice. VAL also implicated in dopamine depleted pathology. Cell-type specific interventions can improve therapeutic duration and efficacy, whereas circuit-specific interventions can improve the specificity of symptomatic relief (yellow boxes).

5.4 Final remarks

Parkinson's disease is a complex, and debilitating disorder that has seen some progress in terms of treatments that produce symptomatic relief. However, the exploration of the circuits underlying symptoms and how they can be leveraged to maximize therapeutic benefits has been limited. While fundamental neuroscience research has often focused on the investigation of cellular and physiological mechanisms of basal ganglia or cerebellar dysfunction, clinical research is often limited to optimization of already existing therapies to tailor them to suit patient needs. The link between taking insights from mechanistic investigations and translating them to clinical solutions that can guide better therapies is often missing. This thesis attempts to bridge this gap by taking optogenetic cellular findings and translating it to paradigms that can directly be used clinically in patients implanted with DBS.

Additionally, by expanding the models of parkinsonian pathophysiology from dopamine depletion-based models to synuclein-based models, we attempt to provide a more comprehensive view of parkinsonian pathology and deficits, which is more representative of what is observed clinically. The findings outlined in this thesis, including the pathophysiology underlying synuclein-induced tremor, and depletion-induced bradykinesia, as well as the therapeutic discoveries that lead to long-lasting motor recovery in parkinsonian mice, provide an exciting foundation for future studies to improve our understanding of this disease and to develop ways to better treat it.

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Appendix A: Gait Analysis Pipeline

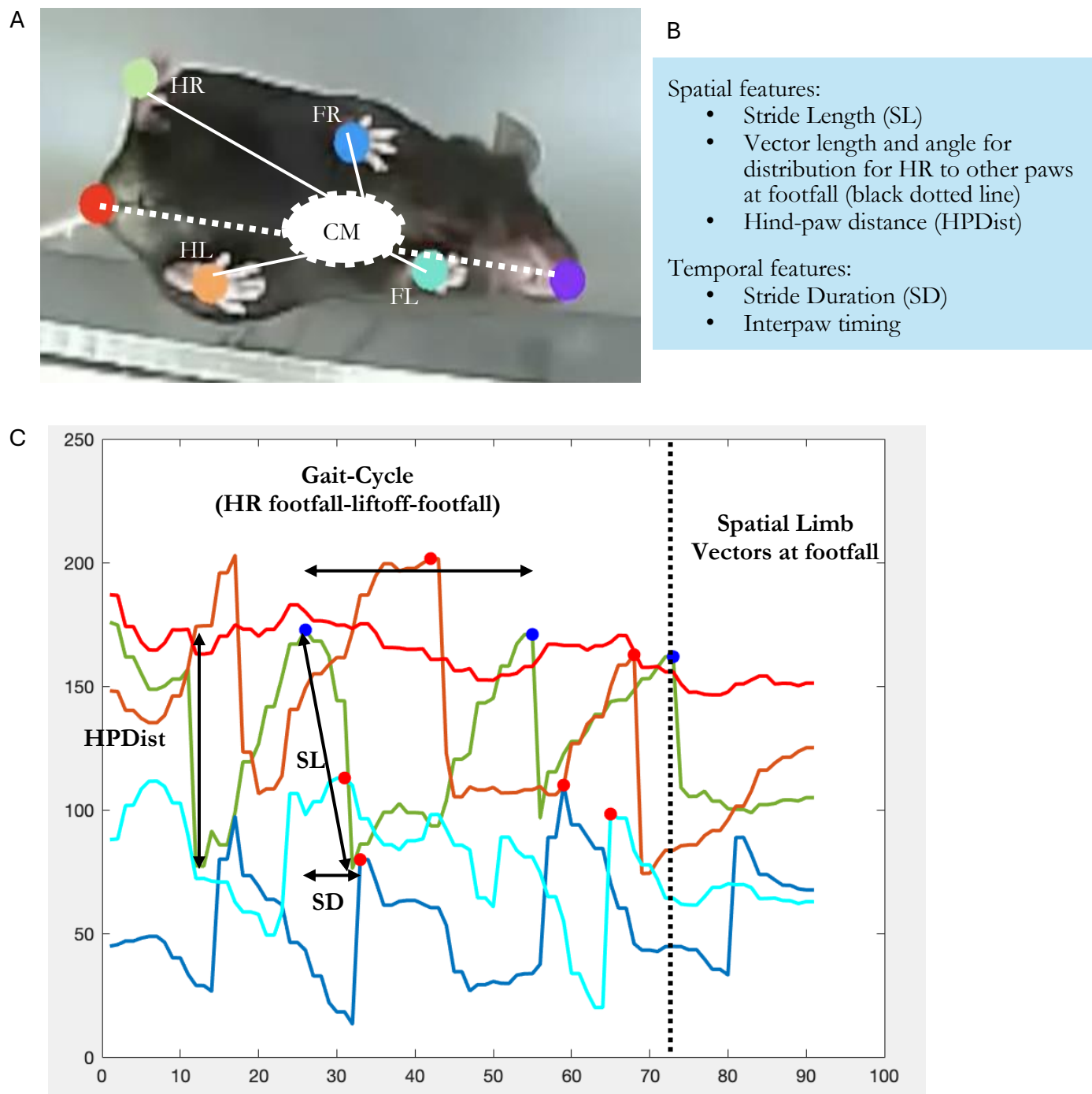


Figure A.1 Gait-analysis pipeline and metrics.

DLC tracking to label body points – tail, nose, and 4 paws. Computation of the center of mass (average X, Y position between nose and tail point). Distance of each paw from CM computed to normalize positions during gait cycle. **B**, Computation of various gait metrics depicted in C. **C**, A bout of locomotion depicted with a sinusoidal gait cycle for each

paw (colors based on A). Gait metrics labeled on graph – Stride duration (SD), Stride length (SL), Hind-paw distance (HPDist). Dotted vertical line represents footfall during which limb vectors computed for each paw relative to the hind right paw (green marker)

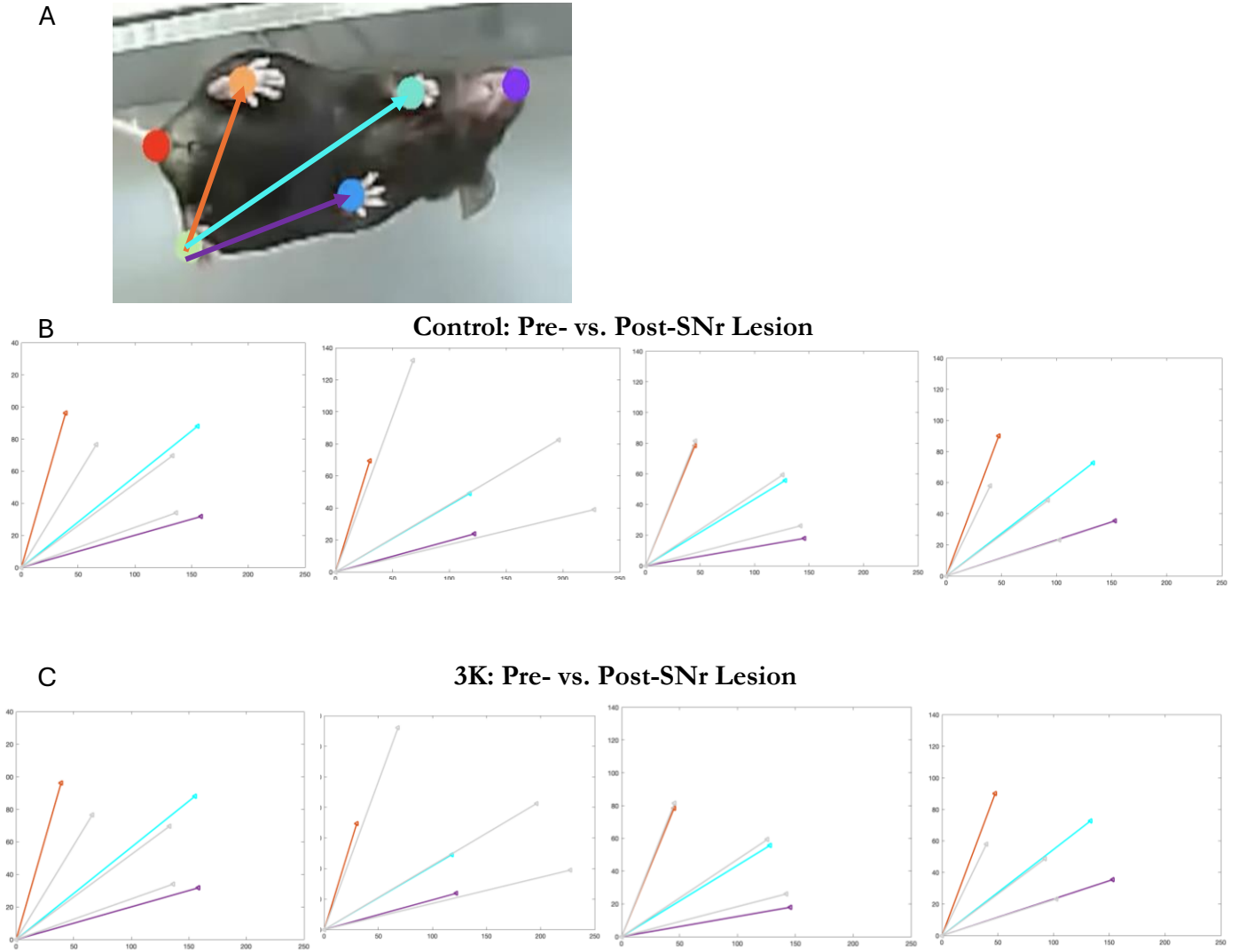


Figure A.2 Effects of SNr lesion on gait.

A, Representative example of limb vectors relative to hind-right paw. Front-Right (purple), Front-Left (light blue), Hind-Left (Orange). **B-C**, Limb vectors computed at footfall before SNr-lesion (colors as shown in A) and after SNr lesion (grey) in the **B**, four control mice and **C**, four 3K mice.

Appendix B: Absence of aSyn in Controls and Depleted Mice, Colocalization with NeuN in 3K mice

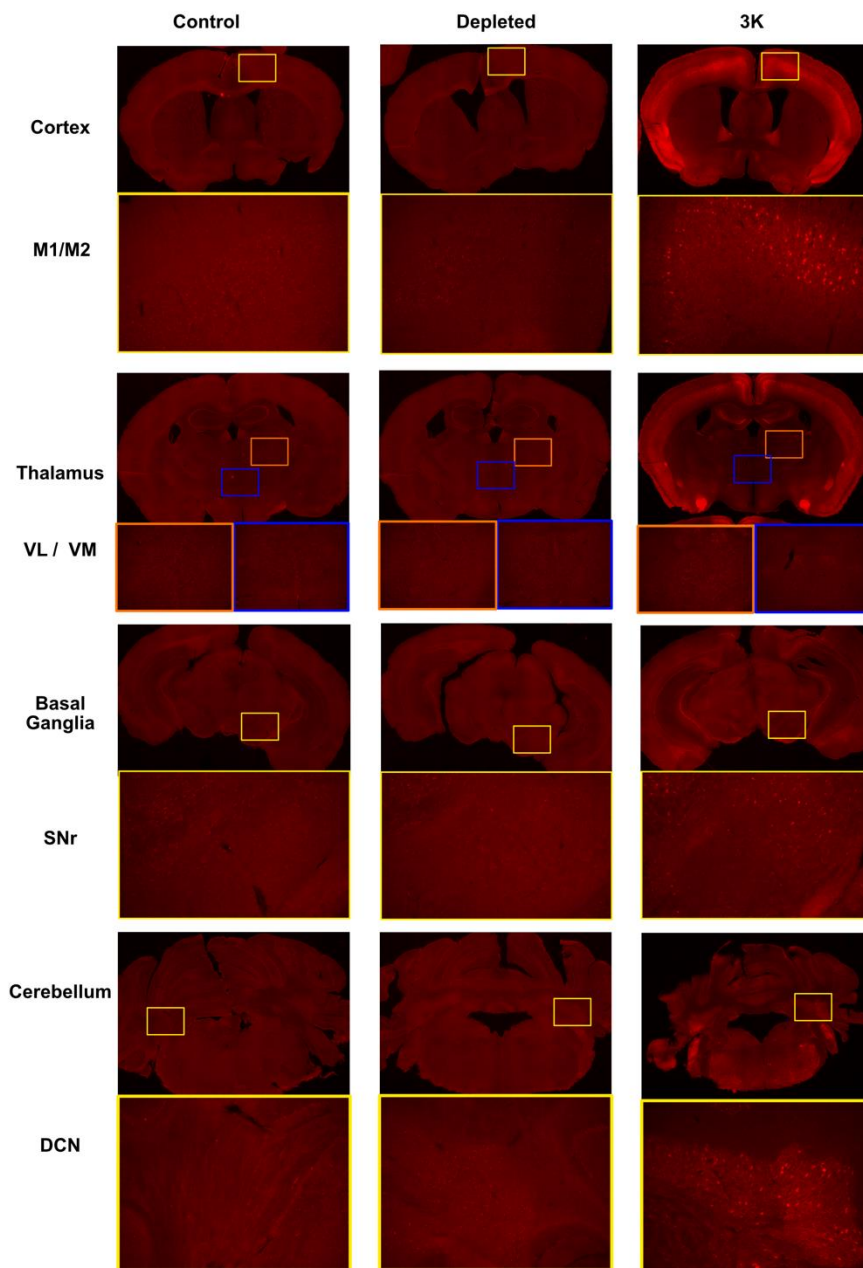


Figure B.1 aSyn Staining in the cortex, basal ganglia, thalamus and cerebellum of a control, 3K and depleted mouse brain

Phosphorylated S129 labeled in red (RFP) in a control, dopamine depleted and 3K mouse brain looking at aSyn aggregation patterns in the cortex, thalamus, basal ganglia, SNr, and cerebellum

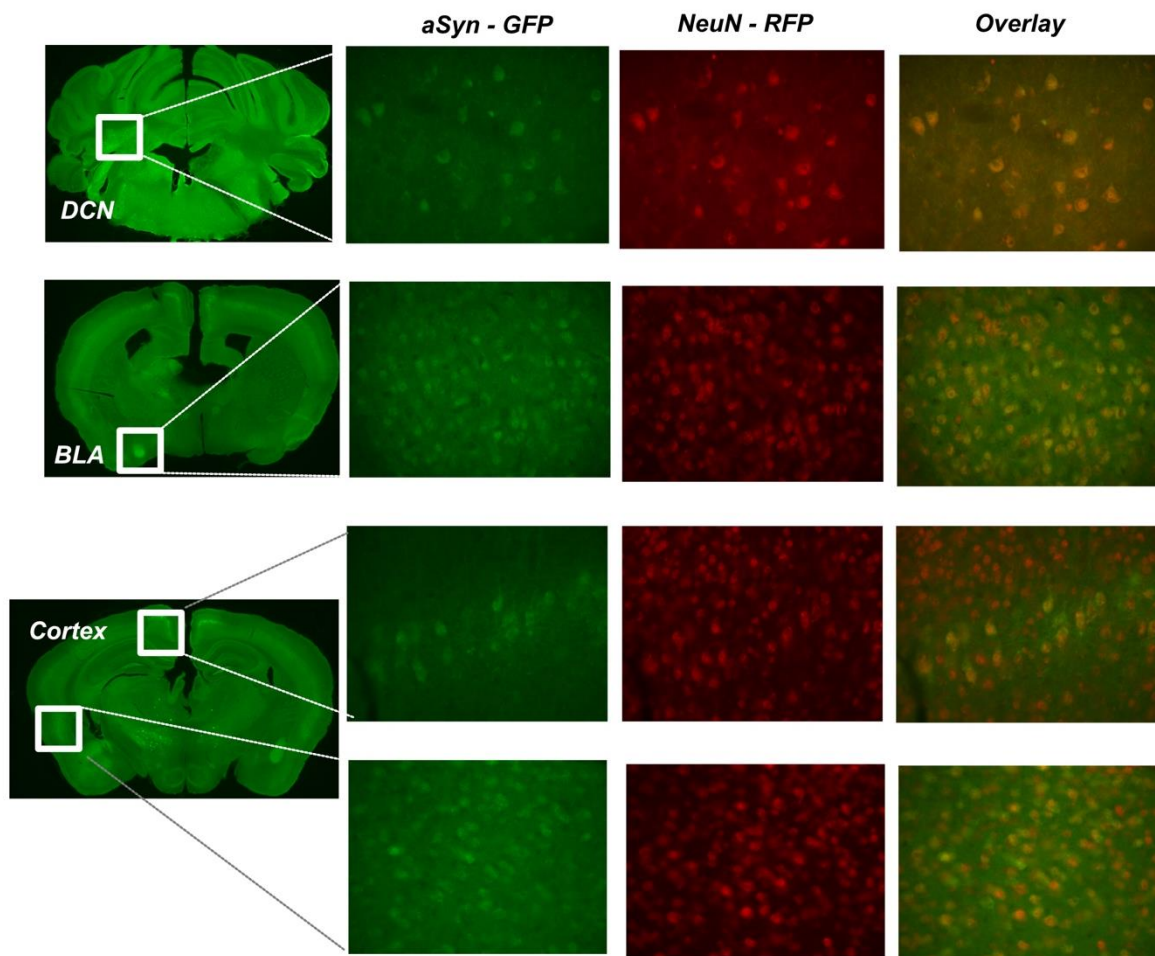
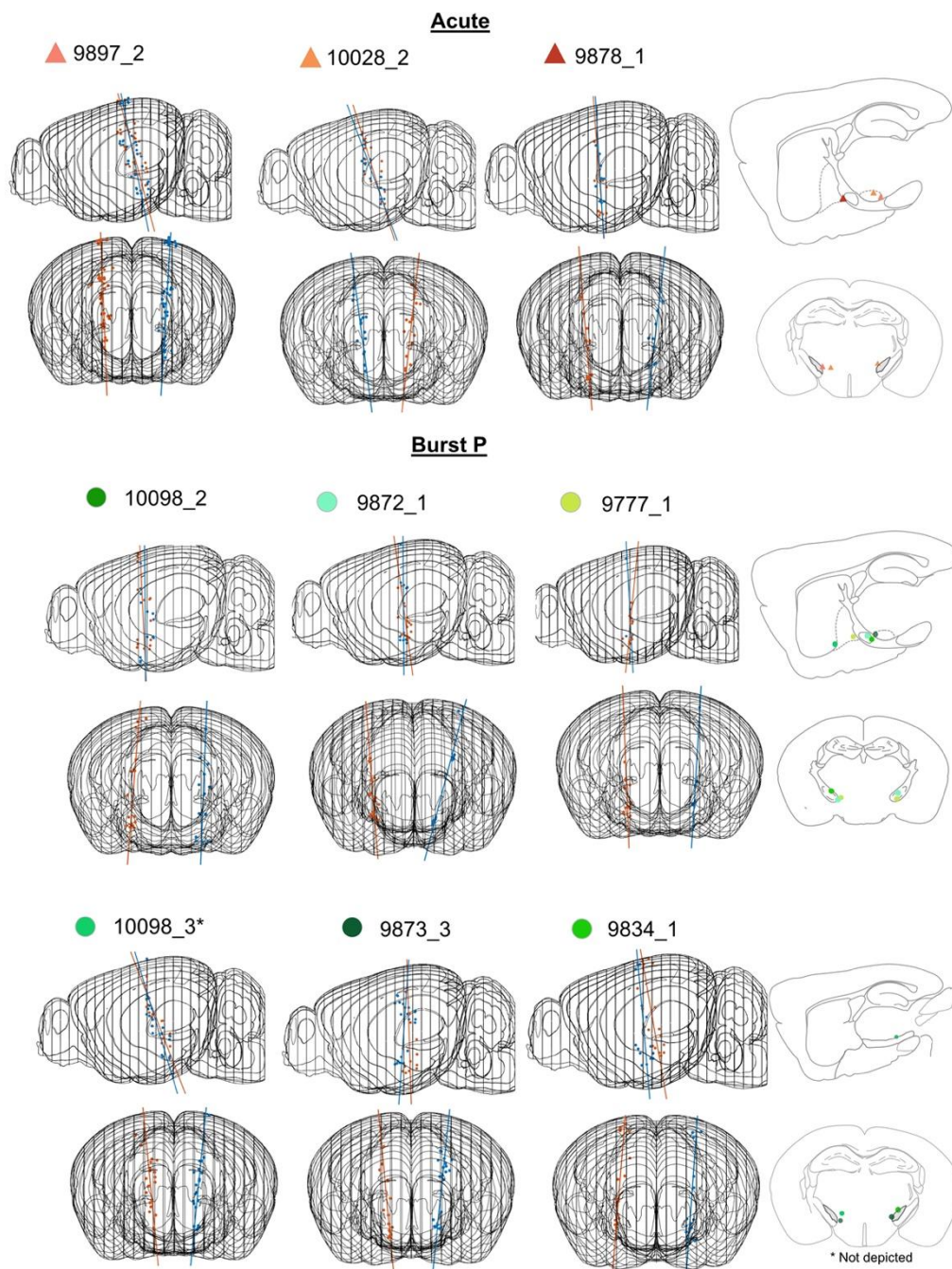


Figure B.2 Co-staining of phosphorylated aSyn and neuronal marker NeuN in the DCN, Cortex and BLA of a 3K mouse.

Overlay channel shows yellow overlapping neurons labelled by the NeuN RFP label and aSyn GFP label suggesting that aSyn is present in some but not all neurons found in the DCN, BLA and Cortex

(Imaging by Arini Bhargava)

Appendix C: Electrode trajectories mapped using AP histology for 3 response groups



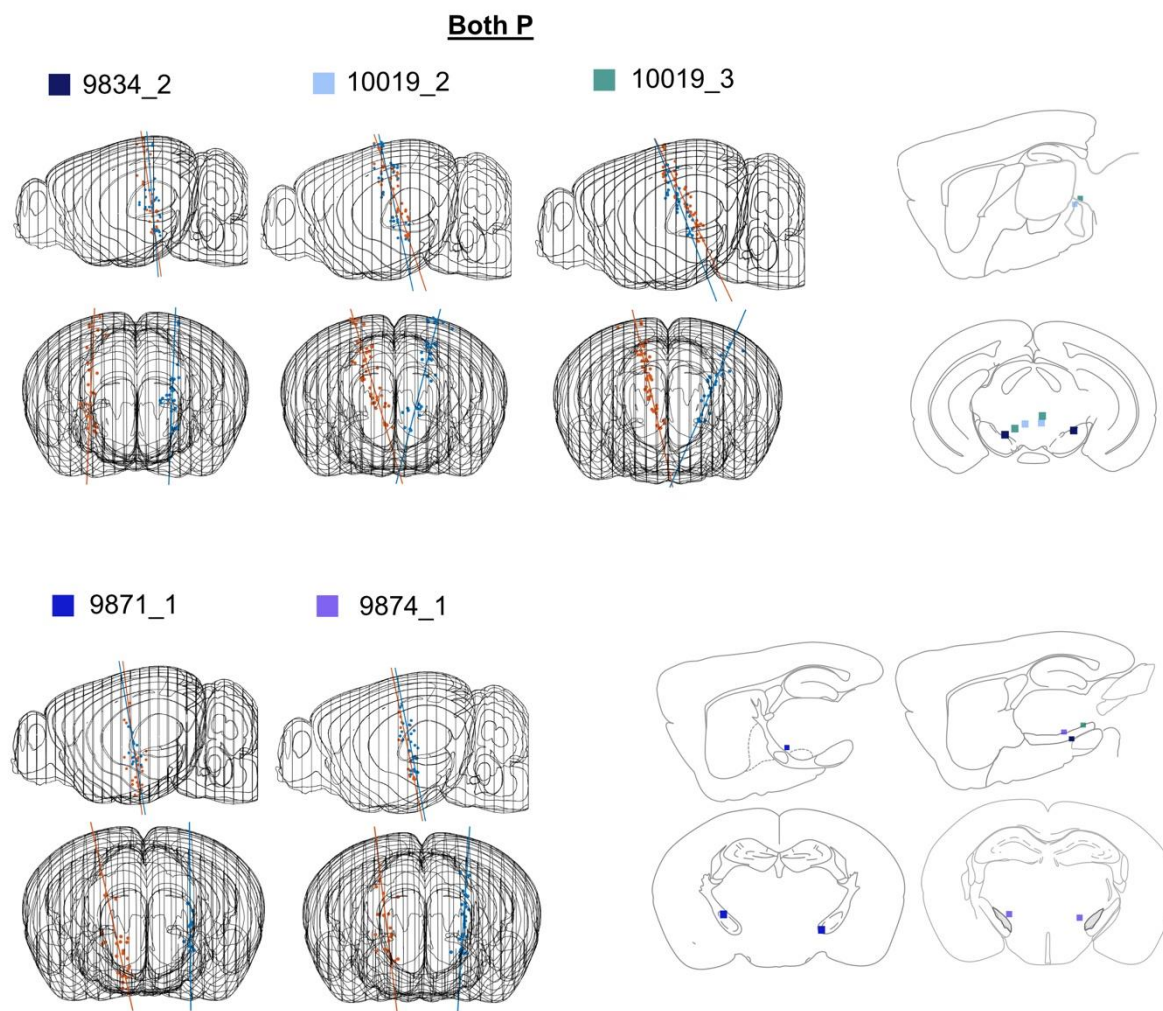


Figure C.1 Electrode trajectories mapped with the AP histology toolbox for all animals in the 3 response groups. AP-Histology is a MATLAB Toolbox that uses the Allen mouse brain atlas to 3D-reconstruct manually labelled probes in brain slices. Atlas reconstructions shown are used for the summary images in Fig. 4.6

Appendix D: SNr In-Vivo Analysis Pipeline

Figure D.1 Processing pipeline for in-vivo data analysis for 64 channel recordings in the SNr.

In vivo data collected using Intan amplifier and analyzed using a combination of Kilosort and Phy for semi-automated spike sorting. Custom MATLAB code was developed for pre-processing and post-processing analyses described below. Spike analysis metrics and definitions for analysis defined on page 2.

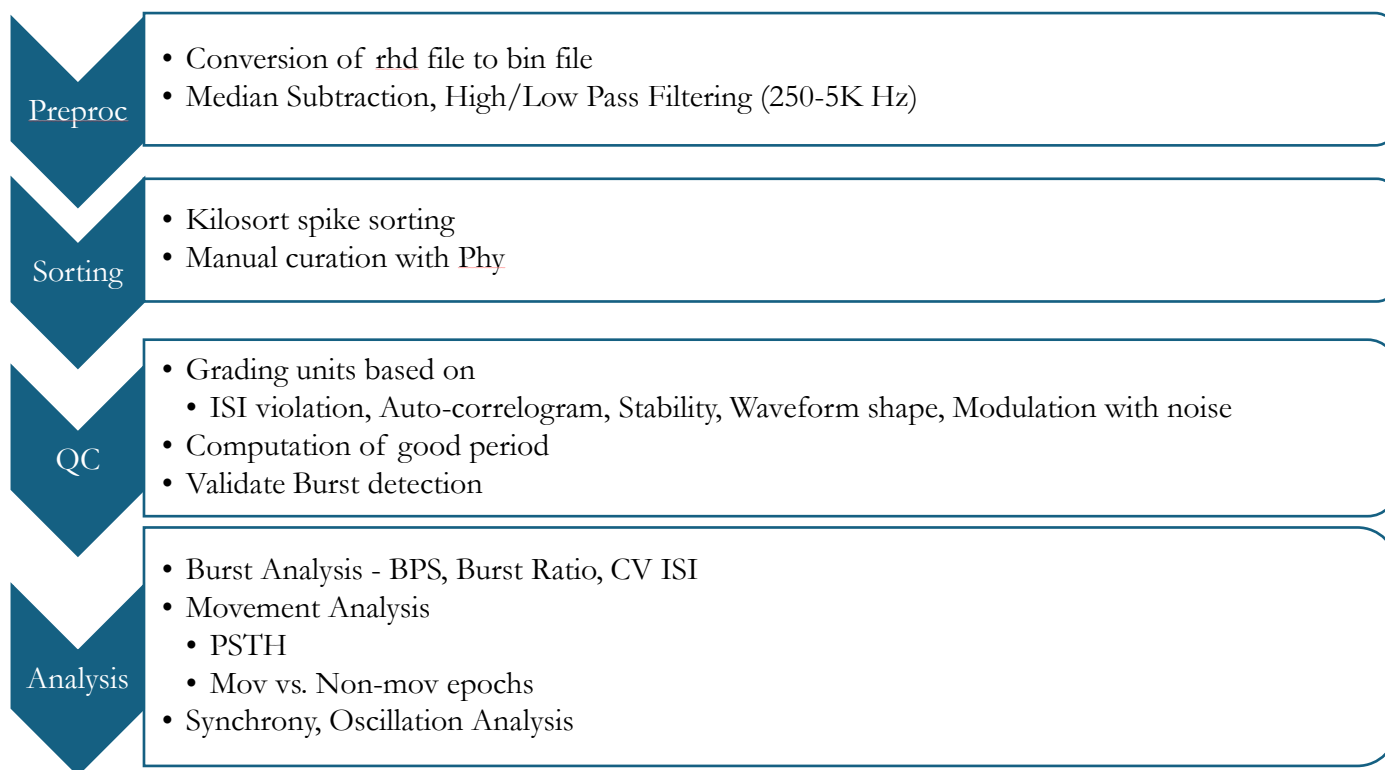
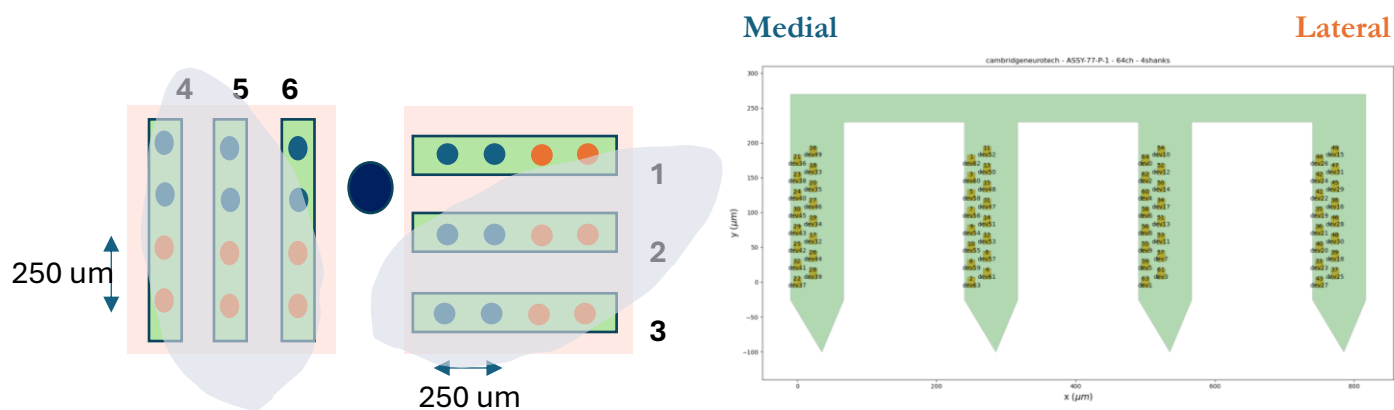


Figure D.2 Spatial Analysis using probe map for 64-channel recording probe to compare activity across medial and lateral SNr.



Metrics of FR Analysis

1. CV ISI

- Coefficient of variation of interspike intervals (measure of irregularity)

2. Burst Analysis:

- Categorical Analysis – Bursty vs Regular Units (based on ISI distribution)
- Surprise Burst Analysis

[Setting selected based on a control spike train with burst detection overlaid]

- Burst 1.5% higher than baseline frequency
- Min. # ISIs = 3
- Surprise cutoff value = 4
- Local length to compute FR = 50 sec
- Highly bursty units = >1 MAD from median baseline burst frequency*

3. Synchrony Analysis

- Population-Level Synchrony measure
- Step 1: Convolve Spike train data to continuous firing signal
- Step 2: Compute population-based synchrony measure for activation of pairs of neurons
- Pair-wise Windowed Cross-correlation Method
- Compute windows of 12 s, zeroed 1st and last 4 seconds of 2nd spike train
- Maximum lag set to 4s
- Computes CCF for 4s window with overlapping windows
- Output: Graph and cross-correlation as a function of the lag between two windows

4. Oscillation Analysis

- Power spectral analysis of spike trains
- Compute fraction of units in delta frequency band (0.5-4 Hz) vs. beta frequency band (7-35 Hz) across conditions
- Repeat spectral analysis with phase shift (more robust analysis due to processing of low-frequency pink noise)